PPG BASED AFIB PREDICTION AT RESOURCE CONSTRAINED ENVIRONMENTS

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DECLARATION

We declare that this is our own work and this Dissertation does not incorporate without acknowledgment any material previously submitted for a Degree or Diploma in any other University or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text. We retain the right to use this content in whole or part in future works (such as articles or books).

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The above candidate has carried out research for the Bachelor of Science Specialized in Computer Science Engineering Dissertation under my supervision. I confirm that the declaration made above by the student is true and correct.

Name of Supervisor: Dr. Chathuranga Hettiarachchi

Signature of the Supervisor: Date:

DEDICATION

We dedicate this dissertation to our families, friends, and mentors whose unwavering support, encouragement, and guidance have been invaluable throughout our academic journey. Their belief in us has inspired us to persevere through challenges and successfully complete this project.

We also dedicate this work to our supervisor, whose expertise, insightful guidance, and continuous encouragement have played a significant role in shaping and completing this research. This achievement would not have been possible without the support and contributions of all those who have accompanied us on this journey.

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ABSTRACT

Atrial fibrillation is the most common sustained cardiac arrhythmia and a major cause of stroke, yet its paroxysmal form is intermittent and frequently escapes detection during brief clinical assessment. Continuous monitoring with low-cost wearable sensors promises earlier diagnosis, but deep learning detectors trained on such signals often fail to generalise to patients they have never seen, because every individual carries a distinct heart-rhythm signature.

This dissertation studies the detection and early prediction of atrial fibrillation from sequences of inter-beat intervals, derived interchangeably from the electrocardiogram or from photoplethysmography, under computational constraints suited to wearable hardware. We propose a patient-aware contrastive learning objective that forms positive training pairs only from segments belonging to the same patient and the same rhythm class. This preserves each patient's own sinus-rhythm structure instead of folding all patients into one shared class prototype, which we argue is the property that drives transfer to unseen individuals.

A deliberately small multi-branch convolutional encoder is pretrained with this objective on the IRIDIA-AF database and assessed under a strict patient-independent protocol, in which no patient appears in both the training and the test data. The proposed method reaches an area under the receiver operating characteristic curve of 0.989 and a sensitivity of 97.4 per cent on unseen patients, while showing 2.6 times lower seed-to-seed variance than supervised contrastive learning. A direct analysis of the embedding geometry shows that per-patient sinus-rhythm cohesion, rather than global class separation, predicts cross-patient generalisation. Finally, the frozen representation extends to a per-patient early-warning score that anticipates the onset of atrial fibrillation without any retraining, supporting practical and individualised cardiac monitoring on resource-constrained devices.

Keywords: Atrial Fibrillation, PPG, Machine Learning, Deep Learning, Resource Constrained Environments

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LIST OF ABBREVIATIONS

Abbreviation	**Description**
AF	Atrial fibrillation
AI	Artificial intelligence
AUCs	Area Under the Curve
AUPRC	Area Under the Precision Recall Curve
AUROC	Area Under the Receiver Operating Characteristic
BCE	Binary Cross-Entropy
CLOCS	Contrastive Learning of Cardiac Signals
CNN	Convolutional Neural Network
CNN–LSTM	Convolutional Neural Network with a Long Short-Term Memory
CT	Computed Tomography
ECG	Electrocardiogram
EEG	Electroencephalography
EMG	Electromyography
HF	High Frequency
HRV	Heart rate variability
IPN	Irregular Pulse Notification
LF	Low Frequency
LF/HF	Ratio between Low and High Frequency
ML	Machine Learning
MRI	Magnetic Resonance Imaging
NPV	Negative Predictive Value
PAF	Paroxysmal AF
PCLR	Patient Contrastive Learning of Representations
pNN50	Percentage of successive Normal-to-Normal intervals that differ by more than 50 milliseconds
PPG	Photoplethysmography
PPV	Positive Predictive Value
RF	Random Forest
RMSSD	Root mean square of successive differences
RNN	Recurrent Neural Networks
RRI	RR-interval
SD1	Standard Deviation 1 (Short-term)
SD2	Standard Deviation 2 (Long-term)
SDNN	Standard Deviation of NN intervals

Abbreviation	**Description**
SimCLR	Simple Framework for Contrastive Learning
SR	Sinus Rhythm
SVM	support vector machine

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CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Atrial fibrillation (AF) is the most common sustained disturbance of the heart's rhythm and already affects more than 37 million people worldwide. It throws the heartbeat into a fast, irregular pattern, raises the risk of stroke roughly fivefold, and feeds into heart failure, dementia, and premature death [6]. The condition also grows more common with age. As populations grow older, the number of people affected keeps climbing and the burden on health systems grows heavier. This dissertation tackles the automatic *detection* and, as a forward-looking goal, *prediction* of *paroxysmal* AF, the earliest and most treatable stage of the condition, from the timing of heartbeats captured by low-cost wearable sensors.

1.1.1 Paroxysmal Atrial Fibrillation

The heart's two upper chambers, the atria, normally contract in a steady, coordinated rhythm. In AF this electrical activity becomes fast and disorganised, the atria stop contracting effectively, and the spacing between consecutive heartbeats turns erratic. This irregular beat-to-beat timing, the sequence of gaps between heartbeats known as the RR-interval (RRI) sequence, is the clearest sign of AF and the signal at the heart of this work [6]. AF is grouped by how long its episodes last: Paroxysmal AF (PAF) starts and stops on its own, usually within 48 hours and always within seven days, whereas the persistent and permanent forms last progressively longer until the arrhythmia becomes continuous [6, 7].

PAF is the best stage at which to step in. AF tends to worsen over time, as each episode causes small changes to the structure and electrical behaviour of the atria. These changes make the next episode more likely, a self-reinforcing cycle often summarised as *AF begets AF*. Left untreated, paroxysmal AF therefore tends to progress toward the persistent and permanent forms, by which point treatments that restore a normal rhythm work much less well. Detecting AF while it is still paroxysmal opens a window for effective treatments like anticoagulant (blood-thinning) drugs to prevent stroke, medication or catheter ablation to control the rhythm, and lifestyle changes that can lower stroke risk and slow the disease [7]. This early window matters all the more because even short, self-stopping episodes carry a stroke risk comparable to that of sustained AF.

The main difficulty is that PAF is, by its nature, brief and unpredictable. Episodes can last from seconds to hours, recur at irregular times, and in about one in three patients cause no symptoms at all [8]. A short test such as a standard resting Electrocardiogram (ECG) or a brief Holter recording will often record only normal rhythm and miss the arrhythmia entirely. The longer a person is monitored, the more likely an episode is captured. This is a strong argument for *continuous, long-term* monitoring during everyday life. And because AF reveals itself so clearly in the timing between heartbeats, analysing these interval sequences is an efficient and clinically meaningful basis for automatic screening.

1.1.2 Cardiac Monitoring in Resource-Constrained Settings

Continuous monitoring matters most precisely where it is hardest to provide. A clinical 12-lead ECG calls for a clinic visit, a trained technician, and electrodes placed across the chest and limbs. Across low- and middle-income countries, rural clinics, and community screening programmes, that kind of infrastructure is scarce or absent altogether; the patients most at risk of undiagnosed AF are often the ones furthest from it. This is the gap everyday wearables can close. A smartwatch or a single-lead patch costs a fraction of clinical equipment, asks little of the wearer beyond putting it on, and keeps recording through ordinary daily life rather than for the few minutes of an appointment.

Two complementary sensing modalities dominate this space; the method in this dissertation is designed to read either. The first is the single-lead ECG, built into chest patches and some smartwatches. It detects each heartbeat as a sharp electrical deflection (the R-peak) and records the interval between consecutive beats, the RR interval. The second is the Photoplethysmography (PPG) sensor, the small optical reader on the back of almost every smartwatch, fitness band, and pulse oximeter. Rather than sensing electrical activity, it shines light into the skin and measures the change in blood volume with each pulse, yielding pulse-to-pulse intervals [9]. Because these pulse intervals closely approximate the ECG's RR intervals, both modalities capture the same beat-to-beat timing and can serve as interchangeable inputs to a single analysis pipeline. Both also operate at negligible power cost, requiring neither full ECG-waveform storage nor continuous wireless transmission. That combination of ubiquitous hardware, an equivalent timing signal, and a minimal energy budget makes lightweight, privacy-preserving, real-time screening realistic at scale.

1.1.3 Machine Learning for Interval-Based AF Analysis

For years, the standard approach was to hand-pick a set of features from the beat-to-beat timing and feed them to a classifier. These features come from Heart rate variability (HRV) analysis and fall into three families. Time-domain statistics summarise how much the gaps between beats vary, such as the Root mean square of successive differences (RMSSD) , Standard Deviation of NN intervals (SDNN) , and Percentage of successive Normal-to-Normal intervals that differ by more than 50 milliseconds (pNN50) . Frequency-domain measures split that variation into low and high-frequency power. Nonlinear descriptors capture its complexity, including sample entropy and the geometry of the Poincaré plot. Paired with a threshold rule or a classical classifier such as a support vector machine (SVM) or Random Forest (RF) , these features reach respectable accuracy. They remain popular for one practical reason: their decisions are easy to interpret [10]. The approach has real limits, though. It relies on features that an expert must design by hand, is easily disturbed by ectopic or mistimed beats, and struggles to capture the subtle, person-specific dynamics that decide the hardest cases.

Deep learning changed this picture by learning discriminative features directly from the raw interval sequence, with no manual feature design. Convolutional Neural Network (CNN) and Recurrent Neural Networks (RNN) , along with hybrids such as Convolutional Neural Network with a Long Short-Term Memory (CNN–LSTM) models, achieve strong AF-detection performance on RRI data [11–13]. Two well-documented weaknesses remain, however. First, these supervised models are data-hungry, requiring large quantities of expertly annotated recordings. Second, and more fundamentally, they generalise poorly across patients: they tend to overfit the physiology of the training cohort rather than learn AF itself [12]. Because every individual carries a distinct cardiac baseline, a model that performs well under conventional evaluation can degrade sharply on patients it has never seen.

These limitations have pushed the field toward *representation learning*, where a model first learns a transferable embedding of the cardiac signal, a compact numerical code that captures its structure, on which a lightweight classifier can then be trained with only a handful of labels. Self-supervised and contrastive methods such as Simple Framework for Contrastive Learning (SimCLR) and supervised contrastive learning [14, 15], together with adaptations tailored to physiological and ECG signals [16, 17], show that a well-structured embedding improves both label efficiency and robustness to unseen data. There is a catch, though. Standard contrastive objectives treat all same-class recordings as interchangeable and so discard the per-patient structure that proves decisive for cross-patient transfer. That individual structure is precisely what a model must retain to generalise to a new patient. Recovering it is the gap this dissertation sets out to close. In parallel, the push toward on-device inference has produced compact convolutional and temporal-convolutional architectures that preserve accuracy under tight compute and memory budgets [18], a marked departure from earlier models built for server-grade infrastructure [11].

1.1.4 From Detection to Prediction

Interval sequences are useful for more than detecting AF that has already begun; they can also signal that an episode is approaching. In the minutes before onset, the autonomic nervous system's control of the heart shifts in measurable ways; these shifts are visible in the beat-to-beat timing recorded by both ECG and PPG [19, 20]. A large Holter study of paroxysmal AF illustrates this clearly. In the half hour before an episode, markers of vagal tone (the calming, parasympathetic branch of the nervous system) rise sharply; the burden of premature atrial contractions (early, out-of-sequence beats) climbs from roughly 4.8% to 8.3% in the final minutes [19]. These measurable precursors make the point clearly: AF is not only *detectable* once present but, to a meaningful degree, *predictable* in advance from the timing alone. Recent deep-learning systems have begun to exploit these early signs for warning [1, 10, 18, 21, 22].

What raises the stakes further is the uneven global distribution of AF burden: a large share of undiagnosed cases sit in precisely those settings where clinical-grade monitoring stays out of reach. Wearable-based screening, from ECG or PPG intervals alike, could bridge this diagnostic gap, but only with models that are at once accurate, robust enough to generalise to patients they have never seen, and lean enough to run on the wearable hardware itself. Those three requirements frame the challenges taken up in the rest of this chapter.

1.2 Problem Statement

Despite the promise of interval-based cardiac monitoring, deploying deep learning models for AF detection and prediction in resource-constrained and clinically diverse settings poses several fundamental challenges.

1.2.1 Cross-Patient Generalisation

Supervised deep learning models trained on RRI dynamics have demonstrated strong population-level performance but generalise poorly to patients whose personal cardiac baseline deviates from the training distribution [12]. This limitation is especially pronounced for PAF, where episode characteristics vary substantially across individuals and approximately one-third of monitored patients are asymptomatic [8]. The severity of this generalisation gap is illustrated empirically by the sensitivity drop from 98.98% under standard cross-validation to 86.04% on fully held-out patients reported by Andersen et al. [12] which is a decline exceeding 12 percentage points. Standard cross-validation protocols permit within-patient data mixing across folds, which inflates reported performance and conceals this fundamental limitation.

1.2.2 Representation Learning Under Inter-Patient Variability

Existing contrastive learning approaches for cardiac signals treat all same-class samples as interchangeable positives, collapsing the rich inter-patient structure of Sinus Rhythm (SR) and AF dynamics into a single shared prototype per class [14, 15]. This formulation is ill-suited for physiological signals where each subject contributes a distinct cardiac baseline. When class differences are partly aligned with subject differences, class-level contrastive objectives discard the individual variation that a model requires to generalise to unseen patients. Specifically, standard supervised contrastive learning folds each patient's distinct physiology into a single shared class cluster, removing the within-subject consistency that a linear probe must rely upon when encountering a new subject during inference.

1.2.3 Subtlety of Pre-Onset Precursors

Predicting an imminent AF episode is inherently harder than detecting an established one. The autonomic precursors that precede onset rising vagal tone and increasing ectopic burden are subtle, vary in their expression across individuals, and unfold against each patient's own fluctuating baseline [19]. A model that has erased per-patient structure in pursuit of clean global class separation is therefore poorly positioned to recognise the small, individual-specific deviations that signal an approaching episode, tightening the link between representation quality and predictive utility.

1.2.4 Computational Constraints

Real-world deployment on wearable hardware imposes strict constraints on model complexity, memory footprint, and inference latency. This is especially true for the PPG-equipped consumer devices that represent the most accessible monitoring platform in low-resource settings. Many existing deep learning models for AF detection are designed for server-grade infrastructure and cannot be deployed directly on edge devices [11], whereas recent work has shown that carefully designed lightweight architectures can retain accuracy on-device [18]. Bridging the gap between high-performing models and resource-constrained deployment is therefore a necessary condition for clinical translation.

1.3 Research Objectives

The primary aim of this research is to build a lightweight system that detects paroxysmal AF from cardiac interval sequences, whether they come from ECG or PPG, and to extend it toward early prediction of an oncoming episode. This sets two demands at once: the system must generalise reliably to patients it has never seen and remain small enough to run on the wearable itself. The specific objectives are as follows:

1. To characterise the geometry of the learned embedding under different contrastive objectives and identify why standard formulations fail to generalise across patients.

2. To design a patient-aware contrastive objective that preserves per-patient rhythm structure while learning discriminative representations that reliably separate AF from SR.

3. To develop a lightweight multi-branch convolutional encoder that runs efficiently on wearable hardware and works on both ECG-derived RR intervals and PPG-derived pulse intervals.

4. To evaluate the approach under a strict patient-independent protocol and benchmark it against state-of-the-art interval-based AF detection methods.

5. To develop a lightweight, per-patient early-warning score for AF onset that works in the learned embedding space, calibrating a personal sinus-rhythm-to-AF axis from one labelled AF episode and one SR episode, then tracking how far later segments drift along it.

1.4 Proposed Approach and Key Contributions

This dissertation proposes a *patient-aware contrastive learning* framework for PAF detection from cardiac interval sequences. The central insight is that what governs cross-patient transfer is not how cleanly classes are separated in the embedding space overall, but how consistently each individual patient is represented within that space. Standard supervised contrastive learning [15] folds each patient's distinct physiology into a single shared class cluster, removing the individual variation a linear probe relies on when it meets a new subject. The proposed objective addresses this by restricting positive pairs to same-patient, same-class segments, explicitly preserving each patient's individual SR baseline while still pushing the two classes apart.

An important empirical finding of this work is what we call the *Binary Cross-Entropy (BCE) paradox*. Binary cross-entropy training produces the cleanest global class separation in the learned embedding space, yet the most disordered per-patient structure and the worst transfer to unseen patients. This shows that clean global class separation is a misleading indicator of cross-patient generalisation, and that per-subject geometric consistency is the real predictor of robust transfer.

The key contributions of this work are:

1. **Patient-Aware Contrastive Objective.** A novel contrastive loss that restricts positive pairs to same-patient, same-class segments. This prevents the encoder from collapsing distinct individual SR baselines into a single shared prototype, the fundamental failure mode of standard supervised contrastive learning on physiological signals organised by subject.

2. **Embedding Geometry Analysis.** A principled analysis of the learned embedding geometry that explains why cross-patient generalisation fails. We show that per-patient SR cohesion, not global class separability, is the key predictor of downstream transfer. The patient-aware objective achieves the highest per-patient SR cohesion among the losses compared (0.850, against 0.800 for supervised contrastive loss and 0.772 for binary cross-entropy), confirming the intended geometric effect.

3. **Lightweight, Modality-Agnostic Encoder Architecture.** A computationally efficient multi-branch 1D-CNN encoder with temporal attention pooling, designed for edge deployment on wearable hardware. Its branches operate at multiple time scales, capturing beat-to-beat fluctuations, medium-range oscillations, and broader trends at once. The same encoder runs unchanged on ECG-derived

RR intervals and PPG-derived pulse intervals, within the limits of resource-constrained inference.

4. **State-of-the-Art Performance Under Strict Evaluation.** On the IRIDIA-AF dataset [23], under a strict patient-independent holdout protocol, the proposed method reaches an Area Under the Receiver Operating Characteristic (AUROC) of 0.989 ± 0.003, outperforming both the supervised contrastive and binary cross-entropy baselines. It also cuts seed-to-seed AUROC variance by $2.6\times$ and $3.4\times$ respectively, and exceeds the patient-holdout sensitivity reported by Andersen et al. [12] by over 11 percentage points.

5. **Towards Early Prediction.** Building on the detection core, we propose a per-patient early-warning scheme that works directly in the learned embedding space. For a given patient, one labelled AF episode and one SR episode are embedded and reduced to their class centroids, which define a personal *SR-to-AF axis*. Each new interval segment is then embedded, referenced to the SR centroid, and projected onto this axis. The result is a single score: how far the current rhythm has drifted from the patient's own sinus-rhythm baseline toward their AF region. A sustained drift gives early warning of an approaching episode. This scheme is well-posed only because the patient-aware objective keeps each patient's SR cluster tight and consistently placed, the very SR-cohesion property that the BCE objective fails to preserve. It also needs only a lightweight, few-shot calibration per patient, with no retraining.

1.5 Scope and Limitations

This research focuses on detecting PAF from cardiac interval sequences in a supervised representation-learning setting. Early prediction is a forward-looking extension, not the principal contribution. We develop and validate the framework on the IRIDIA-AF dataset [23], which provides long-term single-lead ECG recordings from 167 patients with paroxysmal AF. PPG-derived pulse intervals serve as an interchangeable input on the strength of their established physiological correspondence to RR intervals [9]; that correspondence is assumed here rather than validated on PPG recordings. The evaluation protocol is deliberately conservative. We split the data at the patient level and keep the encoder frozen during linear-probe evaluation, so that downstream performance reflects the quality of the representation rather than the capacity of the classifier.

Several directions lie outside this scope: end-to-end fine-tuning, multi-centre prospective validation, validation on PPG-native recordings, and direct on-device deployment benchmarking. Each is an important target for future work, taken up in Chapter 7.

1.6 Thesis Organisation

The remainder of this dissertation is organised as follows.

Chapter 2 presents a comprehensive review of related work, covering traditional HRV-based AF detection and prediction methods, supervised deep learning approaches for RRI and PPG-interval classification, autonomic precursors of AF onset, and recent advances in contrastive and self-supervised representation learning for physiological signals. The chapter culminates in an identification of the research gap that motivates the present work.

Chapter 3 describes the proposed patient-aware contrastive learning framework in full detail, including the dataset and preprocessing pipeline, the patient-aware mini-batch sampling strategy, the contrastive objective formulation, the lightweight encoder architecture, and the end-to-end training procedure.

Chapter 4 presents the experimental results, including the embedding geometry analysis, downstream PAF detection performance, ablation over loss functions, and comparison with prior interval-based AF detection methods.

Chapter 5 provides an evaluation of the proposed system in the context of resource-constrained deployment, examining computational characteristics of the encoder, the cross-modality (ECG/PPG) applicability of the approach, and the practical implications of the patient-independent evaluation protocol.

Chapter 6 discusses the findings, interprets the BCE paradox, analyses the clinical relevance of seed-stable sensitivity, considers the prospects for early prediction from autonomic precursors, and reflects on the generality of the proposed objective to other physiological signals organised by subject.

Chapter 7 concludes the dissertation and outlines concrete directions for future work, including few-shot patient adaptation, prospective early-warning validation on PPG wearables, multi-centre validation, and extension of the patient-aware positive construction to Electroencephalography (EEG) , Electromyography (EMG) , and PPG modalities.

CHAPTER 2

LITERATURE REVIEW

Atrial fibrillation has been approached computationally from two complementary directions: *detection*, which identifies AF in a signal as it is recorded, and *prediction*, which anticipates an episode before it begins. This chapter reviews both, together with the continuous-monitoring infrastructure that makes them clinically useful and the machine-learning paradigms on which modern systems are built. It proceeds from established detection methods, through wearable and edge-based continuous monitoring, to AF prediction, and finally to the representation-learning advances that motivate this dissertation.

2.1 Atrial Fibrillation Detection

Reliable identification of AF in a recorded signal is the foundation on which screening and monitoring systems are built. Detection methods have progressed from rule-based analyses of beat-to-beat timing toward data-driven models that learn directly from raw signals.

2.1.1 Early Methods

Before the widespread adoption of machine learning, automated AF detection relied on explicit characterisation of the irregular ventricular response that defines the arrhythmia. Because AF abolishes the regular atrial pacing of sinus rhythm, the resulting RR-interval series becomes markedly irregular, and early detectors quantified this irregularity directly. Statistical approaches modelled the distribution and transitions of successive RR intervals: Markov-process surrogates of the RR sequence [24], for instance, and tests based on the dispersion of successive interval differences [25]. Related techniques examined the geometry of Poincaré (Lorenz) plots of consecutive intervals, where AF produces a characteristically dense, dispersed cloud rather than the compact cluster of sinus rhythm, while randomness- and entropy-based measures of the RR series were subsequently shown to discriminate AF robustly [26]. These methods are computationally trivial and interpretable, but depend on hand-tuned thresholds, are sensitive to ectopic beats and noise, and offer limited capacity to capture the subtler patterns that separate difficult cases.

2.1.2 AI/ML-Based Methods

The limitations of threshold-based detectors motivated a shift toward machine learning, in which models learn the signal characteristics of AF from labelled data rather than from hand-specified rules.

2.1.2.1 Machine Learning in AF Detection

Machine-learning detectors are commonly grouped into supervised, unsupervised, and semi-supervised approaches [2]. Supervised learning dominates: labelled ECG or PPG segments train classifiers ranging from logistic regression, support vector machines, and random forests to deep neural networks, which have performed strongly on AF classification and cardiac risk stratification. Unsupervised methods instead mine unlabelled recordings for recurring structure, clustering rhythms and flagging anomalies in large ECG archives. Semi-supervised learning occupies the middle ground when labels are scarce [27].

Deep networks have driven most of the recent gains. A Stanford model trained on more than 90,000 single-lead ECGs reached an area under the curve above 0.91 across ten arrhythmia classes [28], and Mayo Clinic researchers trained convolutional networks on over 2.5 million twelve-lead ECGs from more than 720,000 adults, matching expert over-reading and outperforming commercial interpretation [29]. Across studies, reported AF sensitivities span 80–100%, with convolutional architectures at the upper end [27, 28]; Appendix A compares the reported operating points of these architectures. The practical appeal is continuous, scalable screening that tolerates real-world noise. Against conventional pathways that miss up to 85% of AF in annual Holter checks [2], learned detectors offer a more dependable basis for early detection and long-term monitoring.

2.1.2.2 Other AI Methods for AF Detection

Beyond classifiers operating on extracted features, a distinct line of work reformulates AF detection as an image-recognition problem. The one-dimensional ECG or RR-interval series is first transformed into a two-dimensional representation—most commonly a recurrence plot, which encodes the instants at which the signal revisits similar states—and a convolutional network is then trained on the resulting images [30]. This rendering makes the loss of periodicity that accompanies AF visually explicit, letting established image-classification architectures apply directly. A related thread targets robustness to the noise typical of portable recordings: deep belief networks built from stacked restricted Boltzmann machines have sustained detection sensitivities of 96–100% even at very low signal-to-noise ratios [31]. Both illustrate a theme that recurs throughout this chapter, that the choice of signal representation shapes detection performance as much as the choice of classifier.

2.2 Continuous Monitoring

Detecting AF reliably is necessary but not sufficient for clinical impact: because paroxysmal episodes are intermittent and frequently asymptomatic, the detector must be embedded in a system capable of observing the heart over extended periods. This section reviews what continuous monitoring entails, the wearable devices that deliver it, the constraints of running detection at the edge, and the limitations that remain.

2.2.1 What is Continuous Monitoring

Continuous monitoring refers to the uninterrupted, or near-uninterrupted, observation of a physiological signal over hours, days, or weeks, as opposed to the brief snapshot provided by a clinic-based examination. For AF the rationale is direct: diagnostic yield rises with the duration of monitoring, because longer observation increases the probability of capturing a transient episode [32]. Conventional ambulatory tools such as 24- to 48-hour Holter monitors extended observation beyond the resting ECG but remain short relative to the irregular recurrence of paroxysmal AF; patch recorders and implantable loop recorders push the window further at the cost of invasiveness or expense. Consumer wearables have since reframed continuous monitoring as a low-cost, always-available service, shifting the bottleneck from data acquisition to reliable automated interpretation.

2.2.2 Wearables in Continuous Monitoring

Wearable devices such as smartwatches, rings, and wristbands have become popular tools for continuous health monitoring. Their sensors capture vital signs including heart rate, blood pressure, and oxygen saturation [33], and can operate either independently or alongside a smartphone [34]. Beyond general wellness tracking, they support the detection and management of arrhythmias, coronary artery disease, and diabetes [33, 35], and have become practical tools for screening atrial fibrillation in both symptomatic and asymptomatic individuals. The population-scale trials that establish this clinical value, principally the Apple and Fitbit heart studies, are examined later in this section.

Several sensing technologies are built into these devices [36]: accelerometers infer heart rate from the body movements of each contraction through ballistocardiography, while phonocardiography and seismocardiography read cardiac sounds and vibrations [37]. Two modalities dominate AF screening, however, because they recover the beat-to-beat timing that the arrhythmia disturbs, and the rest of this section concentrates on them: photoplethysmography and the single-lead electrocardiogram.

2.2.2.1 PPG in Wearables

Photoplethysmography (PPG) is the technique most commonly built into smartwatches for AF screening. It shines light from a light-emitting diode into the skin and measures the reflected intensity with a photodetector [38]; because that intensity tracks the change in blood volume with each pulse, the resulting waveform is a pulse-pressure trace that supports passive, continuous heart-rhythm monitoring at the wrist. The peak of each pulse aligns closely with the ECG R-wave, so the interval between successive peaks approximates the RR interval and exposes the same timing irregularity that marks AF [39].

This correspondence is what licenses the central modelling assumption of this dissertation, that a model trained on ECG RR intervals can read PPG pulse intervals unchanged. The variability of pulse-to-pulse intervals, termed pulse-rate variability (PRV), has been validated as a surrogate for ECG-derived heart-rate variability in healthy subjects at rest, agreeing closely on the standard time- and frequency-domain indices [9]. That agreement narrows during motion, arrhythmia, and other low-perfusion conditions, so PRV remains an approximation rather than an identity. For the interval-timing features that separate AF from sinus rhythm, though, the two signals carry equivalent information, which is why a single interval-based pipeline can serve either modality.

2.2.2.2 Challenges with PPG in Wearables

Despite the widespread use of PPG, it faces multiple challenges. Poor sensor–skin contact often leads to missing or inconclusive/unclassified signals, while motion artifacts (e.g., from muscular activity) degrade signal quality [40]. PPG accuracy is lower at very high or low heart rates compared to standard ECG-based methods, and ectopic beats can generate irregularities that mimic AF, complicating detection [41].

Physiological factors compound these limits. Skin pigmentation [42], ageing-related changes such as reduced peripheral blood flow and arterial stiffness, and weak signals in elderly populations all degrade detection and raise concerns about generalisability across populations [39].

2.2.2.3 ECG in Wearables

Another major approach to AF detection in consumer wearables relies on recording an ECG. In this method, the back of the smartwatch acts as the positive electrode, while the contralateral fingertip placed on the watch crown serves as the negative electrode, creating a bipolar lead configuration similar to Einthoven's Lead I[33]. The recorded tracing is stored locally and can be analyzed either manually by clinicians or automatically by embedded algorithms. Beyond single-lead ECG, more advanced configurations, such as smartwatch-based wireless six-lead ECGs, have also been explored to improve diagnostic accuracy [39].

2.2.2.4 Challenges with ECG in Wearables

The small electrode spacing on a smartwatch produces low-amplitude P-waves, which makes atrial activity hard to classify reliably [39]. Accurate ECG recordings also demand active user participation, such as holding a finger to the crown, which rules out unattended monitoring [33]; regular charging and consistent cooperation further limit it. These constraints make ECG less practical than PPG for long-term, passive AF screening.

2.2.3 Edge Processing for On-Device Detection

Artificial intelligence (AI) methods operationalize AF detection on resource-constrained wearables and smartphones by (i) preprocessing PPG/ECG signals, (ii) extracting time and frequency-domain features or learning representations directly from raw waveforms, and (iii) classifying rhythm using traditional Machine Learning (ML) (e.g., SVM, random forests) or deep learning (e.g., CNNs/RNNs) [39].

Building on this pipeline, recent work has demonstrated fully edge-deployable AF detection. For example, the *Cardiio Rhythm* algorithm uses a supervised SVM trained on facial and fingertip PPG acquired with an iPhone camera; in a prospective evaluation of 217 cardiology inpatients it achieved 95% sensitivity, 96% specificity, 92% Positive Predictive Value (PPV) , and 97% Negative Predictive Value (NPV) for discriminating AF from sinus rhythm, showing that camera-based smartphone PPG can robustly screen for AF without external hardware [43].

Similarly, a smartwatch CNN (SmartRhythm 2.0) trained on heart rate, activity, and single-lead ECG from 7,500 users and validated against insertable cardiac monitors in 24 patients achieved 97.5% episode sensitivity, 97.7% duration sensitivity, and a 39.9% PPV, illustrating that consumer wearables can deliver high-sensitivity AF surveillance directly on the wrist while leaving room to improve precision [44].

Deploying a detector on a wearable is as much a question of model design as of pipeline engineering. A smartwatch budgets only a few hundred kilobytes of memory and a sliver of energy per inference, so architectures intended for the wrist favour compactness over raw depth. Small one-dimensional convolutional networks, depthwise-separable convolutions that factorise spatial and channel mixing, and temporal convolutional networks whose dilated kernels span long contexts at low parameter cost have all been used to this end. Recent work in this vein performs AF analysis from interval sequences with only thousands of parameters and sub-millisecond inference while retaining the accuracy of far larger models [18], in deliberate contrast to early deep detectors built for server-grade infrastructure [11]. This budget-first philosophy, rather than the post-hoc compression of a large network, is the one the encoder developed in this dissertation adopts.

2.2.3.1 PPG based AFib detection using AI and ML

Two landmark pragmatic trials has been highlighted in the literature to show the feasibility of population-scale AF screening using smartwatch-based photoplethysmography (PPG).

The *Apple Heart Study* enrolled over 419000 participants (mean age 41 ± 13 years; 42% women) who wore an Apple Watch. An Irregular Pulse Notification (IPN) required at least 5 out of 6 consecutive irregular 1-minute tachograms within 48 hours. Only 0.52% of the population received an IPN, though this proportion rose to 3.2% among participants ≥ 65 years. Those receiving an IPN were mailed an ECG patch for confirmatory monitoring; of these, 34% were diagnosed with AF lasting at least

30 seconds, yielding a positive predictive value (PPV) of 84.0% (95% CI, 76.0–92.0), with a slightly lower PPV of 78.0% in those ≥ 65 years [45].

Similarly, the *Fitbit Heart Study* enrolled 455,699 participants (median age 47 years; 71% women). Its algorithm used more continuous PPG monitoring and stricter criteria, requiring at least 30 minutes of irregular rhythm to trigger notification. One percent of participants (3.6% of those ≥ 65 years) received an IPN, after which they were mailed a confirmatory ECG patch. Among returned patches, 32% were diagnosed with AF, yielding an overall sensitivity of 67.6%, specificity of 98.4%, and PPV of 98.2% (95% CI, 95.5–99.5) [46].

Beyond large-scale pragmatic trials, several smaller validation studies have examined smartwatch performance in higher-risk cohorts under controlled conditions. [47] evaluated the Samsung Galaxy Watch Active 2 in 204 participants with either known AF or elevated risk factors (age ≥ 65, hypertension, diabetes, heart failure, or coronary artery disease). Continuous PPG monitoring demonstrated a sensitivity of 87.8% and specificity of 97.4%. When combined with on-demand single-lead ECG, diagnostic performance improved further, with sensitivity 96.9% and specificity 99.3% against a 28-day ECG patch reference standard.

These studies confirm that consumer wearables using PPG can achieve high specificity and clinically meaningful PPV for AF detection at scale.

2.2.4 Limitations of ML-Based AF Detection

Despite this progress, several limitations temper the clinical use of wearable AF detection. The sharpest concern is the cost of errors at screening scale. False positives generate unnecessary follow-up and burden clinicians, while false negatives offer false reassurance; precision and threshold calibration therefore matter as much as headline sensitivity [5, 39, 48]. Inconclusive tracings remain common and still demand expert over-reading. The long-term clinical benefit of smartwatch-detected AF, particularly in asymptomatic individuals, is not yet established [49]. Practical and policy questions compound these, from the limited integration of wearable streams into clinical records to the absence of settled regulatory and reimbursement frameworks [39].

A further limitation is interpretability. Clinicians are reluctant to act on a prediction they cannot inspect. The most accurate detectors, however, are typically opaque deep networks. This tension motivates the wider move toward models whose internal representations can be examined and reasoned about, the subject to which this chapter returns after prediction.

2.3 Prediction of Atrial Fibrillation

Whereas detection establishes that AF is present, prediction seeks to anticipate an episode before it begins, opening the possibility of pre-emptive intervention. Approaches to AF prediction span clinical risk scores, imaging and electrophysiological markers, and machine-learning models operating on interval dynamics. This section first sets out the clinical case for prediction, then traces its evolution from population risk scores to interval-based deep learning, and closes on the requirements of predicting at the edge.

2.3.1 Why Prediction is Needed

The clinical value of prediction follows from the natural history of AF. Each episode promotes electrical and structural remodelling of the atria that makes subsequent episodes more likely, so the disease tends to progress from paroxysmal to persistent and permanent forms unless interrupted. Anticipating an episode therefore creates a window for pre-emptive action—initiating or adjusting antiarrhythmic therapy, prompting vagal manoeuvres, or alerting the patient—that detection after onset cannot provide. This window is physiologically plausible because measurable precursors precede onset: autonomic balance shifts in the minutes beforehand, with rising markers of vagal tone and an increasing burden of premature atrial contractions [19, 20]. The existence of such precursors is what makes short-horizon prediction from interval dynamics feasible, and distinguishes prediction from the related but distinct task of long-term risk stratification.

2.3.2 Early Methods

The earliest attempts to anticipate AF were epidemiological rather than signal-based, estimating an individual's longer-term likelihood of developing the arrhythmia from readily available clinical variables. These population-level instruments answer a different question from the short-horizon, signal-based forecasting pursued later in this chapter: they estimate years-scale risk for cohorts rather than imminent onset for an individual. Nonetheless, they established AF risk modelling as a quantitative discipline and remain the baseline against which machine-learning predictors are measured.

2.3.2.1 Traditional Clinical Risk Scores

The earliest quantitative predictors were clinical risk scores built from routinely available patient characteristics: age, sex, body mass index, blood pressure, smoking status, comorbidities such as diabetes and heart failure, and simple ECG parameters. The Framingham [50], ARIC [51], CHARGE-AF [52], C_2HEST [53], and HATCH [54] models are representative; together they typically reach Area Under the Curve (AUCs) of 0.70–0.78 on their development cohorts [3]. Appendix A tabulates each model with its development- and validation-cohort AUCs. Their weakness is transfer. C_2HEST, for instance, fell from an AUC of 0.75 on its Chinese derivation cohort to 0.59 on a Danish validation set [53, 55]. This unevenness, together with a reliance on demographic rather than signal-level information, is what motivated the move to data-driven predictors.

2.3.3 ML-Based Methods

Data-driven models extend prediction beyond fixed clinical scores, learning from richer sources (electronic health records, cardiac imaging, electrophysiological markers, and interval dynamics) and typically outperforming the conventional scores on the same populations.

2.3.3.1 Risk Scores using ML

Machine-learning models exploit the full richness of electronic health records to sharpen these predictions. Tiwari et al. [56] applied ML to nearly two million patients with over 200 clinical features, reaching an AUC of 0.79; Sekelj et al. [57] reported AUCs up to 0.87 on UK primary-care records, outperforming the conventional scores; and Hill et al. [58] showed that time-varying covariates lifted performance to an AUC of 0.827, against 0.725 for CHARGE-AF on the same population.

2.3.3.2 Cardiac Imaging Approaches for AF Prediction

Structural and tissue abnormalities visible on cardiac imaging are also predictive of AF. Echocardiographic markers such as left-atrial volume and diastolic dysfunction [59–61], atrial and pulmonary-vein morphology on CT [62, 63], and left-atrial fibrosis quantified by late-gadolinium Magnetic Resonance Imaging (MRI) all carry predictive value; fibrosis above 6%, for example, predicted new-onset AF at an AUC of 0.67, rising to 0.80 once combined with clinical variables [64]. Applying ML to Computed Tomography (CT) morphology has pushed AUCs for post-ablation recurrence to 0.78–0.87 [65, 66]. These approaches sit outside the scope of this dissertation, however: imaging demands clinical hardware, is largely confined to small peri-procedural cohorts, and is neither continuous nor wearable, the very constraints that motivate interval-based screening. Appendix A sets out these obstacles in full.

2.3.3.3 Electrophysiological Data Approaches for AF Prediction

Electrophysiological markers tie prediction more directly to the heart's electrical behaviour. On the surface ECG, P-wave duration, dispersion, and amplitude, together with the morphology and frequency of premature atrial contractions (PACs), predict new-onset AF at AUCs between 0.69 and 0.87 [67–69]; frequent PACs on Holter monitoring raise AF risk, although their standalone value can be modest [70]. Invasive intracardiac electrograms add features such as dominant frequency and organisation indices that predict post-ablation recurrence, but their reach is confined to patients already undergoing electrophysiology procedures [71].

Deep learning has sharpened these signals. A landmark Mayo Clinic study trained a CNN on more than 600,000 sinus-rhythm ECGs and predicted new-onset AF at an AUC of 0.87, rising to 0.90 with multiple ECGs per patient [72]; ensembles of linear, nonlinear, and time-frequency HRV features have reached 98.2% accuracy on paroxysmal-AF cohorts [73]; and models that let covariates vary over time improve on static-feature baselines [58]. By reading raw waveforms, such models surface patterns that hand-engineered features miss. Combining clinical, electrophysiological, and imaging modalities is expected to push performance further still [3]. Their opacity is the recurring caveat. The most accurate predictors behave as "black boxes" whose decisions resist inspection, which has spurred interest in explainable-AI methods that expose the features driving a warning [3].

One such interpretable marker is P-wave dispersion, which reflects intra- and inter-atrial conduction abnormality and predicts AF recurrence, especially in women and in patients with higher QT dispersion [74, 75]. Because its physiological substrate, diastolic dysfunction, is also detectable from the pulse waveform, it points to a noninvasive route to early prediction from PPG.

2.3.3.4 Prediction of AF using HRV Metrics and Raw RR Intervals

The line of work closest to this dissertation predicts AF onset directly from RR-interval dynamics and heart-rate variability, evolving from handcrafted-feature pipelines toward end-to-end deep learning. Classical approaches extract HRV features across the time, frequency, nonlinear, and geometric domains [76]. Time-domain measures (SDNN, RMSSD, pNN50) and frequency-domain ratios (Low Frequency (LF) , High Frequency (HF) , Ratio between Low and High Frequency (LF/HF)) capture the autonomic shifts that precede onset, though the linear measures often need long windows that blunt short-horizon prediction. Nonlinear descriptors are more sensitive: a progressive fall in approximate and sample entropy and in short-term fractal scaling marks the destabilisation of sinus rhythm before an episode. Geometric features from the Poincaré plot, principally Standard Deviation 1 (Short-term) (SD1) and Standard Deviation 2 (Long-term) (SD2) , contract in the minutes before onset; kernel-density refinements of this idea rank among the most discriminative handcrafted predictors. Adding low-level ECG morphology to the interval features sharpens accuracy further [77].

End-to-end models remove the manual feature design altogether, sliding short RRI windows through one-dimensional convolutional networks that learn pre-AF dynamics directly. These cut engineering effort and tolerate noise. Two stubborn weaknesses remain, however: low recall on the positive AF class, and limited generalisation given the wide morphological variability of AF across patients. To counter both, [4] recast the interval series as a two-dimensional recurrence plot, a representation that encodes when the signal's trajectory revisits earlier states, and classified it with a ResNet after wavelet denoising. The recurrence structure renders the loss of periodicity before an episode visually explicit; quantification features such as recurrence rate, entropy, and trapping time separate pre-AF from non-AF windows. On the PAF Prediction Challenge Database this approach reported 97% accuracy [78], though it relies on the full ECG waveform and performs only offline screening rather than real-time monitoring.

TABLE 2.1: Model performance on different datasets.[1]

Dataset	Accuracy (%)	AUROC	AUPRC	Predicted time horizon	
				Mean	Median
Test (RRI)	82.7	0.90	0.88	30.8 min	38.0 min
Test (ECG)	82.4	0.95	0.96	32.5 min	43.4 min
External centers	75.0	0.80	0.73	31.8 min	41.3 min

Appendix A reproduces its recurrence-plot construction and quantification metrics in full.

2.3.4 Prediction at the Edge

The recurrence-plot detectors above depend on the full ECG waveform, which a wrist-worn device cannot supply. A more deployable line of work generates the same recurrence representations from RR intervals alone [26], since interval series can be read from the smartwatches and fitness bands people already wear, enabling continuous real-time monitoring in daily life.

WARN is the closest prior work to this dissertation and the clearest demonstration of the idea [1]. It slides a 30-second RRI window every 15 seconds, converts each window to a recurrence plot, and classifies it with an EfficientNetV2 convolutional network into sinus rhythm, a pre-AF regime, and AF. Summing the pre-AF and AF probabilities yields a continuous "probability of danger" that rises as onset approaches; smoothing it with a causal moving average and thresholding the result issues an early warning. A 30-second window proved the best trade-off between sample count and temporal context. On a held-out test set WARN predicted onset on average 31 minutes ahead at 83% accuracy; on external cohorts from other centres it managed roughly 32 minutes ahead at 75% accuracy (Table 2.1). Appendix A details WARN's labelling procedure, network, hyperparameter selection, and error analysis.

A central finding is how little is lost by discarding the waveform. Training the same network on raw ECG rather than RRI raised AUROC and AUPRC by only about 5% and 9%, leaving accuracy and lead time essentially unchanged. Accurate onset prediction is therefore achievable from intervals alone, which is what makes wrist-based early warning practical. The authors further reported a PPG analysis on 554 patients, the largest cohort in their study, where an XGBoost classifier over engineered RRI features reached 88% accuracy.

That PPG result connects to the work of Guo et al., which predicts AF onset from PPG pulse intervals in high-risk individuals [79]. Built on the Huawei Heart Study [80], it extracts time-, frequency-, and nonlinear-domain features from the interval series and trains an XGBoost classifier; the model reached an AUC of 0.94 four hours before onset, improving to 0.97 once enriched with the history of recent intervals and contextual information. Precision improved over the baseline model, though many residual false alarms traced to non-AF atrial arrhythmias such as bigeminy and flutter,

TABLE 2.2: Comparison with previous studies on AF prediction.

Year	Study	Method	Patients (no.)	Window length	Prediction horizon	Accuracy (%)	Sensitivity (%)	Specificity (%)
2012	Mohebbi et al. [81]	RRI, SVM	NR	30 min	onset	96	96	93
2013	Costin et al. [77]	ECG, QRS complexes	75	5 min	onset	90	89	89
2016	Boon et al. [82]	RRI, SVM	53	30 min	onset	80	81	79
2018	Li et al. [83]	ECG, Markov chain	5	2 min	onset	82	86	80
2018	Boon et al. [84]	RRI, SVM	53	5 min	onset	87	86	88
2018	Ebrahimzadeh et al.[39]	ECG, mixture of experts	53	5 min	onset	98	100	96
2021	Guo et al. [79]	RRI, XGBoost	554	1–4 h	onset	88	82	96
2021	Tzoul et al. [85]	ECG, CNN	8	5 min	onset	89	88	89
2022	Grégoire et al. [20]	RRI, CNN	140	300 RRI ($\sim$5 min)	30 beats ($\sim$30 s)	66	80	53
2023	WARN [1]	RRI, CNN	350	30 s	30.8 min before onset	83	95	70
2023		ECG, CNN	350	30 s	32.5 min before onset	82	95	69

underlining how hard it is to separate impending AF from other rhythm disturbances. The principal caveats are transparency and reach: the Huawei dataset is proprietary; the architecture and any edge deployment are only thinly described; and validation was confined to small high-risk cohorts. Appendix A lists the full feature set and the M1/M2 model comparison. Table 2.2 places these interval-based predictors alongside earlier work.

2.4 Representation Learning

The detection and prediction systems reviewed above are predominantly supervised, and their effectiveness is bounded by two factors: the availability of large, expertly labelled datasets, and the ability of the trained model to generalise beyond the patients it was trained on. Representation learning addresses both by separating the problem of *encoding* a signal from the problem of *labelling* it. Rather than optimising an encoder end-to-end for a single classification target, a representation-learning system first learns a general-purpose embedding of the signal, which a lightweight downstream model can then exploit for one or more tasks. This section reviews the shift from supervised to self-supervised learning, clarifies what is meant by a representation and how its quality is assessed, and surveys the application of these ideas to physiological signals.

2.4.1 From Supervised to Self-Supervised Learning

Supervised learning maps signals to labels, and its success in AF detection and prediction has depended on large, expertly annotated datasets. In physiological settings this dependence is costly: expert annotation of long recordings is laborious, rare events such as pre-AF segments are under-represented, and models trained this way tend to absorb the idiosyncrasies of the training cohort, generalising poorly to new patients

[12]. Self-supervised learning offers an alternative by deriving its training signal from the structure of the data itself rather than from external labels: a model first learns a general-purpose representation from large quantities of unlabelled signal, after which a small labelled set suffices to train a downstream classifier. For cardiac monitoring—where unlabelled wearable data are abundant but expert labels are scarce—this decoupling is especially attractive; it reframes the central question from how a segment should be classified to what representation makes classification transfer across patients [15].

Self-supervised methods are commonly grouped into *generative* approaches, which learn by reconstructing masked or corrupted portions of the input, and *contrastive* approaches, which learn by comparing examples. Contrastive Predictive Coding introduced the influential principle of predicting future latent states from past context through a contrastive loss [86]; this principle has since been adapted to time series: Time-Series Temporal and Contextual Contrasting (TS-TCC), for instance, learns representations by enforcing consistency between differently augmented views of a signal across both temporal and contextual dimensions [87]. In the physiological domain these ideas are particularly compelling because the dominant cost is expert annotation rather than data collection—continuous wearable streams are plentiful, whereas beat- or episode-level labels are scarce and expensive—so a pretraining stage that exploits unlabelled signal directly addresses the principal bottleneck identified above.

2.4.2 What a Representation Is

Representation learning shifts the objective from predicting a label to learning an embedding—a vector encoding of the signal in which semantically meaningful structure is laid out geometrically. Once such an embedding is learned, downstream tasks reduce to simple operations within it: classification becomes a linear boundary, similarity becomes distance, and, as exploited later in this dissertation, progression toward an arrhythmic state can be read as movement along a direction in that space. The classical view holds that a good representation disentangles the underlying factors of variation in the data, remaining invariant to nuisance factors while staying sensitive to the factors of interest [88]. For a cardiac interval signal this means an embedding that is insensitive to recording noise, baseline wander, and benign inter-individual differences in resting heart rate, yet responsive to the timing irregularity that distinguishes AF from sinus rhythm. The quality of a representation is therefore judged not only by downstream accuracy but by how faithfully its geometry mirrors the underlying physiology—whether recordings from the same patient cluster coherently, and whether clinically distinct rhythms occupy separable regions.

2.4.3 Evaluating Representations

How a representation is judged is as important as how it is learned. The standard protocol freezes the trained encoder and fits a simple linear classifier—a *linear probe*—

on top, so that downstream accuracy reflects the quality of the representation rather than the capacity of the classifier. Complementary evidence is obtained from nearest-neighbour classification in the embedding space, from transfer to held-out tasks, and from label-efficiency curves that measure accuracy as a function of the number of downstream labels. The intrinsic geometry of the embedding can be examined directly through clustering measures and through the notions of *alignment* (positives mapping to nearby points) and *uniformity* (the embedding making full use of the representation space), which have been proposed as the two properties a good contrastive representation should satisfy [89]. These criteria matter directly for cardiac monitoring, where the practical question is not whether a powerful classifier can fit the training patients, but whether a frozen representation continues to separate rhythms for a patient it has never seen; per-patient consistency of the embedding, rather than aggregate class separability, is therefore the operative measure of quality in this work.

2.4.4 Representation Learning for Physiological Signals

Applied to biosignals, representation learning has progressed from unsupervised reconstruction toward self-supervised pretext tasks tailored to the physiology. Early work used autoencoders and deep metric learning to compress ECG morphology into compact codes, while more recent studies pretrain encoders on large unlabelled ECG corpora and report strong transfer to diagnostic tasks with only a fraction of the labels required by fully supervised training [90]. Time-series-specific frameworks exploit the sequential structure of the signal through temporal and contextual contrasting [87], and the broader trend toward large pretrained "foundation" encoders is beginning to reach cardiology. Two difficulties nonetheless distinguish the physiological setting from the visual one in which these methods were developed: the augmentations that define similarity must preserve diagnostic content rather than merely perceptual identity, and the data are organised by subject, so the encoder can too easily learn to represent *who* produced a signal rather than *what* the signal contains. These difficulties are the subject of the contrastive methods reviewed next.

2.5 Contrastive Methods

Contrastive learning has become the dominant self-supervised paradigm for images and, increasingly, for physiological signals. Its central idea is geometric: an encoder is trained so that representations of semantically similar inputs (positive pairs) are drawn together while representations of dissimilar inputs (negatives) are pushed apart. The power and the difficulty of the approach both reside in a single design choice—what is treated as similar—which this section examines in general, for cardiac signals, and finally in the specific context of paroxysmal AF.

2.5.1 Foundations of Contrastive Learning

The modern formulation trains an encoder with the InfoNCE objective, in which a positive pair must be identified against a set of negatives, the temperature parameter controlling how sharply the loss penalises hard negatives. SimCLR established a simple and influential instance of this framework in which positive pairs are generated by applying two random augmentations to the same example, with all other examples in the mini-batch serving as negatives, optimised through the normalised temperature-scaled cross-entropy (NT-Xent) loss [14]; its performance, however, depends on large batches to supply sufficient negatives. Momentum Contrast (MoCo) removed this dependence by maintaining a queue of negatives encoded by a slowly updated momentum encoder [91]. A complementary line of work dispenses with explicit negatives altogether: BYOL learns by predicting the output of a momentum target network [92], SimSiam shows that a stop-gradient operation alone suffices to prevent representational collapse [93], and Barlow Twins instead drives the cross-correlation between two augmented views toward the identity matrix, reducing redundancy across embedding dimensions [94]. Supervised Contrastive Learning (SupCon) extends the contrastive objective to the labelled setting by treating all samples that share a class label as positives, pulling together entire classes rather than only augmented views of a single instance [15]. Comprehensive surveys of these developments are provided by [95].

2.5.2 Contrastive Learning for Cardiac and Physiological Signals

Adapting contrastive learning to physiological signals requires defining notions of similarity appropriate to the domain. Contrastive Learning of Cardiac Signals (CLOCS) exploits the spatial and temporal structure of the ECG by treating temporally adjacent segments and different leads belonging to the same patient as positives, producing patient-discriminative representations that improve downstream arrhythmia classification and label efficiency [16]. Patient Contrastive Learning of Representations (PCLR) makes patient identity explicit, constructing positive pairs from distinct ECGs of the same individual; the resulting embeddings transfer strongly to a range of downstream cardiac tasks under limited supervision [17]. Collectively, these works demonstrate that incorporating patient identity into the contrastive objective yields more transferable representations than purely instance- or class-based formulations.

A recurring difficulty in transferring contrastive learning from vision to physiological signals is the design of label-preserving augmentations. Image transformations such as cropping and colour distortion have no direct analogue for an RR-interval series, and naive perturbations risk destroying the very timing irregularities that distinguish AF from sinus rhythm; augmentations such as jittering, scaling, temporal masking, and permutation must therefore be chosen so as to preserve diagnostic content. The construction of negative pairs is equally delicate: when negatives are drawn indiscriminately from the batch, segments from the same patient or the same rhythm class may be pushed apart even though they ought to be treated as similar—a phenomenon known as sampling bias. These domain-specific considerations explain why the definition of what counts as a positive or a negative pair, rather than the encoder architecture alone, tends to dominate downstream performance.

2.5.3 Temporality and the Paroxysmal Constraint

A large class of contrastive methods for sequential data derive their positive pairs from *temporal* structure, treating windows that are close in time as views of the same underlying state. Contrastive Predictive Coding predicts future latents from past context [86], TS-TCC contrasts temporally displaced views of a signal [87], and CLOCS explicitly pairs temporally adjacent ECG segments [16]. All of these rest on an assumption of local label stationarity: that two windows close in time belong to the same class.

Paroxysmal AF violates this assumption by construction. Within a single recording the rhythm transitions between sinus rhythm, a pre-onset regime, and AF, and the timing of these transitions is precisely what the task must resolve; two temporally adjacent windows may straddle an onset and therefore belong to different classes. Treating temporal proximity as an invariance would pull such windows together in the embedding, blurring exactly the boundary that detection and early-warning depend on. Consequently, temporality cannot be enforced as a similarity signal for AF representation learning, and a commonly used source of self-supervision is unavailable. This argues for positive pairs grounded in criteria that remain valid across rhythm transitions—namely patient identity together with the rhythm label—rather than in temporal adjacency; it directly motivates the patient-aware pairing strategy adopted in this dissertation.

2.5.4 The Per-Patient Structure Problem

A subtle but consequential issue arises when contrastive objectives are applied to data organised by subject. Class-level objectives such as SupCon treat every same-class segment as an interchangeable positive, collapsing each patient's distinct physiological baseline into a single shared class prototype [15]. For cardiac interval data, where every individual exhibits a characteristic sinus-rhythm signature, this collapse discards precisely the within-subject structure on which a downstream classifier must rely when generalising to a previously unseen patient. Patient-aware formulations such as CLOCS and PCLR mitigate this by organising positives around patient identity [16, 17], though they are designed primarily for label-efficient self-supervised pre-training rather than for the specific problem of preserving per-patient geometry under a known AF/sinus-rhythm label structure. How to retain individual rhythm structure while still learning a discriminative AF-versus-sinus-rhythm boundary therefore remains an open question, and is central to the present work.

2.6 Datasets, Benchmarks, and Evaluation Protocols

Progress in automated AF analysis has been shaped as much by the data on which methods are trained and compared as by the methods themselves. This section reviews the principal public databases for detection and prediction, the evaluation protocols that govern how results should be interpreted, and the metrics appropriate to a highly imbalanced clinical task.

2.6.1 Public Databases for AF Detection

Most benchmark databases for AF detection are distributed through PhysioNet [96]. The MIT-BIH Arrhythmia Database, comprising forty-eight half-hour two-lead recordings from forty-seven subjects, has served as the canonical reference for arrhythmia classification for decades [97], while the companion MIT-BIH Atrial Fibrillation Database provides twenty-five long-term recordings annotated specifically for AF and remains a standard for RR-interval-based detectors. The Long-Term AF Database extends this to longer continuous recordings suited to studying episode dynamics. The 2017 PhysioNet/Computing in Cardiology Challenge marked an important broadening toward consumer-grade data, tasking entrants with discriminating AF, normal, other-rhythm, and noisy classes from roughly eight and a half thousand short single-lead ECGs acquired on a handheld device [98]; its emphasis on noisy, single-lead, short recordings closely mirrors the conditions of wearable screening.

2.6.2 Databases for AF Prediction

Prediction has historically been served by smaller, purpose-built resources. The Atrial Fibrillation Prediction Database (AFPDB), introduced for the PAF Prediction Challenge, pairs recordings taken immediately before an AF episode with episode-free control recordings, enabling the study of pre-onset dynamics [78]; its modest size and short records, however, limit the prediction horizons that can be studied and the statistical power of patient-level evaluation. More recent resources such as IRIDIA-AF supply long-term Holter recordings from a large cohort of paroxysmal-AF patients with multiple records per individual, enabling patient-level partitioning at a scale earlier datasets could not support [23]. The largest cohorts—the Apple, Fitbit, and Huawei heart studies—demonstrate population-scale PPG screening but are proprietary, which constrains reproducibility and independent comparison [45, 46]. Table 2.3 summarises these resources.

2.6.3 Evaluation Protocols and Pitfalls

The choice of dataset is inseparable from the choice of evaluation protocol, and several recurring pitfalls bear directly on the validity of reported results. Because most databases contain many segments per patient, a record- or segment-wise random split

TABLE 2.3: Representative public databases and large-scale cohorts used in automated AF research.

Database / cohort	Primary use	Content	Public
MIT-BIH Arrhythmia [97]	Detection benchmark	48 two-lead recordings of ∼30 min from 47 subjects	Yes
MIT-BIH AFDB [96]	AF detection	25 long-term (∼10 h) two-channel ECG records	Yes
Long-Term AF (LTAFDB) [96]	AF detection	Long-term continuous ECG recordings	Yes
PhysioNet/CinC 2017 [98]	Single-lead AF classification	∼8,500 short (9–60 s) hand-held single-lead ECGs	Yes
AFPDB [78]	AF onset prediction	75 records pairing pre-AF segments with episode-free controls	Yes
IRIDIA-AF [23]	Detection / patient-level	167 paroxysmal-AF patients, long-term Holter, multiple records each	Yes
Apple / Fitbit / Huawei studies [45, 46]	PPG screening at scale	10^5–10^6 participants, wrist PPG with ECG-patch confirmation	No

allows segments from the same individual to appear in both training and test partitions; the model can then exploit patient-specific idiosyncrasies, and reported performance overstates the clinically relevant ability to generalise to new people. A patient-independent (subject-wise) split, in which all data from a given patient fall entirely within one partition, is the only protocol that measures this ability, and the gap between the two can be large—an RRI-based detector reported at 98.98% sensitivity under record-mixed cross-validation fell to 86.04% under patient-independent evaluation [12]. External validation on a dataset from a different centre or device provides the strongest evidence of generalisation, and reproducibility analyses have cautioned that several published pipelines are difficult to replicate under like-for-like conditions [99]. These observations motivate two methodological commitments adopted in this dissertation: strict patient-level data partitioning, and representations explicitly optimised for cross-patient transfer.

2.6.4 Performance Metrics

AF segments are typically far rarer than sinus-rhythm segments, so aggregate accuracy is misleading—a detector that never predicts AF can still appear accurate. Sensitivity and specificity characterise the two error types separately, while positive and negative predictive values express the clinically salient question of how far a positive or negative result can be trusted given the prevalence of the condition. Threshold-independent summaries are preferred for model comparison: the area under the receiver operating characteristic curve (AUROC) measures ranking quality across all thresholds, and under strong class imbalance the area under the Area Under the Precision Recall Curve (AUPRC) is the more informative summary because it is sensitive to performance on the rare positive class. Prediction introduces an additional, temporal axis of evaluation: beyond classification quality, a predictor is judged by its *predicted time horizon*—how far in advance of onset a reliable warning is issued—so that accuracy and lead time must be traded off against one another rather than optimised in isolation.

2.7 Generalisation and the Research Gap

2.7.1 Cross-Patient Generalisation in AF Models

A recurring theme across the detection and prediction literature is the gap between performance under conventional evaluation and performance on genuinely unseen patients. The root cause is a form of domain shift: each individual contributes a distinct cardiac baseline, so a model trained on one cohort encounters, in every new patient, a subtly different input distribution. When patient identity is correlated with the label—as it often is in datasets assembled from diagnosed and control individuals—a model can minimise its training loss by learning to recognise patients rather than rhythms, a shortcut that collapses on deployment. This is why the record-mixed-to-patient-independent decline documented above [12] is not an artefact of any single study but a structural property of subject-organised physiological data, and why robustness to inter-patient variability, rather than peak in-distribution accuracy, is the appropriate target for a clinically useful system.

2.7.2 Research Gap and Summary

The literature reviewed in this chapter supports several conclusions. First, interval-based analysis of RRI and PPG-derived pulse intervals is a clinically meaningful and resource-efficient basis for both AF detection and onset prediction, and is well suited to wearable deployment. Second, deep learning has largely superseded handcrafted-feature pipelines for both tasks, yet supervised models remain data-hungry and generalise poorly across patients. Third, contrastive representation learning offers a route to label-efficient, transferable embeddings, and patient-aware variants confirm that subject identity is a valuable source of structure; at the same time, the paroxysmal nature of the disease rules out temporal proximity as a self-supervisory signal, narrowing the space of admissible pairing strategies. However, no existing work directly treats the preservation of per-patient rhythm geometry as the mechanism governing cross-patient transfer for AF detection, nor connects this representational property to downstream early-warning prediction from the same embedding space. This dissertation addresses that gap through a patient-aware contrastive objective that preserves per-patient sinus-rhythm structure, evaluated under a strict patient-independent protocol and extended toward AF onset prediction.

CHAPTER 3

METHODOLOGY

This chapter sets out the method. At its core sits a *patient-aware contrastive* representation. We pretrain it on detection: telling atrial fibrillation (AF) apart from sinus rhythm (SR) on sequences of cardiac intervals. We then freeze the encoder and read it out with a single linear probe under a strict patient-independent protocol. The same embedding later carries a forward-looking extension to AF onset prediction. The input never changes: a run of inter-beat intervals, either RR intervals (RRI) from the ECG or pulse-to-pulse intervals from photoplethysmography (PPG). Both expose the same beat-to-beat timing; we feed them to one model without distinction.

3.1 Overview

Three stages make up the pipeline. First, a patient-aware contrastive objective pretrains the encoder; it builds positive pairs only from *same-patient, same-class* interval segments, leaving each patient's SR baseline intact rather than dissolving it into a shared class prototype (Section 3.4). Second, we freeze the encoder and train a linear probe on its output embedding to detect AF, a protocol that separates the quality of the representation from the capacity of the classifier (Section 3.5). Third, we measure the geometry of the embedding space directly to explain cross-patient generalisation, then push the representation toward early prediction of onset (Sections 3.6–3.8).

The encoder itself stays deliberately small. Any gain then traces to the training objective rather than to raw capacity. The footprint stays within what wearable hardware can actually run.

3.2 Problem Formulation

We pose AF detection as a representation-learning problem on *subject-structured* data. Every example is a triple (x_i, y_i, p_i): $x_i \in \mathbb{R}^W$ holds a window of W consecutive inter-beat intervals, $y_i \in \{\text{SR}, \text{AF}\}$ gives its rhythm label, and $p_i \in \mathcal{P}$ names the patient who produced it. The encoder f_θ sends each window to an embedding and ℓ_2-normalises it onto the unit hypersphere, $\hat{z}_i = f_\theta(x_i)/\|f_\theta(x_i)\| \in \mathbb{S}^{d-1}$; a downstream classifier reads the rhythm off that embedding.

One feature defines the problem: the class label and the patient identifier entangle. Each patient owns a personal sinus-rhythm signature, and AF dynamics themselves shift from one person to the next. A model tested on patients it met during training can lean on subject-specific cues; the clinically relevant number is performance on *unseen* patients, whose baseline the classifier has never seen. The goal is therefore not to split the two classes globally. It is to learn an encoder whose geometry stays *consistent across subjects*, so that a decision rule fitted on the training patients carries over to new ones. This goal drives both the patient-aware objective of Section 3.4 and the strictly patient-independent evaluation of Section 3.5.

3.3 Dataset and Preprocessing

3.3.1 Dataset

Every experiment draws on IRIDIA-AF [23], a large database of long-term single-lead ECG recordings from 167 patients with paroxysmal AF, sampled at 200 Hz. An expert cardiologist and a specialist cardiac nurse annotated and reviewed each AF episode. Individual recordings run from roughly nineteen to ninety-five hours and arrive as consecutive 24-hour files, the extended, continuous monitoring we need to catch several episodes per patient and to evaluate at patient scale.

3.3.2 Episode Selection and Labelling

From each recording we keep episodes that clear three bars: the AF stretch lasts at least one hour, the SR stretch immediately before it lasts at least four hours, and every beat of the episode sits inside a single continuous file. That four-hour SR rule is deliberately strict. It pushes the start of every SR window well clear of AF onset, which keeps the pre-episode transitional rhythm out of the SR class and leaves a cleaner, more clinically honest negative. Figure 3.1 shows the resulting per-recording layout.

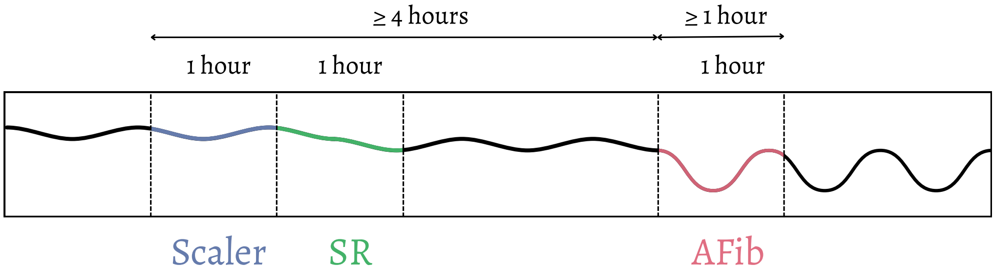

Fig. 3.1: Per-recording episode selection and labelling. The SR class window and the AF class window are each one hour long, drawn respectively from a sinus-rhythm stretch of at least four hours before onset and an AF episode of at least one hour. The first hour of sinus rhythm (*Scaler*) is reserved to fit the per-patient robust scaler of Eq. (3.1) and is kept disjoint from the labelled SR window, so the scaler never sees the segments it later normalises.

We partition the data strictly at the *patient* level into training, validation, and test sets, 119/24/24; after episode-level quality filtering, 154/27/24 episodes survive. No patient crosses a partition boundary. Downstream performance therefore reflects the genuine clinical task (generalising to unseen individuals, not to unseen slices of familiar ones), and the optimistic bias of letting one patient's segments straddle train and test simply cannot arise.

3.3.3 Interval Extraction and Per-Patient Normalisation

We convert each recording into a stream of inter-beat intervals. To strip out inter-individual differences in resting heart rate while keeping the relative variability that separates the two rhythms, we normalise the intervals *per patient* with a robust scaler fitted on that patient's first hour of sinus rhythm, the most temporally stable part of the recording. For an interval RR_i,

$$\widetilde{RR_i} = \frac{RR_i - \text{median}\left(\mathcal{R}^{(p)}_{\text{scale}}\right)}{\text{IQR}\left(\mathcal{R}^{(p)}_{\text{scale}}\right)}, \tag{3.1}$$

where $\mathcal{R}^{(p)}_{\text{scale}}$ is the patient-specific fit window (the first SR hour) and IQR is the interquartile range. We choose the median and IQR over the mean and standard deviation because long-term ambulatory recordings are riddled with ectopic beats and transient artefacts that would distort a z-score. The fit window also stays disjoint from the classification windows, which we draw from a later hour; the scaler therefore never touches the labelled segments it will normalise, closing a subtle leak. We discard beats with $RR < 200\,\text{ms}$ or $RR > 2000\,\text{ms}$ as physiologically implausible.

3.3.4 Segmentation

We slide a window of $W = 200$ beats across the normalised stream at a stride of $S = 50$ beats, so neighbours overlap by 75%. The window runs long enough to expose the beat-to-beat irregularity that marks AF, yet short enough to pin down where a rhythm turns over; the overlapping stride multiplies the training examples and doubles as mild temporal augmentation. We treat both W and S as tunable hyperparameters and fix them on the validation split (Section 3.4.5). Each window becomes one example, labelled SR or AF by the rhythm of its parent segment.

3.3.5 Resulting Dataset

Table 3.1 summarises the dataset the pipeline produces. The strict episode filter is selective: of the 119 training and 24 test patients in the partition, only 68 and 14 respectively contribute an AF episode with its four-hour preceding sinus stretch; these yield 23,989 training and 7,959 test windows at the $W = 200$, $S = 50$ setting. The class balance is mild (AF modestly outnumbers SR), and the test split is the same $3{,}799$ SR / $4{,}160$ AF windows reported throughout Chapter 4.

The patient-level partition is fixed once, before any preprocessing, and the detection and prediction experiments share it; the differing episode requirements of the two tasks merely select different subsets of the same test patients. The detection test set draws on 14 of the 24 test patients, whereas the onset-prediction analysis of Section 3.8 draws on the 11 test patients that meet its own (looser) episode criteria. These

TABLE 3.1: Dataset composition after episode selection and windowing (window size $W = 200$, stride $S = 50$). The patient split is cut at the patient level; counts of windows are measured from the processed data. Validation serves only hyperparameter selection and early stopping, so its per-class window counts are not reported here.

Split	Patients	Episodes	SR windows	AF windows	Total
Train	119	154	10,526	13,463	23,989
Validation	24	27	—	—	—
Test	24	24	3,799	4,160	7,959

two subsets overlap in a single patient and together account for all 24 test individuals, so neither task ever borrows a patient the encoder met during training.

3.4 Patient-Aware Contrastive Pretraining for Detection

Contrastive representation learning trains the detection encoder. What sets the approach apart is how it builds positive pairs: at once *class-aware* and *subject-aware*.

3.4.1 Patient-Aware Mini-Batch Sampling

Each mini-batch samples P patients; every one contributes n SR windows and n AF windows, for a batch of $B = 2nP$. A patient joins a batch only when both its SR and AF pools hold at least n windows. This sampler guarantees that intra-patient, intra-class pairs appear in every batch (so the patient-aware positive set below is never empty), balances the classes across patients, and stops patients with many episodes from dominating the rest. Every objective in this work shares the identical sampler, which pins any performance difference to the loss rather than to how the data arrive.

3.4.2 The Patient-Aware Contrastive Objective

Each sample carries a class label $y \in \{\text{SR}, \text{AF}\}$ and a patient identifier p. For an anchor i with class y_i and subject p_i, the positive and negative sets read

$$\mathcal{P}(i) = \{\, j \neq i \mid p_j = p_i \;\wedge\; y_j = y_i \,\}, \tag{3.2}$$

$$\mathcal{N}(i) = \{\, j \neq i \mid y_j \neq y_i \,\}. \tag{3.3}$$

The positive set is the *intersection* of class identity and subject identity (Figure 3.2); the negative set, by contrast, spans every segment of the opposite class, whatever its patient. These two sets do not exhaust the batch. A segment that shares the anchor's class but comes from a *different* patient belongs to neither $\mathcal{P}(i)$ nor $\mathcal{N}(i)$ and therefore never enters the denominator of Equation (3.4): the loss neither pulls it toward the anchor nor pushes it away. This omission is deliberate. Supervised contrastive learning would instead treat such a pair as a positive [15], collapsing each patient's distinct

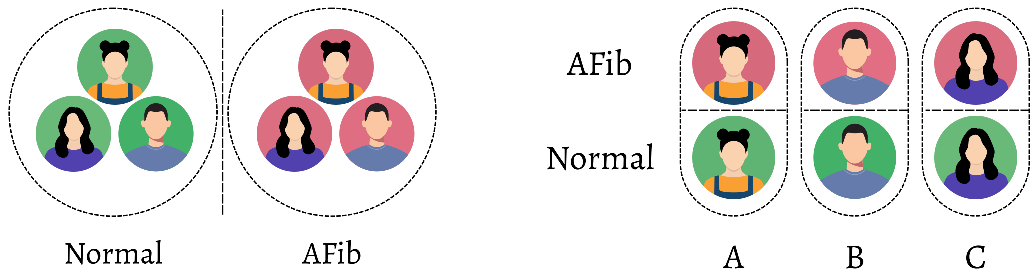

Fig. 3.2: Positive-pair construction: standard versus patient-aware. *Left:* supervised contrastive learning treats every same-class segment as interchangeable, pooling all patients' Normal (SR) windows into one positive set and all patients' AFib windows into another. *Right:* the patient-aware objective restricts positives to the *same* patient, so each individual's own SR and AF windows (A, B, C) form their own positive groups while same-class windows from other patients are excluded—the intersection of class and subject identity in Eqs. (3.2)–(3.3).

sinus-rhythm baseline into a single shared class prototype; treating it as a negative would err the other way, driving apart two windows of the same rhythm and eroding the very class concept the probe must recover. Leaving it unconstrained preserves each patient's own geometry, while the cross-patient negatives still separate the two classes. Holding positives to the same patient is what keeps the within-subject structure that a cross-patient classifier has to lean on.

The per-anchor loss takes the standard InfoNCE form [14],

$$\mathcal{L}_i = -\log \frac{\sum_{j\in\mathcal{P}(i)} \exp(s_{ij})}{\sum_{j\in\mathcal{P}(i)} \exp(s_{ij}) + \sum_{k\in\mathcal{N}(i)} \exp(s_{ik})}, \tag{3.4}$$

where $s_{ij} = \hat{e}_i^\top \hat{e}_j / \tau$ is the cosine similarity between ℓ_2-normalised projected embeddings scaled by a temperature τ, and the batch loss is $\mathcal{L} = \frac{1}{B}\sum_i \mathcal{L}_i$.

Read the construction as one point on a spectrum of positive-pair definitions that all share the InfoNCE form and differ only in $\mathcal{P}(i)$. Hold positives to augmented views of a single instance and you recover instance-discrimination methods such as SimCLR [14]; widen them to every same-class segment and you recover supervised contrastive learning [15]; and unsupervised patient-level methods such as CLOCS [16] and PCLR [17] take any same-patient pair as positive while ignoring the class label entirely. Our objective is the one combination that conditions on both signals at once. It is therefore a generic template for subject-structured data, fitting wherever samples group by subject and the within-subject variation carries information, with SR/AF detection as the instance we build here.

3.4.3 Encoder Architecture

The encoder must stay cheap enough for wearable hardware yet sharp enough to catch multi-scale RRI dynamics. The deployed network holds only $101{,}985$ parameters, under 0.4 MB in single precision, with a further $33{,}024$ confined to a contrastive head used during training alone; Chapter 4 quantifies this footprint in full. Four parts compose the encoder: a multi-scale convolutional backbone, a channel-mixing fusion block, a temporal attention pooling module, and a final projection stage, composed end to end as in Figure 3.3.

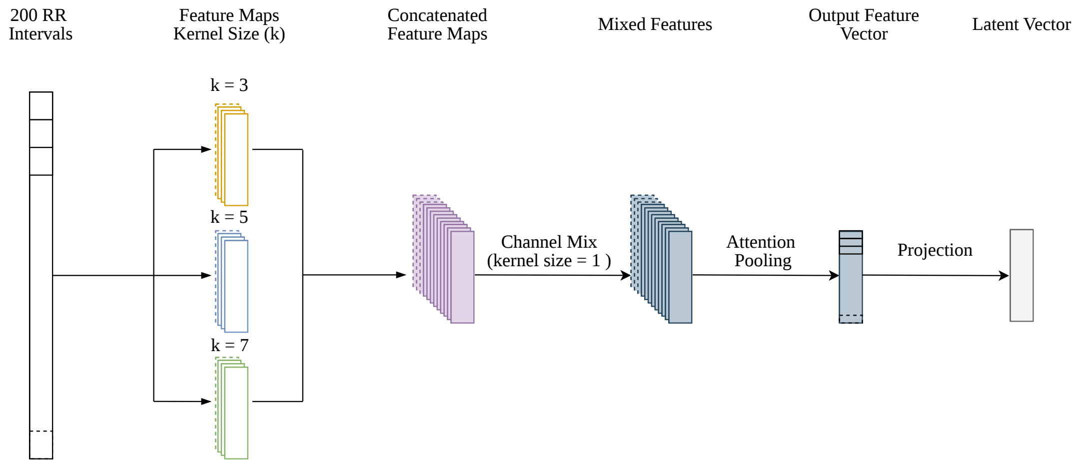

Fig. 3.3: Encoder architecture. A window of 200 RR intervals is processed by three parallel 1D-CNN branches with kernel sizes $k \in \{3, 5, 7\}$; the branch feature maps are concatenated and fused by a 1×1 channel-mixing convolution, collapsed over time by temporal attention pooling into a single feature vector, and mapped by a linear projection to the latent vector $\vec{z}$ used for downstream evaluation. The two-layer contrastive projection head, used only during training, is omitted.

3.4.3.1 Multi-Scale CNN Backbone

Three parallel 1D-CNN branches with kernel sizes $k \in \{3, 5, 7\}$ pick up beat-to-beat jitter, medium-range oscillation, and broad trend at once, following the Inception family's multi-scale processing [100]. Each branch stacks three strided convolution blocks (stride 2) with channel depths of 16, 32, and 64; Group Normalisation [101] with eight groups and a ReLU follow every convolution. We pick Group Normalisation over batch normalisation because the patient-aware sampler yields small, structured mini-batches whose statistics would make batch normalisation unstable.

3.4.3.2 Cross-Branch Channel Mixing

Because the three branches run in parallel, we fuse their outputs before pooling. We concatenate their feature maps along the channel axis into a single 192-channel (3×64) sequence, then apply a pointwise (1×1) convolution that mixes information across all

192 channels at each time step. This recombines the three temporal scales into one shared representation instead of leaving them as separate streams. Group Normalisation (eight groups) and a ReLU follow this convolution as in the backbone; a dropout of 0.12 then regularises the fused features. The block preserves the temporal length and passes a single fused sequence to the pooling stage.

3.4.3.3 Temporal Attention Pooling

Rather than flatten the temporal axis with global average pooling, a temporal attention module scores each time step of the fused sequence on its own [102],

$$\alpha_t = \frac{\exp(w^\top h_t)}{\sum_{t'} \exp(w^\top h_{t'})}, \qquad v = \sum_t \alpha_t h_t, \tag{3.5}$$

where h_t is the fused feature at step t, w a learnable attention vector, and v the resulting pooled representation. This lets the encoder emphasise physiologically salient RRI segments without any manual annotation, an asset for AF episodes whose onset dynamics wander.

3.4.3.4 Projection to the Latent and Contrastive Spaces

A single linear layer, the encoder projection, maps the pooled vector v to the latent vector $z \in \mathbb{R}^{128}$, the encoder output $f_\theta(x)$ used for all downstream evaluation and for the geometry analysis (the final stage of Figure 3.3). During contrastive training *only*, a two-layer MLP projection head g_ϕ maps z to a contrastive embedding $e \in \mathbb{R}^{128}$, following SimCLR [14]. We ℓ_2-normalise both z and e onto the unit hypersphere, which makes cosine similarity and dot product coincide and steadies the contrastive loss landscape [89]. The InfoNCE objective of Equation (3.4) is computed on the projected embedding $\hat{e}$, whereas downstream linear evaluation works on the encoder output $\hat{z}$. Reading out this pre-head representation is standard practice, since the projection head tends to shed information that matters beyond the contrastive objective. The head is discarded once pretraining ends and never runs at inference.

3.4.4 Training Procedure

We implement the encoder in PyTorch and train it on a single NVIDIA GeForce RTX 2080 GPU. The latent dimension and the projection-head dimension both sit at 128. AdamW drives training at a learning rate of 6.8×10^{-3} and weight decay 8.8×10^{-4}, with cosine annealing from that peak down to 10^{-6}. We apply dropout of 0.12 on the projection head, and fix the contrastive temperature at $\tau = 0.07$, the optimum returned by the hyperparameter search of Section 3.4.5. The patient-aware sampler uses $P = 8$ patients and $n = 8$ windows per class per patient ($B = 2nP = 128$). Training runs up to 100 epochs, but early stopping on validation AUROC (patience 10) cuts it off, so

Algorithm 1: Patient-aware contrastive pretraining for AF detection.

Input: episodes grouped by patient; encoder f_θ; projection head g_ϕ; sampler parameters P, n; temperature τ

Output: trained encoder f_θ

while *validation AUROC has not plateaued* **do**

  sample P eligible patients (each with $\geq n$ SR and $\geq n$ AF windows)

  draw n SR and n AF windows from each, forming a batch of $B = 2nP$ windows

  compute contrastive embeddings via the head and ℓ_2-normalise: $\hat{e}_i = g_\phi(f_\theta(x_i))/\|g_\phi(f_\theta(x_i))\|$

  for *each anchor i in the batch* **do**

    $\mathcal{P}(i) \leftarrow \{\, j \neq i : p_j = p_i \wedge y_j = y_i \,\}$

    $\mathcal{N}(i) \leftarrow \{\, j \neq i : y_j \neq y_i \,\}$

    compute $\mathcal{L}_i$ from Eq. (3.4)

  update θ and ϕ with AdamW on $\mathcal{L} = \frac{1}{B}\sum_i \mathcal{L}_i$

the budget bends to each loss instead of being fixed in advance. Algorithm 1 lays out the full procedure.

3.4.5 Hyperparameter Selection

Bayesian optimisation selects the hyperparameters: the Optuna Tree-structured Parzen Estimator (TPE) sampler [103] searches against validation AUROC. We optimise 32 hyperparameters jointly: encoder architecture (kernel sizes, channel depths, and the number of Group Normalisation groups), training configuration (learning rate, weight decay, dropout, and temperature), and data-pipeline parameters (the window size W and the stride S). The whole search lives on the validation split alone; the test split stays sealed until final evaluation. For the loss-function comparison of Section 3.7 we hold the encoder architecture, sampler, and probe fixed across all three objectives and re-tune only the learning rate and, where it applies, the temperature per loss, so that no condition wins on tuning and any gap traces to the objective alone.

3.5 Downstream AF Detection via Linear Probing

To weigh the representation, we freeze the encoder and train a linear probe on the encoder output $\hat{z}$ to call each window SR or AF. The probe is an ℓ_2-regularised logistic regression with inverse regularisation strength $C = 1.0$, fit for up to $5{,}000$ iterations. Frozen-encoder linear probing is the standard readout of representation quality in contrastive learning [14, 15, 89]: since the encoder never updates, probe performance mirrors the geometry of the embedding, not the reach of the classifier. We fit the probe on embeddings of the training patients and test it on the held-out test patients, who stay entirely unseen through pretraining and probe fitting alike. Every experiment repeats over five independent random seeds (each controlling weight initialisation and mini-batch ordering on a fixed patient split), reporting mean $\pm$ standard deviation.

3.6 Embedding-Geometry Analysis

Because the central claim of this work concerns the *structure* of the learned embedding, we measure that geometry directly rather than infer it from downstream accuracy. Every embedding is ℓ_2-normalised, so each lands on the unit hypersphere $\mathbb{S}^{d-1}$. Let $y_i \in \{\mathrm{SR}, \mathrm{AF}\}$ be the class label and p_i the patient identifier, and define the index sets $S^c = \{i : y_i = c\}$ and $S_p^c = \{i : y_i = c,\ p_i = p\}$, with corresponding (unnormalised) centroids μ^c and μ_p^c, their ℓ_2-normalised versions $\hat{\mu}^c$ and $\hat{\mu}_p^c$, and mean class spreads $\bar{d}^c$ and $\bar{d}_p^c$.

3.6.0.0.1 Per-patient class cohesion (primary metric).

For class c,

$$\mathrm{Coh}^c = \frac{1}{|P|} \sum_{p \in P} \frac{1}{|S_p^c|} \sum_{i \in S_p^c} \hat{z}_i^\top \hat{\mu}_p^c, \tag{3.6}$$

the mean cosine similarity between each embedding and its same-patient, same-class centroid; values climb toward 1 as the per-patient class cluster tightens. *SR cohesion* is the primary metric, because sinus rhythm is each patient's individual baseline, the reference a cross-patient probe extrapolates from when it meets a new subject.

3.6.0.0.2 Global class separability.

$$\mathrm{CentDist} = \|\mu^{\mathrm{SR}} - \mu^{\mathrm{AF}}\|_2, \tag{3.7}$$

$$\mathrm{CentSim} = \hat{\mu}^{\mathrm{SR}\top} \hat{\mu}^{\mathrm{AF}}, \tag{3.8}$$

$$C_{\mathrm{glob}} = \frac{\|\mu^{\mathrm{SR}} - \mu^{\mathrm{AF}}\|_2}{\bar{d}^{\mathrm{SR}} + \bar{d}^{\mathrm{AF}}}, \tag{3.9}$$

where C_{glob} is a compactness ratio: the between-class centroid distance over the sum of within-class spreads.

3.6.0.0.3 Per-patient compactness.

$$C_{\mathrm{pp}} = \frac{1}{|P|} \sum_{p \in P} \frac{\|\mu_p^{\mathrm{SR}} - \mu_p^{\mathrm{AF}}\|_2}{\bar{d}_p^{\mathrm{SR}} + \bar{d}_p^{\mathrm{AF}}}, \tag{3.10}$$

the same separation-to-spread trade-off as C_{glob}, but computed inside each patient and then averaged. As $\bar{d}_p^{\mathrm{SR}} + \bar{d}_p^{\mathrm{AF}} \to 0$ this ratio balloons no matter whether the clusters sit consistently across patients; we therefore read it alongside the raw spreads and the per-patient cohesion scores, never on its own. Together these metrics tease apart two things that downstream accuracy alone blurs: how far the classes stand *on average* (global separability) and how *consistently* each patient lands (per-patient cohesion). The latter is the property we expect to govern cross-patient transfer.

3.7 Baselines and Evaluation Protocol

3.7.1 Baselines

We compare the proposed objective against two baselines, each trained under the identical encoder, sampler, and probe. Supervised contrastive learning (SupCon) [15] keeps the InfoNCE form of Equation (3.4) but stretches the positive set to every same-class segment, patient notwithstanding,

$$\mathcal{P}_{\text{SupCon}}(i) = \{\, j \neq i \mid y_j = y_i \,\}, \tag{3.11}$$

so the only thing separating it from the proposed objective in Equation (3.2) is the dropped subject constraint $p_j = p_i$. Binary cross-entropy (BCE) goes the other way and abandons the contrastive structure outright, attaching a sigmoid classification head to the encoder output $\hat{z}$ and minimising

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{B} \sum_i \Big[y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i) \Big], \tag{3.12}$$

where $\hat{y}_i$ is the predicted AF probability. Because all three conditions share the encoder, sampler, and probe, the comparison isolates one thing: the definition of a positive pair.

3.7.2 Evaluation Protocol

The evaluation stays deliberately conservative. We cut every split at the patient level, so the test set holds only unseen individuals; we freeze the encoder through linear-probe evaluation; and we repeat every experiment over five seeds, reporting mean $\pm$ standard deviation. We report seed-to-seed variance outright, since stability across initialisations is a deployment requirement, not a footnote. We set the results against prior RRI-based AF detectors [11, 12], with the caveat that cross-validation protocols which let one patient's segments mix across folds inflate their metrics relative to the patient-independent protocol we adopt here.

3.7.3 Performance Metrics

Write TP, FP, TN, and FN for the per-class confusion-matrix counts. Then sensitivity (recall) is $\mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$, specificity is $\mathrm{TN}/(\mathrm{TN}+\mathrm{FP})$, precision is $\mathrm{TP}/(\mathrm{TP}+\mathrm{FP})$, and F_1 is the harmonic mean of precision and recall. AUROC, the area under the receiver operating characteristic curve, summarises the sensitivity–specificity trade-off across every decision threshold; we report it as the headline, threshold-independent metric. AF-class sensitivity is the primary clinical quantity: a missed AF episode can delay anticoagulation and raise stroke risk. We report accuracy for completeness but lean away from it, because class imbalance makes it misleading.

3.8 Extension to AF Onset Prediction

We reuse the detection representation, untouched, for a forward-looking early-warning scheme that reads straight off the geometry of the embedding space. For a given patient, we embed one labelled SR episode and one labelled AF episode and collapse each to its class centroid, $\vec{\mu}_{\text{SR}}$ and $\vec{\mu}_{\text{AF}}$. Those two centroids fix a personal *SR-to-AF axis* with unit direction

$$\hat{u} = \frac{\vec{\mu}_{\text{AF}} - \vec{\mu}_{\text{SR}}}{\|\vec{\mu}_{\text{AF}} - \vec{\mu}_{\text{SR}}\|}. \tag{3.13}$$

We embed each incoming interval window to $\vec{z}_t$, reference it to the SR centroid, project it onto this axis, and normalise by the axis length for a scalar early-warning score

$$p_t = \frac{(\vec{z}_t - \vec{\mu}_{\text{SR}}) \cdot \hat{u}}{\|\vec{\mu}_{\text{AF}} - \vec{\mu}_{\text{SR}}\|}, \tag{3.14}$$

which gauges how far the current rhythm has drifted from the patient's own sinus-rhythm baseline toward their AF region: $p_t = 0$ sits at the SR centroid, $p_t = 1$ at the AF centroid, and the midpoint $p_t = 0.5$ marks the decision boundary equidistant between them. A climb in p_t across this midpoint toward the AF centroid flags an impending episode; Figure 3.4 sketches the construction in the latent space.

The scheme asks only for a lightweight, few-shot calibration per patient, with no retraining. It holds together precisely because the patient-aware objective keeps each patient's SR cluster tight and consistently placed, leaning directly on the per-patient SR cohesion of Equation (3.6) and sidestepping the cross-patient drift that would otherwise leave the axis ill-defined. This geometric, calibration-based scheme is deliberately indirect. It replaces a long line of earlier attempts to predict onset time directly, by regression, temporal ranking, survival modelling, and patient-invariant representation learning, none of which transferred across patients under strict patient-independent evaluation; Appendices B and C document those attempts and the identity-shortcut mechanism that defeated them.

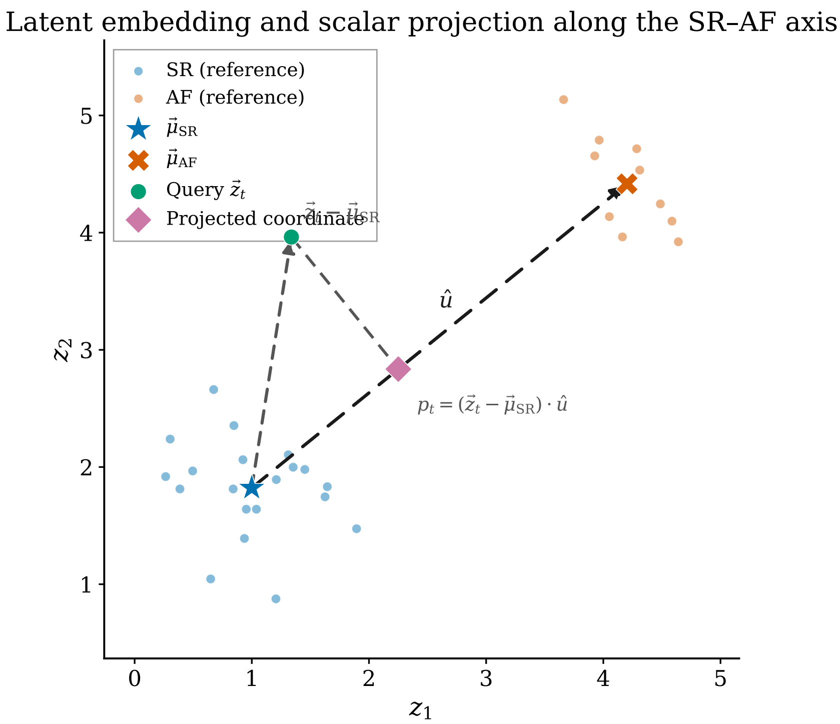

Fig. 3.4: Geometry of the early-warning scheme in the latent space (illustrative two-dimensional sketch). A patient's labelled SR and AF reference embeddings collapse to centroids $\vec{\mu}_{\mathrm{SR}}$ and $\vec{\mu}_{\mathrm{AF}}$, which fix the SR$\to$AF axis with unit direction $\hat{u}$ (Eq. (3.13)). A query window $\vec{z}_t$ is referenced to $\vec{\mu}_{\mathrm{SR}}$ and projected onto the axis; the early-warning score is the position of that projection along the axis (drawn here unnormalised, $p_t = (\vec{z}_t - \vec{\mu}_{\mathrm{SR}}) \cdot \hat{u}$; Eq. (3.14) rescales it by the axis length so that $\vec{\mu}_{\mathrm{SR}}$ and $\vec{\mu}_{\mathrm{AF}}$ map to 0 and 1).

CHAPTER 4

EXPERIMENTAL RESULTS

This chapter reports the empirical findings. We open with the headline detection performance of the patient-aware representation under the strict patient-independent protocol of Section 3.5 (Section 4.1). We then isolate the contribution of the patient-aware constraint by holding the encoder, sampler, and probe fixed and varying only the training objective (Section 4.2), then open up the learned embedding space to explain *why* the constraint helps (Section 4.3) and probe what the attention mechanism actually reads (Section 4.4). We set the results against prior RRI-based detectors (Section 4.5), and finally turn the same frozen embedding forward in time to report a proof-of-concept early-warning analysis for AF onset (Section 4.7). Unless stated otherwise, every number is a mean $\pm$ standard deviation over five independent seeds evaluated on the held-out test patients, who stay unseen through pretraining and probe fitting alike.

4.1 Detection Performance

Table 4.1 reports test-set performance of the proposed patient-aware representation on 4160 AF and 3799 SR windows drawn entirely from unseen patients. The frozen encoder paired with a logistic-regression probe reaches a mean AUROC of 0.989 ± 0.003 and overall accuracy of 0.952 ± 0.011.

TABLE 4.1: Test-set detection performance of the proposed patient-aware representation (4160 AF and 3799 SR windows from unseen patients; mean $\pm$ std over five seeds). Bold marks the primary clinical and headline metrics.

Class	**Precision**	**Recall**	**F1**
SR (0)	0.970 ± 0.009	0.927 ± 0.020	0.948 ± 0.012
AF (1)	0.936 ± 0.017	$\mathbf{0.974 \pm 0.008}$	0.955 ± 0.010
Accuracy		0.952 ± 0.011	
AUROC		$\mathbf{0.989 \pm 0.003}$	

The error profile is clinically favourable. AF recall (0.974 ± 0.008) consistently exceeds SR recall (0.927 ± 0.020) across every seed, an asymmetry that trades a modest rise in false positives for fewer missed AF episodes, the right side to err on for a screening tool. On average 4052 of 4160 AF windows and 3522 of 3799 SR windows are classified correctly. The wider spread in SR recall ($\sigma = 0.020$) tracks genuine patient-level heterogeneity in baseline sinus rhythm rather than model instability. SR precision stays tight at 0.970 ± 0.009; the spread therefore points to a handful of atypical SR baselines among the unseen test patients, not to instability between runs.

4.2 Effect of the Patient-Aware Constraint

Table 4.2 compares the proposed objective against the two baselines of Section 3.7: supervised contrastive learning (SupCon) [15], which widens the positive set to every same-class segment regardless of patient, and binary cross-entropy (BCE), which discards the contrastive structure entirely. The encoder architecture, training budget, sampler, and linear-probe protocol are identical across all three conditions, so any difference traces to the definition of a positive pair alone.

TABLE 4.2: Linear-probe comparison of the three objectives on the fixed patient-independent test split (mean ± std over five seeds). Bold marks the best mean per column. AUROC and its seed variance are the primary differentiating metrics.

Loss	**AUROC**	**AF Prec.**	**AF Rec.**	**AF F1**	**SR F1**
Proposed	$\mathbf{0.989 \pm 0.003}$	$\mathbf{0.936 \pm 0.017}$	0.974 ± 0.008	$\mathbf{0.955 \pm 0.010}$	$\mathbf{0.948 \pm 0.012}$
SupCon	0.983 ± 0.009	0.933 ± 0.017	0.972 ± 0.005	0.952 ± 0.008	0.945 ± 0.011
BCE	0.980 ± 0.012	0.929 ± 0.021	$\mathbf{0.974 \pm 0.009}$	0.951 ± 0.012	0.943 ± 0.016

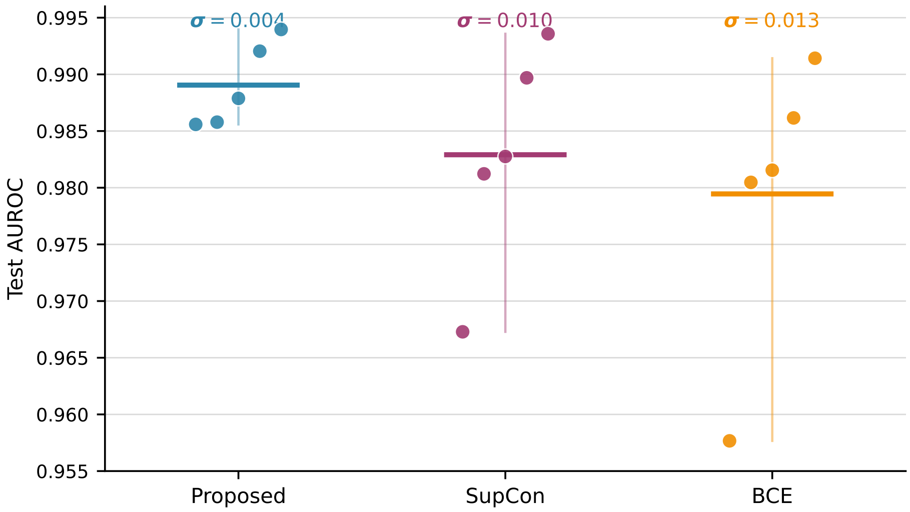

Fig. 4.1: Per-seed test AUROC of the three objectives on the fixed patient-independent split. Each dot is one of the five seeds, the thick horizontal tick marks the seed mean, the thin vertical bar spans the seed minimum to maximum, and the per-group standard deviation is annotated above; the values are those summarised in Table 4.2. The means nearly coincide, but the spreads do not. The proposed objective stays within $[0.986, 0.994]$ ($\sigma = 0.004$), whereas SupCon and BCE each fall away to a low-seed outlier (0.967 and 0.958; $\sigma = 0.010$ and 0.013). This is the seed-stability advantage the section turns on.

Figure 4.1 plots the comparison; the clearest advantage is stability. Seed-to-seed AUROC variance falls by $2.6\times$ relative to SupCon ($\sigma = 0.003$ versus 0.009) and by $3.4\times$ relative to BCE ($\sigma = 0.003$ versus 0.012): the patient-aware objective induces a loss landscape that is markedly more robust to weight initialisation, a property that matters directly when a deployed model must reproduce its behaviour across instances.

The proposed objective also leads on mean AUROC, by $+0.006$ over SupCon and $+0.010$ over BCE, though with only five seeds these mean gaps are not individually significant (Section 4.2.1). The differences in AF F1, SR F1, and accuracy likewise fall within seed noise; we draw no strong conclusion from them and report them for completeness. The headline effect of the patient-aware constraint is thus not a large jump in average accuracy but a representation that generalises slightly better and, above all, far more *consistently*.

4.2.1 Statistical Significance

To confirm these effects are not seed-sampling artefacts, we test the three objectives across the five seeds, pairing the runs by seed. Table 4.3 reports, for test AUROC, the mean difference (proposed minus baseline) with a 95% confidence interval, the paired t-test and Wilcoxon signed-rank p-values, and (for the stability claim) the ratio of seed variances with an F-test.

TABLE 4.3: Significance of the loss ablation on test AUROC across five seeds, paired by seed. ΔAUROC is the mean difference (proposed minus baseline) with a 95% confidence interval; p_t and p_W are the paired t-test and Wilcoxon signed-rank p-values. The variance ratio ($\sigma^2_{\text{baseline}}/\sigma^2_{\text{proposed}}$; larger means the proposed objective is more stable) is tested for equality of variances with a two-sided F-test (p_F).

Comparison	**ΔAUROC (95% CI)**	p_t	p_W	**Var. ratio**	p_F
Proposed vs. SupCon	$+0.006$ $[-0.008, +0.021]$	0.31	0.44	$7.1\times$	0.084
Proposed vs. BCE	$+0.010$ $[-0.009, +0.028]$	0.22	0.19	$11.7\times$	0.035

Two things follow. First, the mean AUROC advantage, though consistently positive ($+0.006$ over SupCon, $+0.010$ over BCE), is not statistically significant at five seeds ($p_t = 0.31$ and 0.22): the paired tests simply lack the power to resolve gaps this small; we accordingly do not rest the argument on mean accuracy. Second, the *stability* difference is both large and detectable. The proposed objective's seed-to-seed AUROC variance is $7.1\times$ smaller than SupCon's and $11.7\times$ smaller than BCE's (the $2.6\times$ and $3.4\times$ reduction in standard deviation reported above), and the reduction relative to BCE is significant even at this sample size ($p_F = 0.035$; $p_F = 0.084$ against SupCon, with the caveat that an F-test on five seeds is itself low-powered). The formal tests thus echo the qualitative reading: the decisive and reproducible contribution of the patient-aware constraint is not a higher mean but a substantially more stable representation.

4.3 Embedding-Geometry Analysis

To explain that stability we measure the geometry of the embedding space directly, using the metrics defined in Section 3.6. Table 4.4 reports per-patient class cohesion (Eq. (3.6)), global class separability (Eqs. (3.7)–(3.9)), and per-patient compactness (Eq. (3.10)), alongside the downstream AUROC. Per-patient SR cohesion is the primary metric: sinus rhythm is each patient's individual baseline, the reference point from which a cross-patient probe must extrapolate when it meets a new subject.

TABLE 4.4: Embedding-space metrics on the test set (mean $\pm$ std over five seeds), grouped into per-patient structure and global class separability. Per-patient SR cohesion is the primary metric; the bottom row repeats the downstream linear-probe AUROC. Bold marks the proposed objective's two decisive metrics, per-patient SR cohesion and AUROC. $\uparrow$ higher is better, $\downarrow$ lower is better; in the lower block, however, greater separation does *not* imply better transfer, which is precisely the BCE paradox discussed below.

Metric	**Proposed**	**SupCon**	**BCE**
Per-patient structure (governs cross-patient transfer)			
SR cohesion $\uparrow$	$\mathbf{0.850 \pm 0.044}$	0.800 ± 0.029	0.772 ± 0.049
AF cohesion $\uparrow$	0.846 ± 0.048	0.921 ± 0.035	0.955 ± 0.019
Per-patient compactness $\uparrow$	3.273 ± 0.433	2.178 ± 1.148	6.396 ± 0.590
Global class separability (a misleading proxy for transfer)			
Centroid similarity $\downarrow$	$+0.238 \pm 0.200$	$+0.155 \pm 0.123$	-0.717 ± 0.029
Centroid distance $\uparrow$	1.034 ± 0.100	1.120 ± 0.095	1.603 ± 0.045
Global compactness $\uparrow$	1.192 ± 0.124	1.313 ± 0.465	2.427 ± 0.226
Downstream AUROC $\uparrow$	$\mathbf{0.989 \pm 0.003}$	0.983 ± 0.009	0.980 ± 0.012

4.3.0.0.1 Per-patient SR cohesion confirms the design. The proposed objective attains the highest SR cohesion (0.850 versus 0.800 for SupCon and 0.772 for BCE), direct evidence that intra-subject positives preserve each patient's own SR baseline rather than collapsing the baselines into one shared class prototype. AF cohesion stays balanced (0.846, against 0.921 for SupCon and 0.955 for BCE), so the SR gain does not come by disordering the AF cluster. The objective holds both classes' per-subject structure together, in contrast to SupCon's lopsided trade.

4.3.0.0.2 The BCE paradox. BCE produces the most cleanly separated class means in the whole comparison (the largest centroid distance 1.603, the most negative centroid similarity -0.717, and the highest global compactness 2.427). Yet it yields the worst downstream AUROC on unseen patients. The reason surfaces only when we look inside each patient. Those well separated class means coexist with the most disorganised per-patient SR placement (0.772 cohesion). A probe trained on seen patients therefore has no consistent SR direction to extrapolate from when it meets a new one. BCE even posts the highest *per-patient* compactness ratio (6.396); as flagged in Section 3.6, this is an artefact of its vanishing within-patient spread: the denominator shrinks toward zero while the tight clusters it measures drift unpredictably from one patient to the next. The lesson is that global class separability is a misleading proxy for transfer; per-subject consistency is what the probe actually has to follow.

4.3.0.0.3 SupCon over-compacts AF and under-aligns SR.

SupCon reaches the tightest *contrastive* AF cohesion (0.921) by pulling all AF segments together irrespective of patient; it pays for this in SR cohesion (0.800 versus 0.850). This is exactly the predicted failure mode: standard supervised contrastive learning trades per-patient SR structure for global AF compactness, the very trade that hurts patients whose sinus-rhythm baseline is atypical.

4.3.0.0.4 Geometry predicts the downstream ordering.

The three objectives rank on downstream AUROC ($0.989 > 0.983 > 0.980$) exactly as they rank on per-patient SR cohesion ($0.850 > 0.800 > 0.772$), and inversely to how they rank on global separability. The embedding-geometry analysis therefore does not merely accompany the accuracy numbers; it explains them, and pins the cross-patient advantage of the patient-aware objective on per-subject geometric consistency rather than on how far apart the class means are pushed.

4.4 Interpretability of the Learned Representation

The geometry analysis pins transfer on per-patient SR cohesion, but not *which part of the rhythm* the encoder reads to build that structure. The temporal attention module of Equation (3.5) gives a direct answer. Its weights α_t are a genuine forward-pass quantity, not a post-hoc saliency estimate, so recovering them needs no approximation: a forward hook captures the per-step softmax weights, which we upsample to the input length and read back against the RR trace. We pair this with the static 1×1 attention-scoring weights and the fused channel-mix activations, to ask not just *where* attention lands but *which channels* put it there.

4.4.0.0.1 Attention settles on the calm stretches of the rhythm.

Across both SR and AF windows the attention weight concentrates on flat, low-variance runs of RR intervals and turns away from the large beat-to-beat swings (Figure 4.2). Measuring this against the multi-scale backbone sharpens it: the per-step activation energy of each convolutional branch is *anti*-correlated with the attention weight, strongly so for the two larger kernels ($r \approx -0.37$, -0.93, -0.80 for $k = 3, 5, 7$; Figure 4.3). The module pools toward rhythm stability, not toward the spiky, high-variability beats that would catch the eye.

4.4.0.0.2 A small cluster of stability channels carries the pooling.

Tallying, across many windows, which channels of the fused representation correlate with the attention weight, the boosting set is dominated by a handful of mid-scale ($k = 5$) channels. One channel, here indexed 116, is the single most frequent positive contributor in both SR and AF windows by a wide margin (Figure 4.4). The dominance is real but not monolithic: channel 116 is the top channel for 5 of the 16 patients, and every one of the 16 picks *some* mid-scale channel from a small redundant set ($\{99, 107, 115, 116, 117\}$), a

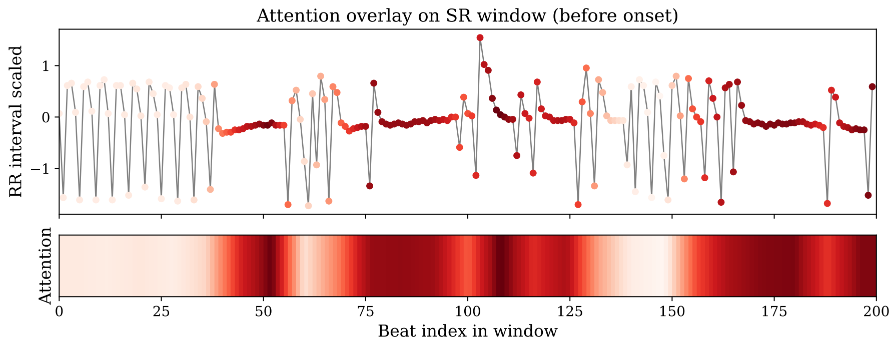

Fig. 4.2: Attention overlay on a pre-onset window. The RR-interval trace is coloured by the upsampled temporal-attention weight α_t (Eq. (3.5)); the strip below repeats the weight as a heatmap. The high-attention mass falls on the flat, low-variability segments rather than on the large oscillations, the qualitative pattern the branch-energy correlations make quantitative.

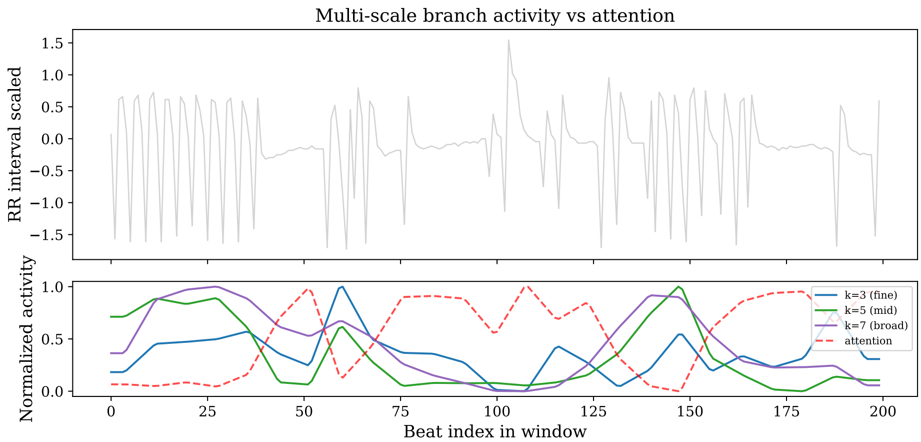

Fig. 4.3: Per-step activation energy of the three convolutional branches ($k = 3, 5, 7$) against the attention weight, over one window (lower panel; the RR trace is above). The attention curve (dashed) rises where branch activity falls, the anti-correlation reported in the text, most pronounced for the two larger kernels.

cluster of near-equivalent units rather than one universal detector. Inspecting channel 116's raw activation, it traces the mirror image of the local RR variance, confirming its role as an anti-variance, rhythm-stability detector. The suppressing set, by contrast, is more class-agnostic and is spread across fine-scale ($k = 3$) channels that take turns damping attention over different stretches of the window.

4.4.0.0.3 The attended channels are not the discriminative ones. The decisive test is whether the channels that drive attention are the channels that drive classification. Projecting the per-patient SR→AF axis ($\vec{\mu}_{\mathrm{AF}} - \vec{\mu}_{\mathrm{SR}}$) back through the encoder-projection weights gives a per-channel "AF-relevance" score, which we correlate against the static attention weight vector. The correlation is $r \approx 0.03$, essentially zero (Figure 4.5). The channels that steer where attention looks are all but unrelated to the channels that define the SR→AF direction the probe reads. Attention's focus on rhythm

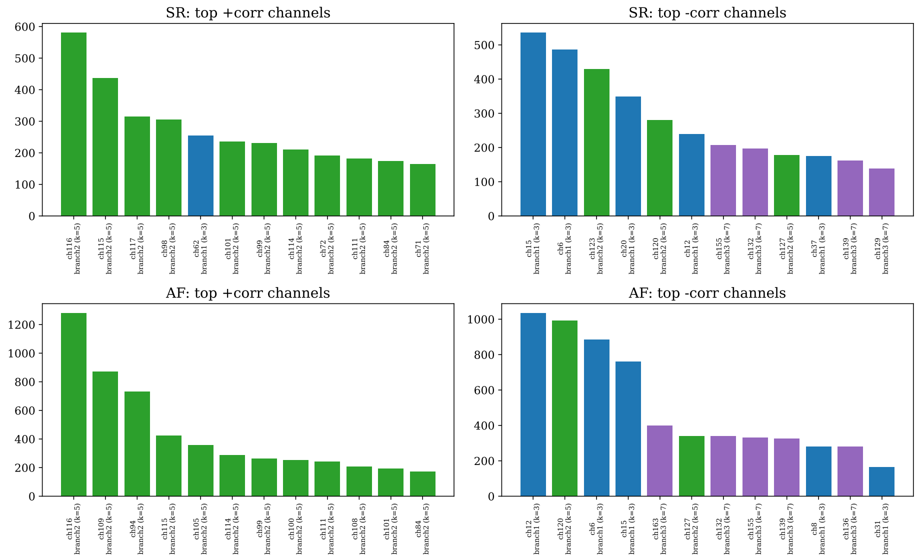

Fig. 4.4: Channels whose fused activation most often correlates with the attention weight, tallied separately over SR and AF windows. A small cluster of mid-scale ($k = 5$) channels dominates the positive (attention-boosting) side in both classes, with channel 116 the most frequent in each; the boosting mechanism is therefore largely class-agnostic.

stability is real and robust; it is simply not the mechanism of class discrimination.

4.4.0.0.4 Causal tests reinforce the decoupling. A correlation between two weight vectors is suggestive but indirect; a causal probe settles it. We compute two input-saliency maps for the early-warning score p_t: a gradient map $|\partial p_t / \partial \mathrm{RR}|$, and a perturbation map that nudges each beat in turn and records the resulting change Δp_t. The two agree at $r = 0.633$, so the per-beat importance they assign is method-robust rather than an artefact of either technique. Both put the leverage on the irregular, high-variability beats (Figure 4.6), the mirror image of where attention concentrates. Overlaying causal sensitivity on the attention weight makes the mismatch explicit: across the window the two are mildly *anti*-correlated ($r = -0.174$, Figure 4.7). The beats the model attends to are not the beats that move its output. Averaged over the population, the gradient sensitivity spreads fairly evenly across the window with only a mild falloff at the trailing edge (Figure 4.8), so the score integrates the whole window rather than keying on the most recent beats.

4.4.0.0.5 Channel and branch ablation locate the discriminative signal. Zeroing parts of the fused representation and re-reading p_t tells the same story from the channel side. Ablating the attention-dominant channel 116 shifts p_t by only -0.015, less than the shift from ablating the fine-scale AF-direction channels 1 ($+0.038$) and 129 ($+0.031$) (Table 4.5). No single channel dominates, so the code is distributed; but the largest causal effects sit on channels aligned with the SR$\rightarrow$AF direction, not on the channel that drives attention. Zeroing whole branches sharpens the point. The

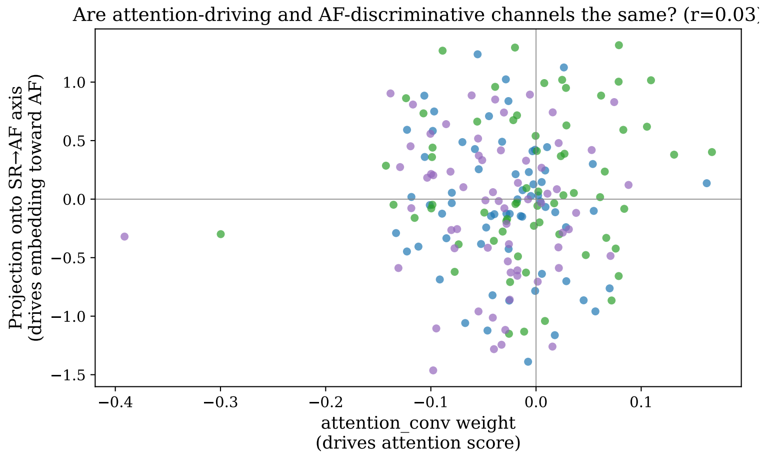

Fig. 4.5: Per-channel attention weight against per-channel AF-relevance (the SR$\to$AF axis projected through the encoder-projection weights). The near-zero correlation ($r \approx 0.03$) shows the two are decoupled: the channels that draw attention are not the ones that separate AF from SR.

score depends most on the fine-scale ($k = 3$) branch (median $|\Delta p_t| \approx 0.19$), then the mid-scale ($k = 5$, ≈ 0.10) and broad ($k = 7$, ≈ 0.06) branches (Figure 4.9). Attention is dominated by mid-scale channels, yet discrimination leans hardest on the fine-scale branch: the salient scale and the discriminative scale are not the same.

TABLE 4.5: Change in the early-warning score p_t when individual fused channels are zeroed, for the attention-dominant channel and five channels aligned with the SR$\to$AF direction. The two largest effects fall on AF-direction channels, not on the attention-dominant one, and no single channel dominates.

Channel	**Role**	Δp_t
116	Attention-dominant	-0.0152
1	AF-direction	$+0.0380$
129	AF-direction	$+0.0312$
164	AF-direction	-0.0075
102	AF-direction	$+0.0029$
91	AF-direction	-0.0005

4.4.0.0.6 The attention shape does not shift before onset. Bucketing windows by their distance to onset and recomputing gross shape statistics (normalised entropy, peak height, weighted centroid, spread, last-quarter mass), every metric stays flat across lead times; entropy hovers between 0.94 and 0.96, close to uniform, at every distance. Whatever pre-onset signal exists does not surface as attention becoming more peaked or shifting in time. The one movement is subtler: channel 116's correlation with attention strengthens from about 0.60 to 0.75 in the final five minutes, a relative shift in *which* channel explains the small departures from uniform, not a change in the distribution's overall shape. This frames the prediction analysis that follows: the early-

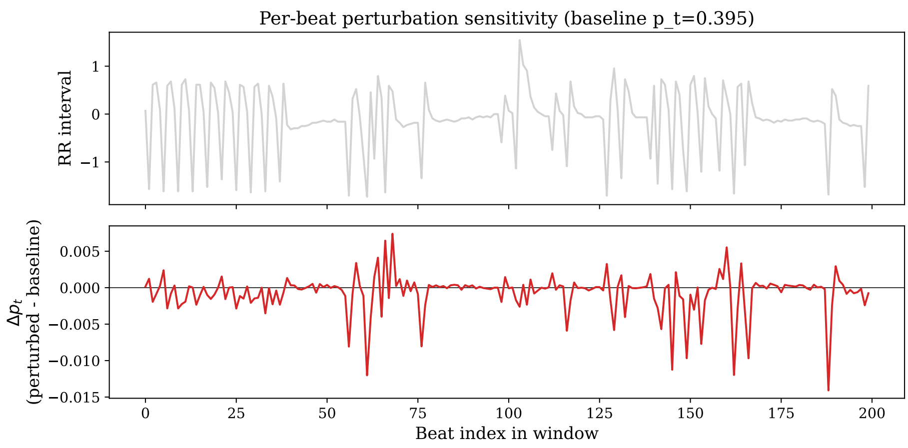

Fig. 4.6: Per-beat perturbation saliency for one window: the change Δp_t produced by nudging each beat (lower panel), against the RR trace (upper). The causal leverage lands on the irregular, high-variability beats, not on the flat segments that draw attention.

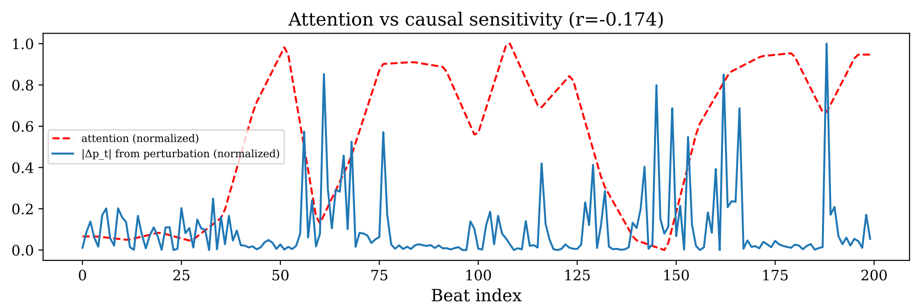

Fig. 4.7: Attention weight (dashed) against per-beat causal sensitivity $|\Delta p_t|$ from perturbation (solid), both normalised. The two are mildly anti-correlated ($r = -0.174$): attention peaks on the calm stretches while causal leverage peaks on the variable beats, a direct causal restatement of the channel-level decoupling.

warning signal, where it exists, lives in the embedding's slow drift along the SR$\rightarrow$AF axis rather than in any visible reshaping of attention.

4.5 Comparison with Prior Work

Table 4.6 places the results beside prior RRI-based AF detectors. Direct numerical comparison is necessarily loose (datasets, episode-selection criteria, and evaluation protocols differ), but the pattern across the table is informative. Methods evaluated under cross-validation that lets a single patient's segments mix across folds report strikingly high figures: Faust et al. reach 98.51% accuracy under tenfold cross-validation [11], and Andersen et al. report 98.98% sensitivity under fivefold cross-validation [12]. The same Andersen model, re-evaluated on fully unseen patients, drops to 86.04% sensitivity, a fall of more than twelve percentage points that exposes how far within-patient mixing inflates the optimistic numbers.

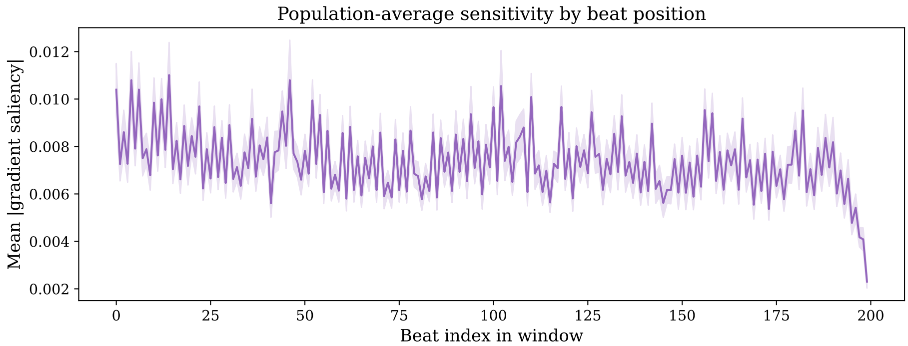

Fig. 4.8: Mean gradient saliency by beat position, averaged over the population (shaded band is the standard error). Sensitivity is roughly uniform across the window with a mild trailing-edge falloff, which indicates the score draws on the whole window rather than a few recent beats.

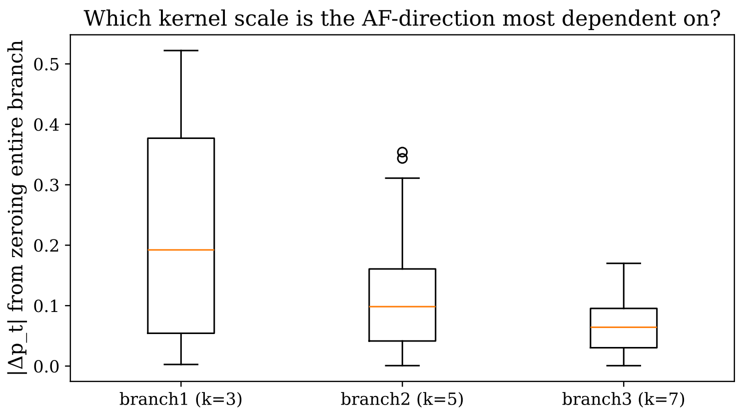

Fig. 4.9: Change $|\Delta p_t|$ from zeroing each entire convolutional branch, across windows. The fine-scale ($k = 3$) branch carries the largest causal effect on the score, ahead of the mid-scale ($k = 5$) and broad ($k = 7$) branches, the reverse of the mid-scale dominance seen in the attention mechanism.

Under the same kind of strict patient-independent holdout, the proposed method reaches $97.40 \pm 0.80\%$ sensitivity, more than eleven percentage points above Andersen et al.'s holdout sensitivity, and achieved with nothing more than a frozen encoder and a linear probe. The comparison is, if anything, conservative on our side: the SR class here is drawn from sinus rhythm at least two hours before AF onset (Section 3.3), so the negatives are unambiguously clean rhythm rather than the transitional pre-episode beats that a looser protocol would admit.

4.6 Computational Efficiency

A central premise of this work is deployment on resource-constrained, wearable-class hardware. We therefore quantify the cost of the model directly. Table 4.7 reports the footprint of the encoder, the only component that runs at inference, since the contrastive projection head is discarded once pretraining ends (Section 3.4.3).

TABLE 4.6: Comparison with prior work on RR-interval AF detection. Direct numerical comparison is limited by differences in dataset and evaluation protocol. †Cross-validation with within-patient mixing inflates the reported metrics.

Study	**Method**	**Validation**	**Sens.**	**Spec.**	**A**
Faust et al. [11]	LSTM	10-fold CV†	—	—	98.
Andersen et al. [12]	CNN+RNN	5-fold CV†	98.98%	96.95%	-
Andersen et al. [12]	CNN+RNN	Patient holdout	86.04%	98.96%	-
Ours	CNN+MLP	Patient holdout	$\mathbf{97.40 \pm 0.80\%}$	$\mathbf{92.72 \pm 2.01\%}$	$\mathbf{95.17}$ $\pm$

TABLE 4.7: Resource footprint of the encoder (the deployment path; the contrastive projection head is discarded after pretraining). Compute is per 200-beat window; latency and throughput are measured single-threaded on CPU.

Resource metric	**Value**
Encoder parameters (deployment)	101,985
Projection-head parameters (training only)	33,024
Total parameters	135,009
Model size on disk	545 KB
Encoder weights (dense fp32)	398 KB
MACs per window	2.13 M
FLOPs per window	4.25 M
CPU latency (1 thread, batch 1)	0.21 ms
CPU throughput (batch 256)	3,268 windows/s

The deployed encoder carries roughly 102 thousand parameters and occupies under 0.4 MB in single precision, the released checkpoint totalling 545 KB on disk. A single 200-beat window costs 2.13 million multiply–accumulate operations (4.25 million FLOPs) and is scored in 0.21 milliseconds on one CPU thread. Since a 200-beat window spans on the order of minutes of recording, the per-window inference cost is negligible relative to real time even without batching, hardware acceleration, or quantisation, leaving a wide margin for continuous on-device monitoring. The footprint is identical across all three training objectives (they differ only in the loss used during pretraining); the accuracy and stability gains of Sections 4.1–4.3 therefore come at no additional inference cost.

4.7 AF Onset Prediction

Finally we turn the frozen detection embedding forward in time and evaluate the early-warning scheme of Section 3.8. For each patient we average the frozen embeddings of their labelled SR and AF *detection* windows into two class centroids, which fix a personal SR→AF axis (Eq. (3.13)). We then embed each window of a separate pre-onset recording, project it onto that axis, and normalise by the axis length to obtain the scalar early-warning score p_t (Eq. (3.14)), so that $p_t = 0$ sits at the patient's own SR centroid, $p_t = 1$ at their AF centroid, and the midpoint $p_t = 0.5$ is the decision boundary half way between the two. Tracking p_t across the hour before annotated onset, we record the *crossing time*, the earliest moment at which p_t crosses this midpoint, together with the axis displacement p_t at that crossing. The scheme asks only for a lightweight per-patient calibration; the encoder itself is never retrained.

Table 4.8 reports the per-patient outcome for the eleven test patients for whom both per-patient calibration windows and a continuous pre-onset recording were available. These patients are held out under the same patient-level partition as the detection experiments (Section 3.3), so the encoder never sees them during pretraining; the only patient-specific information the scheme uses is the lightweight calibration drawn from that patient's own labelled SR and AF windows. The early-warning score crosses the midpoint a median of 50.9 minutes (mean 39.1 ± 21.3 minutes) ahead of annotated onset, ranging from 3.0 to 59.0 minutes. For six of the eleven patients the crossing occurs more than fifty minutes in advance, and for ten of eleven it occurs at least ten minutes ahead.

TABLE 4.8: Per-patient early-warning results on the test split. *Lead time* is the interval between the earliest midpoint crossing ($p_t = 0.5$) of the early-warning score and the annotated AF onset; *axis displacement* is the value of p_t (Eq. (3.14)) at that crossing—the fractional distance travelled along the patient's normalised SR→AF axis, where 0 is the SR centroid and 1 the AF centroid.

Patient ID	**Lead time (min)**	**Axis displacement**
2	58.3	0.351
3	3.0	0.409
25	59.0	0.453
35	33.8	0.490
54	57.7	0.437
98	50.9	0.460
112	29.6	0.502
118	10.8	0.280
133	58.4	0.436
135	10.7	0.422
156	57.7	0.420
Mean ± std	39.1 ± 21.3	0.424 ± 0.060
Median	**50.9**	0.436

Two features of the table are worth drawing out. First, several lead times cluster just below 59 minutes, pressing against the one-hour observation window we monitor before onset. For these patients the early-warning score has already reached the AF-ward half of the axis when monitoring begins, so the reported lead time is right-censored: the true warning horizon is at least as long as the window and plausibly longer. Second, the axis displacement at crossing sits in a tight band just below the boundary (0.424 ± 0.060, spanning 0.280 to 0.502). This is partly by construction (each value is recorded as p_t steps across the 0.5 midpoint and so necessarily lands near it), but the direction is informative: ten of the eleven values fall below 0.5, marking the recorded crossings as predominantly *downward* transitions. The score has therefore already drifted past the midpoint toward the AF region before the monitored hour begins, and fluctuates around the boundary as onset approaches, consistent with the right-censoring seen in the lead times. What makes a single 0.5 threshold meaningful across such different individuals is the per-patient normalisation: because the axis is rescaled so that 0 is each patient's own SR centroid and 1 their AF centroid, the midpoint marks the same relative position, half way between that patient's sinus baseline and their AF region, for everyone. That common operating point is exactly what the per-patient SR cohesion of Section 4.3 underwrites: a tight, consistently placed SR cluster gives the axis a well-defined origin from which drift can be measured the same way for every patient.

4.7.1 Per-Patient Heterogeneity in the Early-Warning Signal

The median lead time of Table 4.8 is a population summary; it conceals a wide spread between individuals. Across the sixteen test patients with enough labelled SR and AF windows to fix a personal axis (the eleven of Table 4.8 are the subset that also provide a continuous pre-onset recording), the p_t trajectories sort into three clear shapes. Classifying each patient by how their trajectory tracks proximity to onset yields six *late-rise* patients whose score climbs as onset nears, five *flat* patients whose score barely moves, and five *noisy* patients whose score wanders unrelated to onset (Figure 4.10).

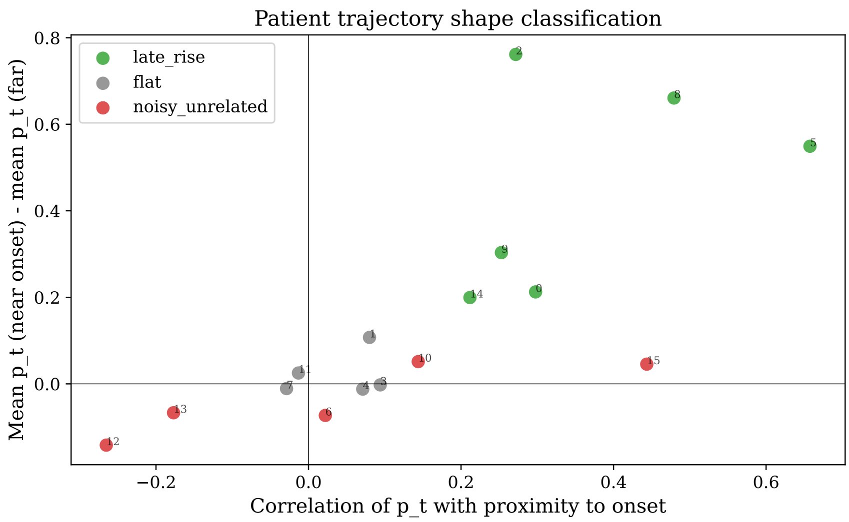

Fig. 4.10: Per-patient classification of the p_t trajectory, by its correlation with proximity-to-onset against the near-versus-far change in score. The sixteen calibrated test patients split into six *late-rise*, five *flat*, and five *noisy* cases, a clean partition rather than a borderline continuum.

Two structural factors explain much of this spread. First, the reliability of p_t is gated by how far apart the patient's SR and AF centroids sit. For three of the sixteen the separation $\|\vec{\mu}_{\text{AF}} - \vec{\mu}_{\text{SR}}\|$ falls below 0.4, and their trajectories swing wildly (excursions past ± 2) because the short axis amplifies embedding noise; the remaining thirteen, with separation above 0.7, give stable trajectories. Second, the calibration is only as good as the reference episodes. Patient 2's SR projections are visibly bimodal, a sign that the episode labelled sinus rhythm is itself contaminated, most plausibly by undetected paroxysmal activity, which drags the SR centroid off its true position. These are calibration-quality failures rather than failures of the score itself; they point to a per-patient calibration screen as a prerequisite for any deployment. Figure 4.11 shows the raw p_t trajectories for five patients and makes both factors visible at once: Patient 2 swings the full width of the axis (past both centroids), the signature of the contaminated baseline, while the others fluctuate around the SR end with only modest, irregular excursions toward AF.

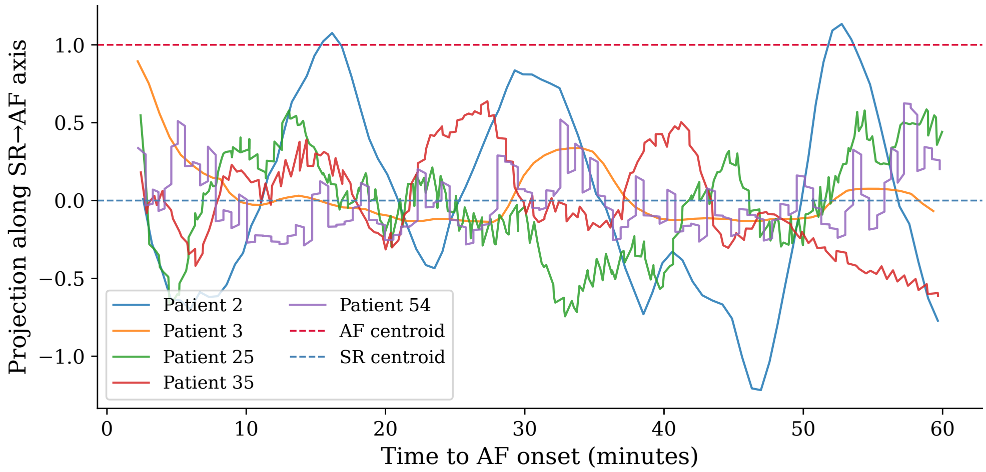

Fig. 4.11: Early-warning score p_t (projection along the SR→AF axis) over the hour before onset, for five test patients. Dashed lines mark the SR (0) and AF (1) centroids. The trajectories are markedly heterogeneous: Patient 2 swings past both centroids, betraying a contaminated SR baseline, whereas the others drift around the sinus end with small, noisy movements toward AF.

4.7.2 Discrimination by Lead Time and Subgroup

To quantify the signal without committing to a threshold, we ask how well p_t in a given lead-time window separates from a far-from-onset background (windows more than 45 minutes before onset), scored by AUROC. Pooled across all patients, discrimination peaks at 0.66 in the 5–10 minute window and decays to chance (0.45–0.48) beyond 15 minutes (Figure 4.12). Split by subgroup, this pooled figure is exposed as a dilution.

Restricted to the six late-rise patients, the score reaches an AUROC of 0.86 at 5–10 minutes (patient-level bootstrap 95% CI $[0.73, 0.99]$); the other ten patients sit at chance at every lead time tested (Figure 4.13). With only six patients the estimate is necessarily wide, which is why we bootstrap it, but its lower bound still clears the conventional 0.7 good-discrimination mark. The early-warning signal is thus real but patient-stratified: strong in a recognisable subgroup, undetectable in the rest.

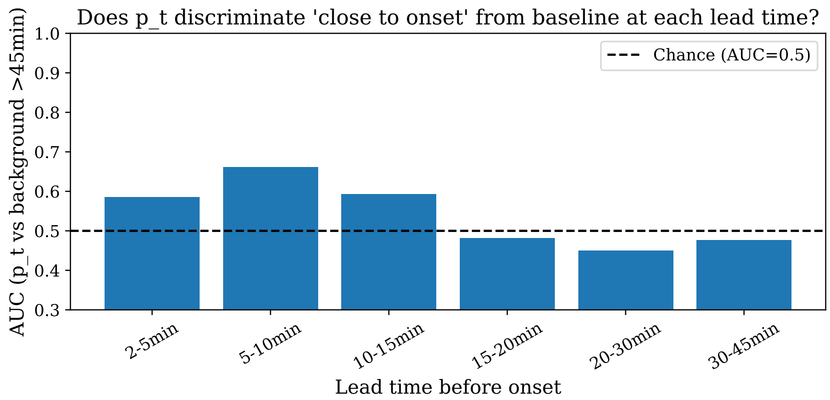

Fig. 4.12: AUROC for separating each lead-time window from a far-from-onset (> 45 min) background, pooled over all calibrated patients. Discrimination peaks at 0.66 in the 5–10 minute window and falls to chance beyond 15 minutes; the dashed line marks chance (0.5).

4.7.3 From Axis Crossing to an Alarm Rule

Turning the trajectory into an actual alarm exposes the limits of the midpoint rule. A fixed absolute threshold ($p_t > 0.5$, sustained) fires for only 18.8% of episodes, and recalibrating it upward toward the AF-centroid level removes the signal entirely, since pre-onset rhythm by definition never reaches full AF. A relative rule does better: firing when p_t rises a fixed margin above the patient's own early-window baseline, held for a few minutes, detects 46.2% of episodes in the high-separation cohort at zero false alarms (measured on each episode's own far-from-onset segment as a negative control). Scaling the required rise by each patient's baseline variability, a per-patient z-score, dominates the flat-margin rule along the Pareto frontier in the low-false-alarm regime (Figure 4.14); the two rules favour complementary patients, the flat rule those with steady baselines and the personalised rule those with noisy ones. OR-combining them lifts detection to 61.5% at a 7.7% false-alarm rate. One limit is structural: the earliest measurable crossing sits around 2–3 minutes before onset, set by the window length, so sub-two-minute warning is simply not observable under this scheme. These alarm rules are exploratory, tuned and tested on the same single dataset; we report them to chart the route from a geometric score to a usable warning, not as a validated detector.

These prediction results are a proof of concept rather than a fully validated forecasting system. They reuse the frozen detection embedding without any predictive training, run on a single dataset, and depend on per-patient calibration episodes whose quality, as the heterogeneity analysis showed, gates the result. We return to their clinical reading and limitations in the Discussion. Their value here is to show that the patient-aware representation, built for detection, already carries enough per-subject geometric structure to support a forward-looking early-warning signal: a median horizon approaching an hour, and strong discrimination (AUROC 0.86) in an identifiable subgroup of patients, at no extra training cost.

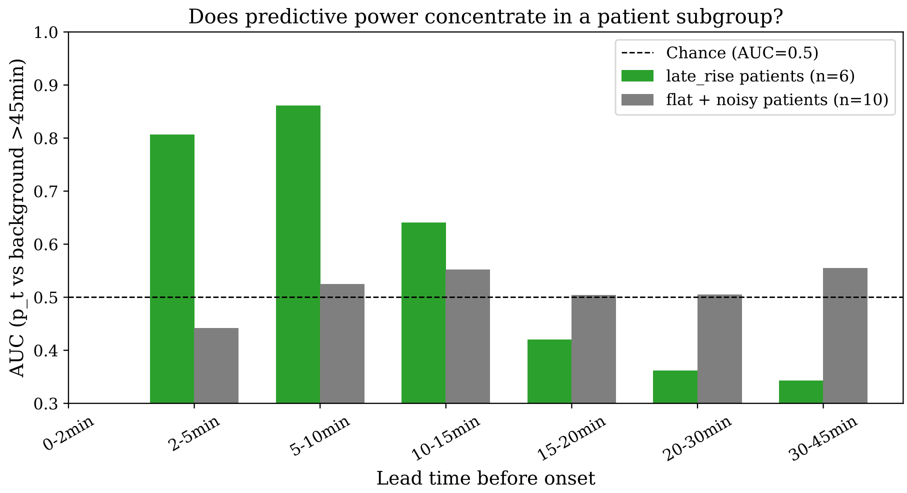

Fig. 4.13: AUROC for separating each lead-time window from a far-from-onset (> 45 min) background, split by trajectory subgroup. The six late-rise patients reach 0.86 at 5–10 minutes; the remaining ten stay near chance at every lead time, which is why the pooled curve peaks at only 0.66.

4.8 Summary of Findings

Four results stand out. First, the patient-aware representation detects AF in unseen patients at an AUROC of 0.989 ± 0.003 with clinically favourable sensitivity-weighted errors. Second, against matched SupCon and BCE baselines its decisive edge is stability (a $2.6\times$ and $3.4\times$ reduction in seed-to-seed AUROC variance), and the embedding-geometry analysis traces that edge to the highest per-patient SR cohesion in the comparison, exposing along the way a BCE paradox in which the cleanest global class separation accompanies the worst cross-patient transfer. Third, the same frozen embedding extends to AF onset prediction, flagging impending episodes a median of roughly fifty minutes ahead at a consistent operating point across patients. That signal is markedly patient-heterogeneous, reaching an AUROC of 0.86 in an identifiable late-rise subgroup while staying at chance in the rest. A companion interpretability analysis shows the attention mechanism learns a robust rhythm-stability detector, decoupled ($r \approx 0.03$) from the direction that actually drives discrimination. Fourth, these gains carry essentially no inference cost: the deployed encoder is a compact model of about 102 thousand parameters (under $0.4\,$MB) that scores a 200-beat window in 0.21 milliseconds on a single CPU thread, comfortably within a wearable budget. Together these findings support the central claim of this work: that per-subject geometric consistency, not global class separability, is what governs robust generalisation to unseen patients.

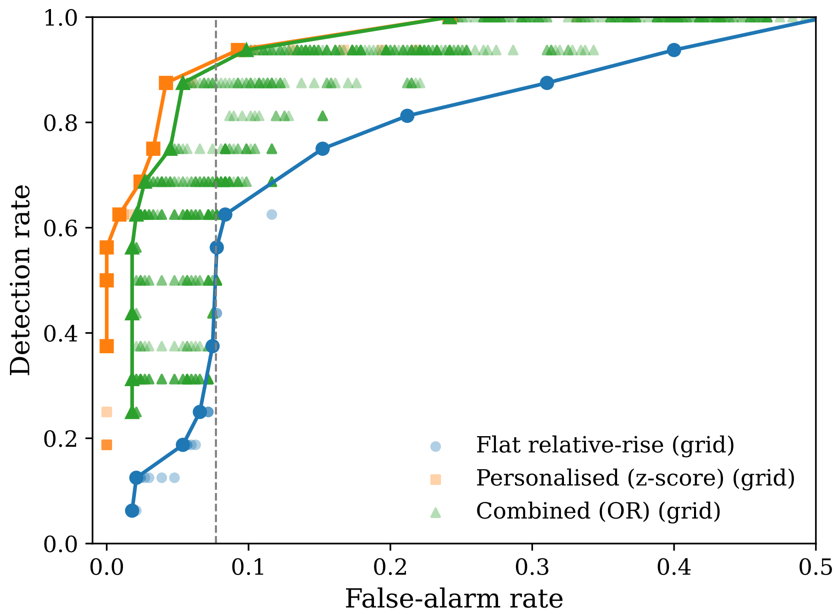

Fig. 4.14: Detection rate against false-alarm rate for the early-warning alarm rules, swept over their parameter grids (high-separation cohort; false alarms measured on each episode's own far-from-onset segment). The personalised z-score rule dominates the flat relative-rise rule along the Pareto frontier in the low-false-alarm regime, and OR-combining the two reaches 61.5% detection at a 7.7% false-alarm rate.

CHAPTER 5

EVALUATION

Chapter 4 reported *what* the patient-aware representation achieves. This chapter asks a different question: whether the system, as designed, is fit for the deployment setting that motivates it—continuous cardiac screening on low-cost wearable hardware in resource-constrained environments. Accuracy alone does not answer that. A model that generalises to unseen patients is of little use if it cannot run within the compute budget of a wearable, if it is locked to the one sensing modality it was trained on, or if its headline numbers were obtained under a protocol that flatters it. We therefore evaluate the approach along three deployment-facing axes: the computational characteristics of the encoder (Section 5.1), its applicability across the ECG and PPG sensing modalities (Section 5.2), and the practical implications of the strict patient-independent evaluation protocol (Section 5.3). Throughout we separate what this work has *validated* from what it has *designed for but not yet measured*, and defer the broader interpretation of the findings to Chapter 6.

5.1 Computational Characteristics for Edge Deployment

The encoder was deliberately kept small (Section 3.4.3): a compact multi-scale 1D-CNN backbone, a cross-branch fusion, a temporal attention module, and—at training time only—a two-layer projection head. This restraint serves two ends at once. It pins any performance advantage on the training objective rather than on raw capacity, and it keeps the deployed model within reach of wearable-class hardware.

5.1.1 Parameter and Memory Budget

Table 5.1 gives the parameter budget measured directly from the trained checkpoint, broken down by component of the architecture in Section 3.4.3. The deployed encoder—the multi-scale backbone, the 1×1 cross-branch fusion, the temporal attention pooling, and the $192 \rightarrow 128$ encoder projection that produces the embedding $\hat{z}$—totals $101{,}985$ parameters. The contrastive projection head adds a further $33{,}024$, but it exists only for the contrastive objective and is discarded once the encoder is frozen for linear probing (Section 3.5); it never runs at inference. The full model thus holds $135{,}009$ parameters—squarely on the order of 10^5—of which roughly three-quarters sit on the inference path.

A budget of this size translates into a small memory footprint. At full `float32` precision the deployed encoder occupies about 398 KB of weights—consistent with the 545 KB on-disk checkpoint, which stores the full model in `float32`—and post-training `int8` quantisation would bring this to roughly 100 KB. Either figure fits comfortably within the on-chip memory of microcontroller-class wearable processors, and the activation memory is smaller still: the strided convolutions downsam-

TABLE 5.1: Parameter budget of the encoder, measured from the trained checkpoint and grouped by architectural component (Section 3.4.3). The projection head is used only during contrastive training and is absent at inference.

Component	Parameters	On inference path?
Multi-scale CNN backbone (3 branches, $k \in \{3, 5, 7\}$)	39,648	Yes
Cross-branch 1×1 fusion (channel mixing)	37,440	Yes
Temporal attention pooling	193	Yes
Encoder projection ($192 \to 128$)	24,704	Yes
Deployed encoder	**101,985**	—
Contrastive projection head (two-layer MLP)	33,024	No (training only)
Full training model	**135,009**	—

ple the temporal axis from 200 beats to 25 across the three blocks (measured shapes $200 \to 100 \to 50 \to 25$), so the largest intermediate tensor—the fused 25-step, 192-channel representation—holds only 4,800 values. The model never needs to materialise a large feature map. Profiling the forward pass confirms the compute cost is equally modest: a single 200-beat inference window incurs approximately 2.1 million multiply–accumulate operations (about 4.3 MFLOPs), dominated by the convolutional backbone and the 1×1 fusion.

5.1.2 Architectural Properties That Aid Edge Inference

Beyond its size, three structural choices make the encoder well-behaved on constrained hardware. First, it is *feed-forward and recurrence-free*. Unlike the LSTM and CNN–RNN detectors that define much of the prior art [11, 12], it carries no hidden state across time steps, so a fixed 200-beat window incurs a fixed, fully parallelisable amount of computation with no sequential dependency to serialise. Second, the strided convolutions shrink the sequence as it deepens, concentrating compute in the early, low-channel layers and keeping the deeper layers cheap. Third, the backbone uses Group Normalisation rather than Batch Normalisation [101]: Group Normalisation computes its statistics within each sample and is therefore unaffected by batch size, which matters directly for streaming edge inference where windows arrive one at a time (batch size one). A Batch-Normalised encoder would have to either cache running statistics or behave inconsistently at batch size one; the design sidesteps that entirely.

Taken together, these properties place the model firmly in the regime of recent lightweight on-device AF architectures [18] rather than the server-grade models it is benchmarked against. We are careful, however, not to overstate the claim. The parameter count, memory footprint, and per-window compute reported above are measured from the trained model, and they establish the architectural case for edge readiness; but this work does *not* report measured on-device latency, throughput, or energy on a specific wearable target, nor does it carry out the `int8` quantisation that the ~ 100 KB figure presupposes. That profiling—and the accompanying quantisation-versus-accuracy trade-off—is deferred to the deployment work outlined in Chapter 7.

5.2 Cross-Modality Applicability: ECG and PPG

A central premise of this dissertation is that the same pipeline can serve both the single-lead ECG patches that stream RR intervals and the far more ubiquitous PPG sensors that yield pulse-to-pulse intervals. The evaluation of that premise is partly a matter of design and partly a matter of acknowledged untested transfer.

5.2.1 Why the Design Is Modality-Agnostic

The case for cross-modality use rests on the input representation. The encoder never sees a waveform; it consumes a sequence of inter-beat intervals (Section 3.2). Because PPG pulse-to-pulse intervals approximate ECG RR intervals to within a slowly varying pulse-transit-time offset [9], both modalities expose the same beat-to-beat timing dynamics that carry the AF signature. Two further elements of the pipeline reinforce this modality-independence. The per-patient robust scaler of Equation (3.1) centres and scales each patient's intervals by their own median and IQR, absorbing constant or slowly drifting offsets between the two modalities rather than treating them as signal. And the patient-aware objective itself needs only a subject identifier and a class label per segment (Section 3.7)—never any morphological feature that an ECG provides and a PPG trace does not. The learning mechanism is, by construction, blind to which sensor produced the intervals.

5.2.2 What Remains Untested

This is a design property, not a validated result, and the distinction matters. Every experiment in this work was conducted on IRIDIA-AF [23], which is single-lead *ECG*; no PPG data entered training or testing. The cross-modality claim therefore rests on the established physiological correspondence of the two interval streams, not on measured PPG performance. That correspondence is real but imperfect. PPG is more susceptible to motion artefact, poor perfusion, and ambient-light interference than ECG, and these corrupt the very beat-detection step that the pipeline depends upon, producing missed or spurious pulses that distort the interval sequence. The pipeline offers partial protection—the physiologically implausible-beat filter (< 200 ms or > 2000 ms, Section 3.3) discards a class of detection errors, and the robust scaler resists the remaining outliers—but it does not address motion artefact at its source. An honest evaluation must therefore record cross-modality applicability as a well-motivated hypothesis the architecture is built to support, whose empirical confirmation on paired or PPG-native data is left to future work (Chapter 7).

5.3 Implications of the Patient-Independent Protocol

The headline numbers of Chapter 4 are only as meaningful as the protocol that produced them. We argue here that the protocol is conservative by design, that it mirrors the deployment scenario, and that its remaining limitations bound the strength of the conclusions.

5.3.1 The Protocol Measures the Deployment Task

Every split is cut at the patient level (Section 3.3), so the test set contains only individuals the model has never encountered. This is not a stylistic choice but a faithful rendering of the deployment task: a wearable screening tool is, by definition, applied to new patients. The cost of relaxing it is documented in the literature we benchmark against—Andersen et al.'s sensitivity falls from 98.98% under cross-validation to 86.04% on held-out patients once within-patient mixing is removed [12]. Our protocol forecloses that inflation from the outset. The frozen-encoder linear probe reinforces the point: because the encoder does not update during evaluation (Section 3.5), the reported performance reflects the quality of the representation a new patient would actually meet, classified by a probe of minimal capacity. This also mirrors a realistic deployment path—a fixed encoder shipped on-device, adapted to a site or cohort only through a lightweight linear readout, or, for the early-warning extension, through the per-patient calibration of Section 3.8.

5.3.2 Stability as a Deployment Property

A consequence of this work that a single accuracy figure obscures is the $2.6\times$ and $3.4\times$ reduction in seed-to-seed AUROC variance over the SupCon and BCE baselines (Section 4.2). We treat this as a first-class deployment metric rather than a statistical footnote. A medical screening tool must behave consistently from one trained instance to the next; a model whose performance swings with its random seed is a liability in a regulated setting, regardless of its mean. That the patient-aware objective tightens this variance—and, as the geometry analysis showed (Section 4.3), does so by enforcing a more consistent per-patient embedding structure—is therefore a property of direct practical value, not merely a favourable number.

5.3.3 A Deliberately Hard Benchmark, and Its Limits

The evaluation is conservative in a further respect: the SR negatives are drawn from sinus rhythm at least two hours before AF onset (Section 3.3), so the negative class is unambiguously clean rhythm rather than the transitional pre-episode beats a looser protocol would admit. The model is thus asked to separate AF from *clean* SR, a sharper and more clinically honest test than one that lets borderline rhythm leak into the negatives.

These strengths are real, but they bound rather than eliminate the threats to validity, and an honest evaluation must name them. The results rest on a single dataset from a single centre, recorded with a single ECG device, and the test partition comprises 24 patients; robustness across centres, devices, and larger cohorts is therefore assumed, not demonstrated. The detection probe is fit once on the training patients and applied unchanged to all test patients, with no per-patient adaptation—a conservative choice for measuring representation quality, but one that leaves the gains available from few-shot personalisation unexamined. And the early-warning analysis of Section 4.7, while encouraging, rests on eleven calibrated patients and on lead-time measurements that are right-censored at the one-hour observation window; it is, moreover, strongly patient-heterogeneous and gated by calibration quality, with the discriminative signal concentrated in a subgroup whose SR and AF centroids are well separated (Section 4.7.2). Any deployment would therefore need a per-patient calibration-quality screen rather than uniform application, and the analysis should be read as a feasibility result, not a validated forecaster. We return to these limitations, and the work needed to close them, in Chapters 6 and 7.

5.4 Summary

Evaluated as a deployable system rather than a benchmark entry, the approach holds up on the axes that matter for resource-constrained screening, with clearly marked boundaries. Its deployed encoder holds $101{,}985$ parameters ($135{,}009$ including the training-only projection head)—about 398 KB at full precision, near 100 KB after `int8` quantisation—and a single inference window costs only about two million multiply–accumulate operations; this, together with its feed-forward, batch-size-one-friendly structure, suits streaming edge inference, though measured on-device latency and energy remain to be profiled. Its interval-only input makes it modality-agnostic by design across ECG and PPG, though PPG transfer is so far argued from physiology rather than measured. And its patient-independent, frozen-probe protocol measures the genuine deployment task and yields the seed-stable behaviour a clinical tool requires, within the limits of a single dataset and a modest test cohort. The recurring theme is a system engineered for deployment whose deployment-readiness this chapter substantiates analytically and architecturally, while marking precisely where empirical confirmation has yet to be supplied.

CHAPTER 6

DISCUSSION

Chapter 4 established *what* the patient-aware representation achieves and Chapter 5 asked whether it is fit to *deploy*. This chapter steps back from both to interpret *why* the findings take the shape they do, and what they imply beyond the confines of a single dataset. Five threads run through it: the mechanism behind the BCE paradox and what it says about model selection under patient shift (Section 6.1); the clinical reading of seed-stable, sensitivity-weighted detection (Section 6.2); what the interpretability analysis reveals about the attention mechanism, and why a faithful attention map can still mislead (Section 6.3); the prospects for turning a detection embedding into a genuine early-warning system (Section 6.4); and the generality of the patient-aware construction to other physiological signals organised by subject (Section 6.5).

6.1 Interpreting the BCE Paradox

The most conceptually important result of this work is also its most counterintuitive. Binary cross-entropy produced the cleanest global class structure of any objective compared (the largest centroid distance 1.603, the most negative centroid similarity -0.717, and the highest global compactness 2.427 in Table 4.4). Yet it transferred worst to unseen patients. The resolution lies in what a cross-patient linear probe actually has to do. Fit on the training patients, the probe learns a single SR$\rightarrow$AF direction and offset; at test time it must place a never-before-seen patient's sinus baseline on the correct side of that boundary. For this to succeed, every patient's SR cluster must sit in a *consistent* place relative to the decision surface. BCE has no incentive to provide this. Its loss is minimised the moment the training classes become linearly separable, by whatever direction does the job most cheaply.

When class membership is partly correlated with subject identity (as it is whenever a cohort's AF and SR segments come from overlapping but non-identical sets of patients), the easiest separating direction can exploit subject-correlated cues that split *this* cohort's classes but carry no meaning for a new individual. The symptom is written into the geometry: BCE's per-patient SR cohesion is the lowest in the table (0.772), so each patient's sinus baseline lands in an idiosyncratic spot even as the class means are pushed far apart. The probe is left without a stable origin to extrapolate from, and a new patient's SR can fall on the wrong side of a boundary that looked crisp on the training cohort. BCE's chart-topping *per-patient* compactness ratio (6.396) is the same pathology in another guise: as flagged in Section 4.3, it is an artefact of a vanishing within-patient spread (the denominator collapses toward zero while the tight clusters it measures drift unpredictably from one patient to the next), not evidence of useful structure.

The patient-aware objective removes the shortcut by construction. Restricting positives to same-patient, same-class segments forces the encoder to make AF and SR separable *within each subject*, which is the one form of separation guaranteed to remain valid for a new subject. That is why it attains the highest per-patient SR cohesion (0.850) while keeping AF cohesion balanced (0.846), and why the three objectives rank on downstream AUROC ($0.989 > 0.983 > 0.980$) exactly as they rank on per-patient SR cohesion and inversely to how they rank on global separability. Read through the lens of representation-learning theory, the patient-aware positive enforces a *conditional* form of the alignment that contrastive methods are known to optimise [14, 89]: it aligns same-class samples conditioned on subject, leaving the across-subject variation intact rather than averaging it away.

The practical lesson outlives this dataset. Global class separability, however measured, is a misleading criterion for choosing models or losses when the deployment distribution differs from training by subject. A practitioner who tunes for the cleanest training-set separation will be steered toward precisely the objective that transfers worst. Per-patient class cohesion, and per-patient SR cohesion in particular, is the diagnostic that tracks transfer, computable without any test labels.

6.2 Clinical Relevance of Seed-Stable, Sensitivity-Weighted Detection

The headline advantage of the patient-aware objective is not a large jump in mean accuracy but a representation that behaves *consistently*: a $2.6\times$ and $3.4\times$ reduction in seed-to-seed AUROC variance over SupCon and BCE (Section 4.2). It is worth dwelling on why this is a clinical property rather than a statistical nicety. A screening model is periodically retrained (on a new data release, a new site's cohort, a corrected label set), and should reproduce its behaviour, not gamble it on a random seed. In a regulated medical setting, run-to-run variance is itself a hazard: two instances of the same architecture that disagree on borderline patients are difficult to validate, monitor, and certify, whatever their average performance. By tying the encoder's behaviour to a stable per-patient geometry rather than to whichever separating direction a given initialisation happens to find, the patient-aware objective makes reproducibility a structural feature, and the variance reduction is its measurable footprint. This is why Chapter 5 treated stability as a first-class deployment metric (Section 5.3) rather than a footnote.

The second clinically salient feature is the asymmetry of the errors. AF recall (0.974) consistently exceeds SR recall (0.927) across every seed (Section 4.1): the model trades a modest rise in false positives for fewer missed AF episodes. For a screening instrument this is the right direction in which to err. A missed paroxysmal episode can delay anticoagulation in a patient at elevated stroke risk [6], whereas a false positive triggers a confirmatory review that carries far smaller cost. The combination that matters for triage is therefore not a marginally higher but seed-volatile accuracy; it is high, *stable* sensitivity paired with a benign error profile, exactly the regime the patient-aware representation occupies. The wider spread in SR recall ($\sigma = 0.020$) is itself informative rather than alarming: as Section 4.1 argued, it tracks genuine patient-level heterogeneity in baseline sinus rhythm (a handful of atypical SR baselines among unseen patients) rather than model instability, since SR precision stays tight at 0.970 ± 0.009.

6.3 What the Attention Mechanism Does, and Does Not, Explain

The interpretability analysis of Section 4.4 returns a result worth dwelling on, because it cuts against an intuitive reading of attention. The temporal attention module learns a clear, robust, and physiologically sensible behaviour: it pools toward the stable, low-variability stretches of the rhythm, driven by a small redundant cluster of mid-scale channels that act as anti-variance detectors. The natural temptation is to treat this as *the* explanation for how the encoder tells AF from SR. The geometry says otherwise. The channels that steer attention are essentially uncorrelated ($r \approx 0.03$) with the channels that define the SR→AF direction the probe actually reads. Attention and discrimination are decoupled. This is not a correlational artefact: causal probes tell the same story. Per-beat saliency, from a gradient map and a perturbation map that agree at $r = 0.633$, places the leverage on the irregular beats attention turns away from (the two are anti-correlated at $r = -0.174$), and branch ablation shows the score leaning hardest on the fine-scale convolutional branch even though attention is dominated by the mid-scale one (Section 4.4).

Two lessons follow. The first is methodological. A faithful, forward-pass attention map can be both genuine and beside the point: here it reports where the model looks, not what separates the classes, a caution that echoes the broader finding that attention weights need not coincide with feature importance [104]. An interpretability story built on the attention overlay alone would have been confidently wrong about the mechanism. The second lesson refines the prediction account. Because the pre-onset signal never surfaces as a reshaping of attention (its gross shape is flat across lead times, Section 4.4), the early-warning drift must live in the same distributed, discrimination-aligned directions that carry detection, which is exactly what the axis-projection score reads. The attention mechanism is a clean diagnostic of what the encoder finds salient, while the SR→AF axis is the diagnostic of what it finds discriminative; keeping the two apart is what let the analysis avoid a tempting but false conclusion.

6.4 Prospects for Early Prediction from Autonomic Precursors

The early-warning analysis of Section 4.7 is the most speculative part of this work and, for that reason, the most interesting to interpret with care. Its striking feature is that a frozen embedding trained *only to tell AF from SR* already carries enough forward-looking structure to flag impending episodes a median of roughly fifty minutes ahead, with no predictive training whatsoever. The most plausible reading is physiological. The autonomic and ectopic precursors that precede paroxysmal onset—

vagal-tone HRV markers that rise over the half hour before an episode, and premature-atrial-contraction burden that climbs from roughly 4.8% to 8.3% in the final minutes [19, 20]—perturb exactly the beat-to-beat interval dynamics the encoder was trained to read. A representation that separates established AF from clean SR appears, as a by-product, to place pre-onset rhythm on a continuum between the two, which the per-patient SR$\rightarrow$AF axis reads as progress toward an episode.

That this works at all leans directly on the detection contribution. The early-warning score is comparable across very different individuals only because the axis is normalised per patient and anchored at a tight, consistently placed SR centroid, the very per-patient SR cohesion the patient-aware objective maximises (Section 4.3). An objective that scattered each patient's SR baseline, as BCE does, would give the axis an unreliable origin and the score no common meaning across patients.

The honesty the rest of this dissertation has tried to keep must extend here too. The result is a feasibility signal, not a forecaster. The lead times are right-censored at the one-hour observation window; the analysis covers eleven calibrated patients on a single retrospective dataset; and, as Section 4.7 noted, the recorded crossings are predominantly *downward*, meaning the score is already in the AF-ward half of the axis when monitoring begins and fluctuates around the boundary rather than rising cleanly through it. The signal is also far from uniform across patients. The subgroup analysis (Section 4.7.2) splits the cohort cleanly: an AUROC of 0.86 at 5–10 minutes in a late-rise subgroup, against chance-level discrimination in the remainder, with the divide tracking calibration quality, the separation between a patient's SR and AF centroids and the cleanliness of their reference episodes. A population-uniform forecaster is therefore the wrong target; the realistic version is patient-stratified, applied only where calibration is sound. Turning this into a deployable early-warning system would demand more than reusing the detection embedding: a principled alarm rule with hysteresis and smoothing to suppress spurious crossings, in the spirit of the non-anticipative moving-average gating used by dedicated onset predictors [1]. Our exploratory relative-rise and personalised-threshold rules (Section 4.7.3) are a first step in that direction, combining to 61.5% detection at a 7.7% false-alarm rate in the high-separation cohort, though they remain tuned and tested on a single dataset. Beyond the alarm rule, a deployable system would also want objective training for the prediction task rather than zero-shot reuse [21, 22], and prospective evaluation against a clinically defined warning horizon. That objective training is not a small undertaking, and the reason is instructive: a long line of experiments that trained directly for onset prediction, documented in Appendices B and C, failed to generalise across patients precisely because they had to fight the patient-identity confound that the detection embedding sidesteps by construction. What the present result *does* establish is that the patient-aware representation is a credible starting point for such a system: the geometry that prediction needs is already present, at no extra cost, in a model built for detection.

6.5 Generality to Other Subject-Structured Signals

Nothing in the patient-aware construction is specific to RR intervals or to atrial fibrillation. Its only requirements are that the data be organised by subject and that the positive set be definable from a subject identifier and a class label, metadata that virtually every physiological dataset already carries. The failure mode it corrects is equally general: wherever class differences are partly confounded with subject differences, an unconstrained objective can separate the training cohort's classes through subject-correlated shortcuts that do not transfer, and a class-only contrastive objective will average away the inter-subject structure a new individual needs.

This makes the approach a natural candidate for other domains marked by strong inter-subject variability (EEG-based seizure detection and brain–computer interfaces, EMG-based gesture recognition, and PPG-native cardiac analysis among them), where the gap between within-subject and cross-subject performance is a recurring obstacle, and where contrastive methods for physiological signals have already taken hold [16, 17]. Equally transferable is the diagnostic that accompanies the objective: the per-subject cohesion metrics of Section 3.6 offer a label-free way to ask whether a learned representation has preserved individual structure or collapsed it, and thereby to anticipate cross-subject transfer before a single test label is seen.

The scope of the claim should be stated precisely. The patient-aware constraint helps *because* subject identity is a confound an unconstrained objective would otherwise exploit. Where class is cleanly independent of subject (where every individual exhibits both classes with identical within-subject geometry), the shortcut does not exist, and standard supervised contrastive learning [15] would suffice. The contribution of this work is to identify the broad and practically common regime in which that independence fails, to explain why its failure breaks cross-patient generalisation, and to supply a minimal, almost cost-free modification of the positive set that repairs it.

CHAPTER 7

CONCLUSION

This dissertation set out to build a system for paroxysmal atrial fibrillation detection (and, as a forward-looking extension, early onset prediction) that generalises to genuinely unseen patients from cardiac interval sequences. It had to stay light enough to run on the wearable hardware that makes large-scale screening feasible in resource-constrained settings. Its guiding finding is a single, transferable principle: what governs robust cross-patient generalisation is not how cleanly classes are separated in aggregate, but how consistently each individual is represented within the embedding space. The remainder of this chapter summarises the contributions that follow from that principle, the limitations that bound them, and the work that would carry them toward clinical deployment.

7.1 Summary of Contributions

The work makes five contributions, answering the objectives set out in Chapter 1.

1. **A mechanism for cross-patient generalisation failure.** The embedding-geometry analysis of Section 4.3 identifies per-patient SR cohesion, not global class separability, as the property that tracks downstream transfer, and exposes the *BCE paradox*: binary cross-entropy attains the cleanest global class structure yet the most disordered per-patient geometry and the worst transfer to unseen patients. This reframes a long-standing generalisation gap as a question of per-subject, rather than aggregate, structure.

2. **A patient-aware contrastive objective.** Restricting positive pairs to same-patient, same-class segments prevents the encoder from collapsing distinct individual SR baselines into one shared prototype, the failure mode of standard supervised contrastive learning on subject-structured signals. The objective attains the highest per-patient SR cohesion in the comparison (0.850 versus 0.800 for SupCon and 0.772 for BCE), confirming the intended geometric effect at minimal additional cost.

3. **A lightweight, modality-agnostic encoder.** A multi-branch 1D-CNN with temporal attention pooling, whose deployed form holds $101{,}985$ parameters (about 398 KB at full precision) and costs roughly two million multiply–accumulate operations per inference window (Chapter 5). Being feed-forward, recurrence-free, Group-Normalised, and fed only inter-beat intervals, it is built to stream on microcontroller-class hardware and to operate identically on ECG-derived RR intervals and PPG-derived pulse intervals.

4. **State-of-the-art detection under a strict protocol.** On IRIDIA-AF [23] under patient-independent holdout, a frozen encoder with a linear probe reaches an AUROC of 0.989 ± 0.003 and a sensitivity of $97.40 \pm 0.80\%$, exceeding the patient-holdout sensitivity of Andersen et al. [12] by more than eleven percentage points. It also cuts seed-to-seed AUROC variance by $2.6\times$ and $3.4\times$ over the SupCon and BCE baselines, a stability that Section 6.2 argued is itself a deployment property.

5. **A per-patient early-warning scheme.** Reusing the frozen detection embedding, a personal SR$\rightarrow$AF axis calibrated from a patient's own labelled episodes yields a scalar score that flags impending onset a median of roughly fifty minutes ahead (Section 4.7). This feasibility result shows that the detection representation already carries forward-looking, per-subject structure.

7.2 Limitations

The boundaries of these claims, set out in full across Chapters 4 and 5, are worth restating plainly. All experiments rest on a single dataset from a single centre, recorded with a single ECG device, with a test partition of twenty-four patients; robustness across centres, devices, and larger cohorts is therefore assumed, not demonstrated. The cross-modality premise is a design property argued from the physiological correspondence between PPG pulse intervals and ECG RR intervals [9]; no PPG data entered training or testing. The detection probe is fit once and applied unchanged to every test patient, leaving the gains available from per-patient personalisation unexamined. The early-warning analysis is a feasibility study on eleven calibrated patients with right-censored lead times, not a validated forecaster. And while the parameter, memory, and compute budgets are measured from the trained model, this work does not report on-device latency, throughput, or energy on a specific wearable target, nor does it carry out the `int8` quantisation its $\sim$100 KB footprint presupposes.

7.3 Future Work

These limitations map directly onto concrete next steps.

- **Few-shot patient adaptation.** The frozen-probe protocol deliberately measures representation quality without personalisation; a lightweight per-patient adaptation of the linear readout, or of the SR$\rightarrow$AF axis, is the obvious lever for recovering the headroom it leaves untouched, and fits the on-device calibration path of Section 3.8.

- **On-device deployment benchmarking.** Carrying out post-training `int8` quantisation, measuring latency, throughput, and energy on representative microcontroller-class hardware, and quantifying the accompanying quantisation–versus–accuracy trade-off would convert the architectural case for edge readiness (Chapter 5) into a measured one. A controlled comparison of the frozen linear probe against end-to-end fine-tuning belongs here too.

- **Multi-centre, multi-device, and PPG-native validation.** Prospective evaluation across sites, devices, and especially on PPG-acquired intervals would test the cross-modality and cross-cohort assumptions head-on, including the motion-artefact robustness that ECG data cannot probe.

- **From feasibility to forecaster.** Elevating the early-warning scheme would require objective training for the prediction task rather than zero-shot reuse of the detection embedding, a principled alarm rule with smoothing and hysteresis [1], and prospective assessment against a clinically meaningful warning horizon. Appendices B and C record the direct prediction attempts that preceded this scheme and the patient-identity shortcut that defeated them, marking out the pitfalls any future forecaster will have to clear.

- **Beyond cardiac intervals.** Because the patient-aware positive needs only a subject identifier and a class label, the construction transfers to other subject-structured physiological signals, among them EEG, EMG, and PPG-native streams (Section 6.5), where the same confound between class and subject identity is known to undermine cross-subject transfer.

7.4 Closing Remarks

The through-line of this dissertation is that a small change in *what counts as a positive pair*—same patient, same class—repairs a failure mode that no amount of global class separation can fix. The change is almost free: it adds no parameters, no inference cost, and no annotation beyond a subject identifier the data already carry. Yet it reorders the embedding geometry in the way that matters for deployment, replacing brittle, seed-dependent class separation with a stable per-subject structure that a new patient can be measured against consistently, and from which a forward-looking onset signal can be read for free. Physiological machine learning for resource-constrained, real-world screening asks two things of a model at once: that it be trusted on patients it has never seen and that it run on the hardware in a person's pocket. For that setting, the reframing from global separability to per-subject consistency is the contribution we hope proves the most durable.

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APPENDIX A

SUPPLEMENTARY DETAIL ON REVIEWED METHODS

Chapter 2 compresses several prior systems into a sentence or two each, so that the narrative stays on the thread that motivates this dissertation. This appendix restores the detail behind those summaries: the reported-performance tables, the two recurrence-plot pipelines that the chapter names but does not unfold, the engineered-feature predictor from the Huawei Heart Study, and the survival-analysis background. Each section is keyed to the point in the review where the corresponding work is first cited.

A.1 Reported Performance of Learned Detectors and Risk Scores

A.1.1 Neural-Network Detectors

Three families of network recur in the detection literature, and their reported operating points diverge sharply. A deep neural network trained on more than 90,000 single-lead ECGs reached an area under the curve above 0.91 across ten arrhythmia classes [28]. Deep belief networks, stacked from three-layer restricted Boltzmann machines, hold sensitivities of 96–100% even when the signal-to-noise ratio drops to a few decibels; this robustness is what recommends them for noisy portable recordings [31]. Convolutional networks report lower headline sensitivity in this particular comparison; even so, they scale to the largest cohorts and match expert over-reading at that scale [29]. Table A.1 collects the figures reported in the systematic review [2].

TABLE A.1: Reported performance of neural-network architectures for AF detection [2].

Metric	DNN	DBN	CNN
Overall sensitivity	86.1%	73.4–100% (-5 to 10 dB) 96.4–100% (-5 to 5 dB)	79%
Overall accuracy	AUC 0.91	75–99.5%	79.4%

A.1.2 Traditional Clinical Risk Scores

The traditional risk scores share one weakness: they transfer poorly off the cohort that produced them. Table A.2 lists each model with its input variables, the AUC reported on its development set, and the AUC obtained on an independent validation cohort. The decline is starkest for C_2HEST, which fell from 0.75 on a Chinese derivation set to 0.59 on a Danish validation set [53, 55]. HATCH moves in the opposite direction across two Taiwanese cohorts, a reminder that transfer depends as much on cohort composition as on the score itself.

TABLE A.2: Traditional clinical AF risk scores and their predictive performance [3].

Risk score	Variables	Original study (AUC)	Validation study (AUC)
FHS [105]	Age, sex, BMI, SBP, hypertension treatment, PR interval, murmur, CHF	4,764 US patients, AUC 0.78 (95% CI 0.76–0.80)	49,599 US patients, AUC 0.73 [106]
CHARGE-AF [52]	Age, ethnicity, height, weight, BP, smoking, antihypertensive use, DM, CHF, MI	18,556 US patients, AUC 0.765 (95% CI 0.748–0.781)	114,475 Netherlands patients, AUC 0.74 (95% CI 0.73–0.74) [107]
C_2HEST [53]	CAD/COPD, hypertension, age, CHF, hyperthyroidism	471,446 Chinese patients, AUC 0.75 (95% CI 0.73–0.77)	1,047,330 Danish patients, AUC 0.59 (95% CI 0.59–0.59) [55]
HATCH [54]	Hypertension, age, stroke/TIA, COPD, CHF	670,804 Taiwanese patients, AUC 0.716 (95% CI 0.71–0.72)	692,691 Taiwanese patients, AUC 0.77 [108]

A.2 Limitations of Cardiac Imaging for AF Prediction

Imaging markers predict AF well inside the cohorts that produce them, and machine learning applied to cardiac CT morphology has pushed AUCs for post-ablation recurrence to 0.78–0.87 [65, 66]. Five obstacles nonetheless keep imaging out of continuous, population-scale screening; they are the reason this dissertation works from interval series rather than images [3].

- **Small, procedural cohorts.** Most imaging studies draw on small or peri-ablation populations, which restricts how far their findings generalise.
- **No systematic population use.** Unlike clinical and electronic-health-record data, imaging has not been folded into large prospective AF risk models.
- **Cost and feasibility.** Routine CT or MRI for asymptomatic screening is impractical on grounds of cost, limited availability, and, for CT, radiation exposure.
- **Selection bias.** Imaging cohorts are usually patients with an existing cardiovascular indication, which can overstate predictive performance relative to the general population.
- **Data complexity.** Preparing imaging data for machine learning needs specialised pipelines, a higher technical barrier than ECG- or record-based approaches.

A.3 Recurrence-Plot Prediction from the Full ECG

The chapter notes that [4] recast the interval series as a two-dimensional recurrence plot and classified it with a ResNet, reporting 97% accuracy on the AFPDB [78]. The full pipeline runs in two stages, data preparation and data input, and is worth setting out because it fixes the vocabulary that the WARN pipeline in Section A.4 reuses.

ECG signals of varying length are first cut into fixed ten-second segments. Each segment is min-max normalised to suppress inter-database scale differences,

$$x_i' = \frac{x_i - \min(x)}{\max(x) - \min(x)}. \tag{A.1}$$

Raw ECG carries baseline drift, powerline interference, and muscle artefact, which surface as low-frequency trends and high-frequency fluctuations. An adaptive wavelet denoiser removes them. The Daubechies db5 wavelet drives the multi-level decomposition, chosen for its resemblance to the QRS complex and its energy concentration in the low band. A soft-threshold rule is then applied to the wavelet coefficients before reconstruction, with threshold

$$T = \delta\sqrt{2\log N}. \tag{A.2}$$

The denoised one-dimensional series is then embedded as a recurrence plot. For a series $x = x(i)$, $1 \leq i \leq N$, the m-dimensional state-space reconstruction is

$$u_m(i) = \{x(i),\, x(i+\tau),\, \ldots,\, x(i+(m-1)\tau)\}, \tag{A.3}$$

with $1 \leq i \leq N-(m-1)\tau$, time delay τ, and embedding dimension m; this study sets $\tau = 3$ and $m = 3$. The recurrence matrix follows from thresholding the pairwise state distance,

$$R_{i,j} = \Theta(\varepsilon - \|u_m(i) - u_m(j)\|)\,, \tag{A.4}$$

where $\|\cdot\|$ is the Euclidean norm, ε is set to 10% of the largest pairwise distance, and Θ is the Heaviside step,

$$\Theta(x) = \begin{cases} 1, & x > 0 \\ 0, & x \leq 0. \end{cases} \tag{A.5}$$

This maps the distance matrix to a binary recurrence matrix of size $[N-(m-1)\tau] \times [N-(m-1)\tau]$.

From that matrix the authors derive a panel of recurrence-quantification metrics that read off point density, diagonal structure, and vertical structure. With M the matrix dimension and $l_{\min}$ a minimum line length of two, the recurrence rate counts the fraction of recurrent points,

$$\mathrm{REC} = \frac{1}{M^2} \sum_{i,j=1}^{M} R_{i,j}, \tag{A.6}$$

determinism gives the share of recurrent points lying on diagonals,

$$\mathrm{DET} = \frac{\sum_{l=l_{\min}}^{M} l\, p(l)}{\sum_{i,j=1}^{M} R_{i,j}}, \tag{A.7}$$

and the entropy of the diagonal-length distribution measures structural disorder,

$$\mathrm{ENTR} = -\sum_{l=l_{\min}}^{M} p(l) \ln p(l). \tag{A.8}$$

Trapping time captures how long the trajectory lingers near a state,

$$\mathrm{TT} = \frac{\sum_{v=v_{\min}}^{M} v\, p(v)}{\sum_{v=v_{\min}}^{M} p(v)}, \tag{A.9}$$

laminarity gives the share of points in vertical structures,

$$\mathrm{LAM} = \frac{\sum_{v=v_{\min}}^{M} v\, p(v)}{\sum_{i,j=1}^{M} R_{i,j}}, \tag{A.10}$$

and the mean diagonal length averages the lines that exceed $l_{\min}$,

$$L_{\mathrm{mean}} = \frac{\sum_{l=l_{\min}}^{M} l\, p(l)}{\sum_{l=l_{\min}}^{M} p(l)}. \tag{A.11}$$

The recurrence plots feed a ResNet built from three parts: convolutional layers, improved residual blocks, and fully connected layers. The first stage extracts the quantification features above (recurrence rate, entropy, trapping time, and the longest diagonal), and the residual blocks deepen the representation before classification. Table A.3 reports the separation the authors found between pre-AF and non-AF windows. Recurrence rate, trapping time, and entropy rise in pre-AF windows, and trapping time gives the cleanest split, with means of 0.726 against 0.605 at $p = 0.00037$.

This pipeline depends on the full ECG waveform and screens offline rather than in real time, the two constraints that the wrist-deployable line of work sets out to remove.

A.4 The WARN Early-Warning Pipeline

WARN [1] is the closest prior work to this dissertation, and the chapter reports only its headline lead time and accuracy. This section reconstructs the pipeline end to end,

TABLE A.3: Statistical comparison of recurrence-quantification features between pre-AF and non-AF windows [4].

Feature	**Pre-AF (mean $\pm$ SD)**	**Non-AF (mean $\pm$ SD)**	**p-value**
Recurrence rate	0.544 ± 0.182	0.479 ± 0.127	0.00095
Determinism	0.999 ± 0.033	0.998 ± 0.035	0.814
Entropy	0.462 ± 0.169	0.441 ± 0.141	0.042
Trapping time	0.726 ± 0.028	0.605 ± 0.023	0.00037
Laminarity	0.999 ± 0.016	0.997 ± 0.015	0.712
L_{max}	0.369 ± 0.125	0.324 ± 0.129	0.008
L_{mean}	0.172 ± 0.136	0.166 ± 0.128	0.078

from signal acquisition through the early-warning decision and external validation.

A.4.1 Signal Preparation and Pre-AF Labelling

WARN acquires lead-II ECG at 128 Hz with 10-bit resolution, applies a 0.5–40 Hz band-pass filter to suppress noise, and detects R-peaks with the Pan–Tompkins algorithm, whose error rate sits near 1% [109]. Each recording is labelled into three regimes: sinus rhythm, pre-AF, and AF. The pre-AF label is the delicate one, and the authors derive it in five steps.

1. From the annotated AF onset, slide a five-minute window backward in time with a 30-second overlap.
2. Inside each five-minute segment, apply a secondary 30-second window with a five-second step to extract smaller samples.
3. Run Pan–Tompkins on lead II within each 30-second window to recover the R–R intervals.
4. For each 30-second segment, compute the coefficient of variation of the RRI and build a histogram over the parent five-minute window.
5. Track the histogram backward until its median drops below 0.7.

The 0.7 cutoff is the intersection of the coefficient-of-variation distributions for AF and sinus rhythm in the training set. The pre-AF span runs from the start of that final window to AF onset. These spans vary in length across patients, and across separate onsets within one patient, because the heart's morphology and electrical substrate shift over time [110].

A.4.2 Recurrence Plots and the Classifier

Each 30-second RRI segment becomes a 224×224 recurrence plot, sized for standard image-classification networks. The plot stores the pairwise distance between states,

$$R_{ij} = \|x(i) - x(j)\|, \tag{A.12}$$

with $\|\cdot\|$ the Euclidean norm, following Takens' time-delay embedding [111]. The embedding parameters differ by signal: $m = 2$ and $\tau = 3$ for RRI, and $m = 5$ and $\tau = 6$ for ECG. Both follow established criteria, with m taken at the first local minimum of the mutual-information function [112] and τ the smallest value for which the share of false nearest neighbours falls below 10% [113]. Read side by side, the plots make the rhythm change visible: high recurrences thin out from sinus rhythm to AF as periodicity weakens, while the pre-AF state shows cross-shaped regions left by the intermittency of the transition.

The recurrence plots feed EfficientNetV2, a 479-layer convolutional network released by Google in 2021 [114, 115]. It accepts 224×224 inputs, and its final fully connected layer is rewritten for three-class output, producing $P(\mathrm{SR})$, $P(\text{pre-AF})$, and $P(\mathrm{AF})$.

The authors swept the sampling window from 10 seconds to 5 minutes. A 30-second window won, trading sample count against temporal context: longer windows starve the model of training samples, while shorter ones lose the dynamics that signal an approaching episode. Table A.4 reports the sweep.

TABLE A.4: Effect of sampling-window length on WARN accuracy [1].

Length (s)	Samples ($\times 10^6$)	Accuracy
10	2.3	0.70
30	0.8	0.74
60	0.4	0.72
120	0.2	0.69
300	0.1	0.66

At the chosen window, the authors compared EfficientNetV2 against the two architectures most common in arrhythmia prediction, a one-dimensional CNN and an LSTM. WARN reached a mean validation accuracy of 0.74 with a low cross-fold standard deviation of 0.03, ahead of both baselines. Table A.5 gives the ten-fold breakdown.

TABLE A.5: Ten-fold cross-validation accuracy for WARN and two benchmark networks [1].

Model	1	2	3	4	5	6	7	8	9	10	Average
EfficientNetV2	0.78	0.71	0.73	0.73	0.73	0.71	0.75	0.75	0.76	0.71	0.74
LSTM	0.64	0.61	0.63	0.61	0.61	0.58	0.60	0.63	0.57	0.60	0.60
1D-CNN	0.71	0.68	0.67	0.71	0.69	0.67	0.68	0.67	0.70	0.70	0.69

A.4.3 From Class Probabilities to an Early Warning

WARN turns the class probabilities into a single running risk, the probability of danger,

$$P(\text{danger}) = P(\text{pre-AF}) + P(\mathrm{AF}), \tag{A.13}$$

the chance that the current window already sits in the pre-AF or AF regime. This signal is noisy, so a non-anticipative moving average smooths it, averaging every sample up to and including the current one without looking ahead. A binary alarm then fires when the smoothed signal crosses a probability threshold. Two hyperparameters govern the behaviour: the moving-average window length and that threshold.

Both were tuned on 60-minute sequences drawn per patient, one before AF onset and one during sinus rhythm. The 60-minute horizon was set because more than 70% of pre-AF spans last under an hour. Sinus-rhythm samples were taken at least two hours before any onset, so that the median coefficient of variation of their RRIs matched the population sinus-rhythm value. The two objectives pull against each other: a low threshold lengthens the predicted horizon but costs accuracy, and pushing accuracy to its maximum of 88.3% leaves the horizon short. The authors therefore searched for settings where accuracy, sensitivity, specificity, and F1 all cleared 80% while the mean and median horizons both exceeded 30 minutes. Thirty-four combinations qualified. Among those they took the smallest moving-average window to spare on-device compute and memory, then picked the combination that maximised accuracy, F1, and the horizons. The result is a window of seven samples, about 1.5 minutes, with a threshold of 0.57. At that setting WARN scored 86.7% accuracy, 87.5% F1, 93.3% sensitivity, and 80% specificity, with mean and median horizons of 31.4 and 36.3 minutes.

A.4.4 Test Performance, Errors, and External Validation

On the held-out RRI test set WARN reached 83% accuracy at the 0.57 threshold, with strong AUROC and AUPRC; Table 2.1 reports these alongside the ECG and external-centre results. The error analysis is instructive. Of four false negatives, one was a sudden onset after a very stable sinus period, and the other three carried tachycardia, bradycardia, unstable baselines, or noise. Of 22 false positives, 13 involved premature atrial contractions, five premature ventricular contractions, six unstable baselines, four sinus tachycardia, and one atrial flutter. Fifteen of the 22 records were heavily noisy, which underlines how much clean electrode contact matters. Table A.6 lists the cases.

Noise is not the whole story. Some false positives may mark moments when the heart approached a transition to AF without crossing it, since stress or stimulants can push cardiac dynamics toward that tipping point. The pattern is sharpest for the 13 records with premature atrial contractions, which are well-known precursors of AF [116].

WARN was validated externally on three centres: Clínica y Maternidad Suizo Argentina (53 patients, 24-hour ECG), Groupe Hospitalier Privé Ambroise Paré–Hartmann in France (250 patients, 24-hour ECG), and the open-access AFPDB from PhysioNet (75 patients, 30-minute ECG) [78]. The validation drew 73 patients in total: 8 from Argentina, 25 from France, and 40 from PhysioNet. The PhysioNet set held 50 sinus-rhythm controls and 25 AF patients, recorded from 30 minutes before onset to 5 minutes after. After excluding 5 AF records with episodes under one minute and 2 con-

TABLE A.6: Misclassifications by WARN on the test dataset [1].

False negatives	
Sample	Observation
1	Very stable rhythm, only SR
2	Tachycardia, bradycardia, and noise
3	Tachycardia, unstable baseline, and noise
4	Unstable baseline
False positives	
Sample	Observation
1	Atrial flutter
2, 6, 14, 19	Noise and unstable baseline
3, 8	Multiple PVCs and PACs
4, 15	PACs and noise
5	PACs and sinus tachycardia
7	Sinus tachycardia with PACs and noise
9, 10, 16	Noise
11, 17, 22	PVCs, PACs, and noise
12	Sinus tachycardia, multiple PACs, and noise
13	Sinus tachycardia and unstable baseline
18	Multiple PACs and noise
20	PACs, unstable baseline, and long RR intervals
21	PACs

trols corrupted by AF or artefact, the balanced external set held 20 AF and 20 control records.

Performance on PhysioNet was lower, at 0.7 accuracy, 0.76 AUROC, and 0.79 AUPRC, with a shorter mean horizon of 12.9 minutes. Two factors explain the gap: WARN was trained only on patients already diagnosed with AF, and the 30-minute PhysioNet recordings cap how far ahead a warning can reach. Reproducibility caveats on this benchmark also complicate any direct comparison [99].

Finally, the authors retrained the network on raw ECG rather than RRI. AUROC and AUPRC rose by 5.5% and 9.1%, while accuracy and mean horizon stayed essentially level. The gain is modest given how much richer continuous ECG is than discrete RRI, which is the central evidence that accurate onset prediction needs only the interval series, the property that makes wrist-based early warning practical. On a separate PPG cohort of 554 patients, the largest in the study, 17 engineered RRI features fed to an XGBoost classifier reached 88% accuracy.

A.5 PPG-Based Prediction in the Huawei Heart Study

The PPG predictor of [79] forecasts AF onset in high-risk individuals and underpins the chapter's claim that pulse intervals can carry a prediction signal. It was built on the Huawei Heart Study [80], a large PPG-based screening programme whose dataset is proprietary, which limits reproducibility and head-to-head comparison.

The authors extracted 17 features from the RRI series, spanning the time, frequency, and nonlinear domains, plus one feature carried over from the upstream detector.

- **Heart-rate features:** minimum, mean, and median RR interval, and the skewness of the interval series (SkRR).
- **Heart-rate-variability features:** coefficient of variation of the intervals (CVRR), SDNN, an SD ratio, and pNN50.
- **Entropy and dynamics:** sample entropy, Shannon entropy, approximate entropy, a Poincaré-stepping feature, a frequency-domain feature, an interval-difference feature, and the PPG monitoring time.
- **User-defined:** AFprob, the AF probability emitted by the prior Huawei detection model (M0).

After comparing random forests, support vector machines, gradient boosting, and XGBoost, the authors trained an XGBoost classifier (M1) on 554 patients. It reached an AUC of 0.94 four hours before onset. They then optimised it into M2 by adding two feature groups and tuning hyperparameters. HRR features combine the current interval features with the recent history of those features, so the model sees trends rather than instantaneous values. CHRR features add contextual information on top, widening the temporal and situational view. M2 reached an AUC of 0.97, accuracy 0.93, precision 0.91, F1 0.86, sensitivity 0.85, and specificity 0.97 at the same four-hour horizon, ahead of M1 and the other baselines. Table A.7 reports both models across two test splits.

TABLE A.7: Predictive ability of the primary (M1) and optimised (M2) AF models [5].

	Test 1		Test 2	
	M1	M2	M1	M2
Accuracy	0.912 (0.910–0.913)	0.935 (0.933–0.936)	0.891 (0.890–0.893)	0.934 (0.933–0.936)
Precision	0.864 (0.860–0.868)	0.914 (0.910–0.917)	0.840 (0.836–0.844)	0.912 (0.909–0.916)
Sensitivity	0.775 (0.772–0.781)	0.821 (0.817–0.825)	0.730 (0.725–0.735)	0.813 (0.809–0.827)
Specificity	0.958 (0.957–0.959)	0.974 (0.973–0.975)	0.950 (0.948–0.951)	0.973 (0.972–0.974)
F1 score	0.818 (0.815–0.821)	0.865 (0.862–0.868)	0.781 (0.778–0.784)	0.865 (0.863–0.868)
AUC	0.942 (0.933–0.943)	0.971 (0.970–0.981)	0.932 (0.923–0.935)	0.972 (0.971–0.982)

M2 cut the false-positive rate relative to M1, though many residual false alarms traced to atrial arrhythmias other than AF, such as bigeminy and flutter; separating impending AF from these disturbances stays hard. Three limitations bound the work. The dataset is proprietary, the architecture and any edge deployment are described only thinly, and validation was confined to small high-risk cohorts, so transparency, reproducibility, and broad generalisability all remain open. The study also gives little account of why each feature was chosen or how it shaped a prediction.

A.6 Survival Analysis for Disease Prediction

Survival analysis, or time-to-event analysis, models the duration until a specified event occurs [117]. It was developed for clinical studies of time to death; it now serves prediction of disease onset, recurrence, and progression. Where ordinary classification asks whether an event will occur, survival analysis asks when, which is the framing that the early-warning task in this dissertation ultimately shares.

Two functions describe the process. The survival function gives the probability of staying event-free beyond a given time, and the hazard function gives the instantaneous risk at a given time. The Cox proportional-hazards model is the classical workhorse for identifying risk factors, and its time-dependent extensions let covariates vary over the follow-up. Parametric alternatives such as accelerated failure-time models, with Weibull or log-normal baselines, estimate the actual time remaining rather than a relative risk.

Machine learning has since reshaped the field [118]. Random survival forests capture nonlinear interactions among covariates without strong parametric assumptions, and support-vector approaches perform well on engineered physiological features. Deep survival models extend the idea further: DeepSurv recasts the Cox model with a neural network to learn nonlinear risk functions [119], and DeepHit models the joint distribution of event times while supporting competing risks. For longitudinal and streaming data, dynamic models such as Dynamic-DeepHit combine recurrent networks with attention to update the risk estimate as new observations arrive [120].

APPENDIX B

EXPLORATORY EXPERIMENTS ON AF ONSET PREDICTION

The detection contribution of this dissertation did not arrive in a straight line. Before the patient-aware contrastive objective took its final form, we spent a long exploratory phase trying to predict *how close* a stretch of sinus rhythm sat to the next episode of atrial fibrillation, rather than simply classifying the rhythm in front of us. That prediction line largely failed under strict patient-independent evaluation, and the shape of its failure is what pointed us at the detection formulation that succeeded. This appendix documents the prediction experiments and their outcomes; Appendix C then dissects the mechanism that defeated them and the invariance methods we tried against it. Both appendices report negative results. We include them because the negative evidence is load-bearing: it is the reason the final method preserves per-patient sinus-rhythm structure instead of pooling every patient into one class.

B.1 Task Formulation and Evaluation Protocol

The prediction task took a window of inter-beat (RR) intervals drawn from sinus rhythm and asked a model to estimate the time remaining until atrial-fibrillation onset. We framed this three ways over the course of the work: direct regression onto the minutes-to-onset, a three-class coarsening of that target into Near, Mid, and Far bins, and a within-patient ranking of segments by their proximity to onset. The IRIDIA-AF database supplied the recordings. In the baseline phase each patient contributed a sinus-rhythm window of ninety minutes before every annotated onset and an hour of atrial fibrillation per episode. We cut each window into segments of ten sub-windows of one hundred RR values, so a single segment spanned a thousand intervals, roughly two minutes of rhythm. A RobustScaler standardised the intervals and onset times were normalised to the unit range.

One evaluation principle governed everything that follows; it is the principle the naive experiments learned to respect the hard way: train and test patients must be disjoint. A model that has seen any segment from a patient can memorise that patient's baseline and recover an inflated score. We therefore report patient-grouped splits (GroupKFold or an explicit patient holdout) as the only honest measure of whether a temporal signal transfers to a new individual. Section C.1 of Appendix C quantifies exactly how large the gap between the two protocols became.

B.2 Foundations That Carried Forward

Two early results survived the cull and became the backbone of the detection work; they deserve a note here. First, contrastive predictive coding pretraining worked once we replaced within-batch negatives with cross-batch InfoNCE; the within-batch variant failed to converge, while the cross-batch objective gave a stable next-step accuracy around seventy per cent on held-out patients. Second, a linear head over those frozen representations separated atrial fibrillation from sinus rhythm at roughly 92.7% macro-F1 on unseen patients. That number is the seed of this dissertation. It told us the representation carried a strong, transferable *rhythm-state* signal even though, as the rest of this appendix shows, it carried almost no transferable *time-to-onset* signal.

B.3 Direct Regression and Time-Bin Classification

The first serious attempts predicted onset time directly; they exposed the central problem at once. A single patient is easy: fit a regressor to one person's segments and the ordering is almost perfect. Across the cohort that ease evaporates. Figure B.1 audits ninety patients individually. The best cases reach a Spearman ρ near 0.97 with visibly diagonal predicted-versus-actual plots, but a long tail collapses to flat lines at the patient's mean onset time, with ρ at or below zero. Ten patients qualified as excellent, twenty-one as poor. The audit makes the danger of pooling concrete: a single global model averages a bimodal population and inherits the failures.

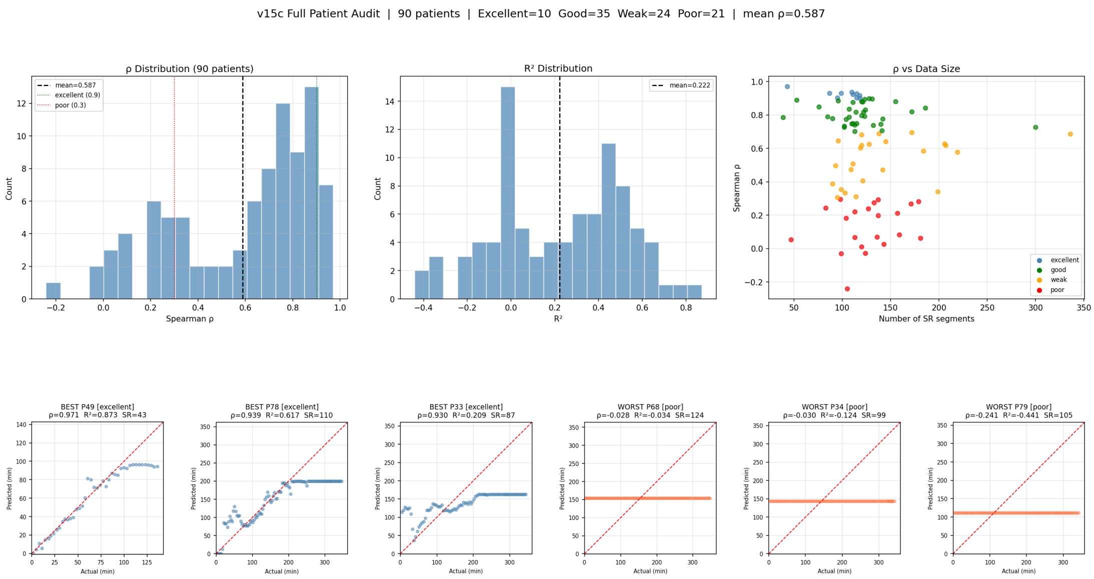

Fig. B.1: Per-patient audit of direct RR-to-onset regression across ninety patients. Top row: distributions of Spearman ρ and R^2, and ρ against the number of sinus-rhythm segments, coloured by quality tier. Bottom row: predicted against actual onset time for the three best and three worst patients. Strong patients track the diagonal; weak patients regress to a flat prediction at their own mean, which is what pooling across patients inherits.

Pooling confirmed the fear. A single regressor trained on all patients reached a global ρ of only 0.154 with a mean absolute error near 18.8 minutes. Restricting training to the excellent tier and holding patients out did not rescue it; the global ρ fell slightly negative, to -0.027, at a mean absolute error of 19.3 minutes. Filtering the noisy patients out, in other words, was not enough, because the failure was not noise but transfer. The three-class coarsening told the same story from a different angle. A classifier over frozen contrastive features reached a macro-F1 of 37.8% against a three-class chance floor of 33.3%. Table B.1 collects the direct-modelling results.

TABLE B.1: Direct time-to-onset prediction under a patient-held-out protocol. The three-class chance floor is 33.3%; a Spearman ρ near zero indicates no transferable ordering.

Setting	Split	Metric	Value
Single patient (best case)	Within patient	Spearman ρ	$\sim$0.97
Regression, all patients pooled	Patient held-out	Global ρ / MAE	0.154 / 18.8 min
Regression, excellent tier only	Patient held-out	Global ρ / MAE	-0.027 / 19.3 min
Three-class time bins (CPC features)	Patient held-out	Macro-F1	37.8%

B.4 Temporal Ranking and Cox Survival Modelling

Direct regression collapsed to the mean, so the next idea was to stop asking for absolute minutes and to ask instead for an ordering: which of two segments sits closer to onset. This line also gave the project its early working name, ST-CoxNet, because it paired a contrastive encoder with a survival head. The encoder was a multi-scale residual network with squeeze-and-excitation attention, trained jointly with a contrastive-predictive-coding objective and a temporal ranking loss that used a Gaussian kernel over normalised time-to-onset distances. A Cox proportional-hazards head, in the survival-analysis tradition set out in Appendix A, then read a hazard off the frozen representation.

The first version, v13, computed its ranking comparisons across the whole batch, mixing segments from different patients. It surfaced the failure that governs both of these appendices. The cross-patient ranking taught the encoder to tell patients apart rather than to place segments in time: the similarity gap between near-onset and far-onset pairs stayed small, the within-sinus-rhythm Spearman correlation sat near zero, and the Cox head reached a concordance index of only about 0.55, barely above the 0.5 of a coin toss. High segment overlap, ninety per cent at a stride of twenty, made the identity shortcut easier still. A numerical footnote from this run is worth recording: the temporal loss produced not-a-number values from a $0 \times (-\infty)$ term in the log-softmax, fixed by masking the diagonal of the log-probability matrix before the multiply.

The diagnosis pointed at an obvious repair, and v14 applied it. Restricting the ranking loss to comparisons *within* a patient, with a patient-balanced sampler that drew sixteen patients and thirty-two segments each, lifted the within-sinus-rhythm correlation from about 0.28 at the first epoch to 0.47 by the fiftieth. The encoder was now learning genuine within-patient temporal structure. Yet the Cox concordance barely moved, from roughly 0.55 to 0.553, and the survival head overfit within a dozen epochs. The ordering the encoder had learned lived inside each patient and would not cross to a held-out one. That split, strong within-patient structure beside near-random cross-patient transfer, is the exact pattern every later experiment reproduced, and Appendix C takes up why it proved so stubborn.

B.5 Multimodal and Sequence-Level Models

If a single short segment carried too little signal, perhaps richer inputs or longer context would help. We added fifteen hand-engineered heart-rate-variability features alongside the learned representation. We also replaced single-segment classification with trajectory models that read a run of consecutive segments. The gains were real but small. They plateaued fast. Global normalisation of the multimodal features actually hurt, dropping macro-F1 to 34.0%. Per-patient normalisation, which z-scores each feature within a patient and so strips the baseline offset, recovered the signal to 42.8%. A Transformer over sixteen-segment trajectories edged that to 43.5%, the best result in the family; a lighter gated-recurrent model reached 41.0% while overfitting less. Table B.2 lists them.

TABLE B.2: Three-class time-bin macro-F1 under a patient-held-out protocol (chance 33.3%). Per-patient normalisation is the single largest lever; architecture is not.

Model	Normalisation	Macro-F1
CPC features only	Global	37.8%
CPC + HRV	Global	34.0%
CPC + HRV	Per-patient	42.8%
Sequence Transformer	Per-patient	**43.5%**
Sequence GRU	Per-patient	41.0%

The training curves settle the question of whether capacity was the bottleneck. Figure B.2 tracks the sequence Transformer: training macro-F1 climbs past 0.90 while validation macro-F1 stays pinned in the high thirties, a hair above chance, for every epoch. Heavy regularisation, dropout at 0.4, label smoothing, and weight decay at 0.05, slowed the divergence but never closed it. A half-million-parameter Transformer and an eighty-thousand-parameter recurrent network overfit to the same validation ceiling. The problem was never model size.

B.6 Dataset Redesign for Longer Context

The plateau suggested the data recipe itself was the limit rather than the model. Two minutes of RR intervals may simply not contain the slow autonomic drift that precedes an episode, and a ninety-minute pre-onset window with a short stride produced heavily overlapping, near-redundant segments. We drafted a redesign that widened the sinus-rhythm window from ninety minutes first to two and a half hours and then to six, relaxed the minimum atrial-fibrillation duration so that more patients survived the filtering, and increased the stride to two hundred intervals to cut the overlap. Relaxing the requirements paid off in cohort size on paper: the baseline recipe admitted

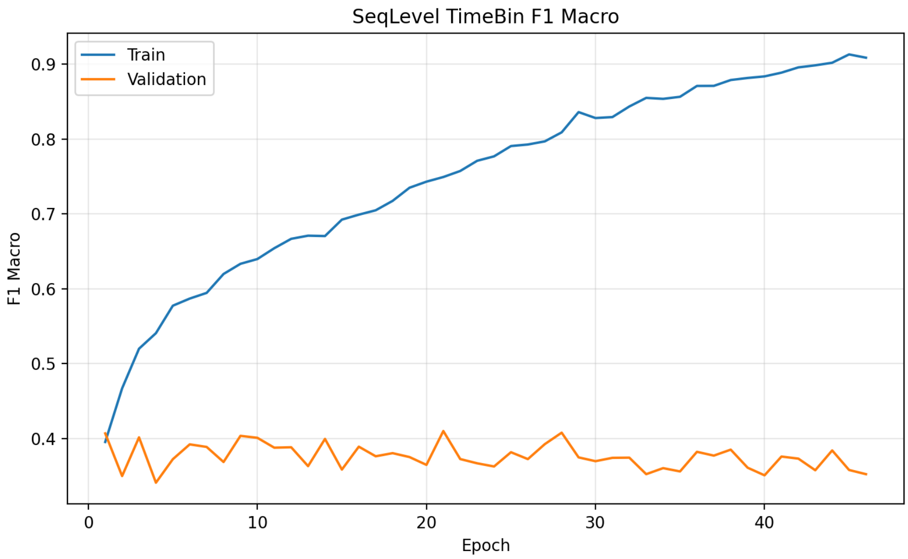

Fig. B.2: Training and validation macro-F1 for the sequence-level Transformer across epochs. Training accuracy climbs toward 0.90 while validation stays near the three-class chance floor of 0.33, the signature of a model memorising patient-specific structure that does not transfer to held-out patients.

128 episodes from 102 patients, while the two-and-a-half-hour window admitted 164 episodes from 127 patients. This line was planning and partial preparation rather than a completed set of runs. It stands as a direction the negative results opened, not a result in itself. Its premise, that longer context and cleaner sampling matter more than architecture, is consistent with everything the completed experiments showed.

APPENDIX C

PATIENT-IDENTITY SHORTCUTS AND THE LIMITS OF INVARIANCE

Appendix B showed that onset prediction succeeds within a patient and fails across patients. This appendix explains why. It also records the methods we deployed against the cause once we understood it. The short version is that the models were not learning when an episode would arrive; they were learning who the patient was. Every attempt to strip that identity out of the representation, or to condition around it, produced at best a weak and unstable improvement. That failure is instructive, because it is the exact failure the detection method in Chapter 3 was designed to avoid by construction.

C.1 Diagnosing the Failure: Leakage and Identity Shortcuts

The shortcut first announced itself in the temporal-ranking experiments of Appendix B, where a cross-patient ranking objective learned patient identity instead of time. The cleanest evidence for it, though, came from a deliberately simple baseline. A random forest on the engineered features reached 82.2% cross-validation accuracy when segments were split without regard to patient, and 34.6% when the same features were split by patient group. That 47-point collapse has only one explanation: with segments from a patient present on both sides of the split, the forest memorised the patient's baseline and read the answer off identity rather than off any pre-onset dynamics. Under an honest patient-grouped split the memorised cue vanishes and performance drops to chance. The lesson generalises well beyond the forest; it is why every table in Appendix B reports a patient-held-out protocol.

Per-patient normalisation is the natural response. It helped in every setting we tried, which is telling in itself. Z-scoring each feature within a patient removes the baseline offset that identity rides on and exposes whatever relative change remains. It lifted the random forest from 34.6% to 42.5% and the multimodal network from 34.0% to 42.8%, gains of roughly eight points in both cases. The consistency matters more than the magnitude: relative change within a patient carries a real if faint temporal signal, whereas absolute interval values across patients carry almost none. A variance decomposition of the engineered features put a number on the confound. The mean intraclass correlation across the heart-rate-variability features was 0.274, moderate rather than overwhelming, so patient identity does not swamp the features outright; it is enough, though, to hand a segment-level split an easy shortcut, which is why the patient-grouped protocol is not optional.

How faint that signal is comes through in Figure C.1. For each engineered feature we computed the within-patient Spearman correlation against time-to-onset and plotted the distribution over patients. Every feature centres near zero, with means between -0.15 and $+0.07$ and a wide spread that straddles both signs. No single feature orders segments by proximity to onset in any patient-independent way. The temporal signal is present, but weak, inconsistent in direction, and swamped by between-patient variation.

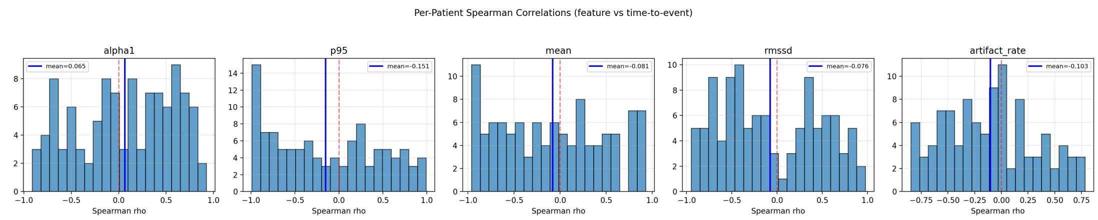

Fig. C.1: Distributions of within-patient Spearman correlation between five heart-rate-variability features and time-to-onset, one histogram per feature over the patient cohort. Every mean sits near zero and the spread crosses both signs, so no feature provides a consistent, patient-independent ordering toward onset.

We also asked whether the learned contrastive representation captured more temporal signal than these simple statistics. It did not. Figure C.2 compares the median within-patient correlation of the best heart-rate-variability features against the strongest dimensions of the contrastive encoder and context. The engineered features win: pNN50 alone reaches a median magnitude near 0.56, above every learned dimension, and the encoder's mean magnitude of 0.19 trails the feature set's 0.43. For onset timing specifically, the representation we would eventually prize for detection was no better than a handful of textbook heart-rate-variability numbers.

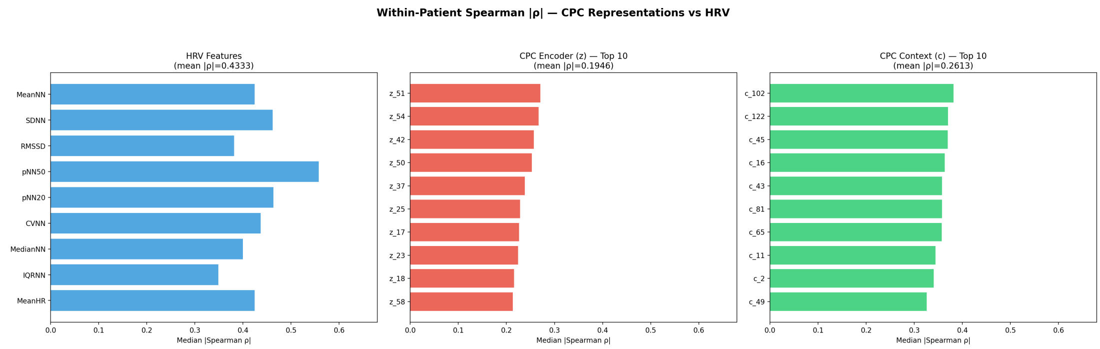

Fig. C.2: Median within-patient Spearman magnitude against time-to-onset for engineered heart-rate-variability features (left) versus the top contrastive encoder (centre) and context (right) dimensions. The best hand-crafted feature, pNN50, exceeds every learned dimension, so the pretrained representation held no extra time-to-onset signal over simple statistics.

C.2 Domain-Invariance Objectives

If the models leaned on patient identity, the textbook remedy is to train it out. We built a regression network on the excellent-tier patients with a strict patient-level split and attached one of three invariance mechanisms to the shared representation: a gradient-reversal layer with an adversarial patient classifier, a maximum-mean-discrepancy penalty that aligns feature distributions across patients, and instance normalisation inside the encoder. Each ran with relative-RR and raw-RR inputs, giving the six variants in Table C.1. A nearest-centroid probe measured how well the resulting embedding hid patient identity from a held-out set, where an area under the curve near 0.5 means identity is no longer recoverable.

TABLE C.1: Domain-invariance variants on the excellent-tier onset-regression task, patient-held-out. A higher ρ is better; an unseen-patient probe AUROC near 0.5 indicates identity has been removed. No variant achieves both at once.

Variant	Method / input	MAE (min)	Global ρ	Unseen AUROC
v21a	GRL, relative RR	18.40	0.271	0.490
v21b	GRL, raw RR	19.34	-0.004	0.440
v21c	MMD, relative RR	18.24	**0.301**	0.544
v21d	MMD, raw RR	18.86	-0.093	0.472
v21e	InstanceNorm, relative RR	19.38	-0.144	0.571
v21f	InstanceNorm, raw RR	19.38	0.091	0.549

The pattern is discouraging in a specific way. The relative-RR variants order segments a little better than chance, with the maximum-mean-discrepancy run reaching a global ρ of 0.301, but their probe scores sit near or slightly above 0.5, so the ranking gain does not come from a cleaner, identity-free representation. A separate consolidated probe made this explicit. Linear patient classifiers trained on the seen patients still recovered identity far above the random baseline of about 0.023, while their ability to flag unseen patients ranged from 0.41 to 0.53, at or below chance. Across gradient reversal, maximum-mean-discrepancy alignment, and instance normalisation, we never produced an embedding that both dropped identity and transferred an onset ordering.

C.3 Conditional and Adaptive Models

The opposite strategy is to stop fighting patient variation and to model it instead, giving each prediction a few segments of context from the same patient to calibrate against. We tried a spread of these. A pairwise-ranking objective learned to order two segments from the same patient by proximity to onset; training pairwise accuracy climbed to 87% while validation stalled at 51%, a single point above the coin-flip floor, so the ordering it learned was patient-specific and did not carry over. Conditional models built a support-set context, by mean or by attention, and predicted a query segment from the conditioned representation. Feature-wise linear modulation generated per-query scale and shift parameters from that context, in one-stage and two-stage forms, the latter pretraining an invariant representation before fine-tuning the regressor. Table C.2 gathers the family.

Read down the ρ column and the story is monotonous in the worst way: everything lands between 0.19 and 0.25, every R^2 hovers around zero, and the identity probes stay

TABLE C.2: Conditional, ranking, and modulation variants on the excellent-tier onset task, patient-held-out. Every variant lands at a weak positive ρ with an R^2 near or below zero and an identity probe near chance.

Variant	**Approach**	**MAE (min)**	**Global** ρ	R^2
v22a	Conditional mean context	17.58	0.251	0.056
v22c	Conditional attention context	17.47	0.126	-0.275
v23	Pairwise ranking	—	0.187	—
v24	GRL + cross-patient contrastive	18.44	0.240	—
v25	Two-stage GRL + contrastive	19.08	0.199	-0.082
v26	FiLM, mean context	18.03	0.212	-0.015
v27	FiLM, attention context	17.63	0.230	0.007

near chance throughout. Some variants nudged the embedding geometry, the mean-context modulation lifted its unseen-patient probe to 0.58, but none of that reshaping translated into an onset prediction that a clinician could trust on a patient the model had never seen. Conditioning on patient context was, at best, a marginal and unstable win over doing nothing.

C.4 Lessons That Shaped the Detection Objective

Two conclusions came out of this long phase, and both are structural rather than incidental. The dominant bottleneck was cross-patient domain shift and baseline variability, not model capacity, and leakage control was non-negotiable, since a segment-level split could inflate a chance-level task to 82%. The methods that ought to have removed the identity shortcut, gradient reversal, distribution matching, instance normalisation, all failed to produce a representation that both forgot the patient and kept a transferable signal.

That last finding reframed the whole project. The invariance methods failed because they tried to *erase* patient structure, and in a physiological signal the patient's own sinus-rhythm geometry is not noise to be averaged away; it is the reference against which a rhythm change is legible. A class-only contrastive objective makes the same mistake in gentler form, collapsing every patient's sinus rhythm into one shared prototype. The patient-aware constraint at the centre of this dissertation inverts that instinct. It forms positive pairs only from segments sharing both patient and rhythm class, which preserves each patient's individual sinus-rhythm structure rather than folding it into a population mean. The negative results in these two appendices are what taught us that preserving per-patient structure, not discarding it, is the property that transfers to unseen patients, and Chapter 4 shows that this is exactly the property that carries the detection model to an AUROC of 0.989 on patients it has never seen. The onset-prediction signal that defeated every model here survives, faintly, as the per-patient early-warning score of Section 4.7, read for free off that same frozen representation.